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# MetaDex Scanner Reference

Informatica MetaDex Scanner Reference
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# Preface

Refer to the *Informatica® Enterprise Data Catalog MetaDex Scanner Reference* for information about the conversion paths and modules that you can use to perform tasks such as configuring necessary parameters for metadata scanning in the MetaDex tool. This guide is written for administrators.

# Informatica Resources

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To find online support resources on the Informatica Network, visit https://network.informatica.com and select the eSupport option.

CHAPTER 1

# Conversion paths reference

## Advanced Database Objects Primary/Foreign Key addition to EDC

This conversion path extracts information about Primary and Foreign Keys from a database.

### List of modules

This chain calls the following modules:

- “General” on page 171
- “Database object reader” on page 132
- “Entry Point Detector” on page 166
- “Advanced PL/SQL Converter” on page 73
- “SQL Evaluator” on page 292
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “Graph Property Adder” on page 172
- “HTML Visualizer” on page 178
- “Assets Corrector” on page 78
- “Context Mapper” on page 130
- “Enterprise Data Catalog Writer” on page 156

### Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**Database object reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlDatabase | Database name |
| SqlHost | SqlHost |
| SqlSchema | Schema name |
| SourceModificationDate | Last source modification date |
| SqlExternalTable | Name of External Table |
| SqlTrigger | Name of the SQL Trigger |
| SqlExternalTablePath | File path of External Table |

**Entry Point Detector**

This module does not contain any graph properties.

**Advanced PL/SQL Converter**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedure | SQL Procedure name, if exists |
| SqlPackage | SQL Package name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |
| SqlView | Name of SQL View |
| GraphUri | VOLATILE: Fully qualified URI of Graph |
| SourceLocationFilename | Source file name |
| SourceLocationLineNumber | Position of SourceStatementText according to SourceCompleteText |

| Property name | Description |
|---|---|
| SourceLocationColumnNumber | Position of SourceStatementText according to SourceCompleteText |
| SourceLocation | DEPRECATED: Location of the source |
| SourceStatementText | Source Text |
| SourceCompleteText | Source Complete Text |
| SqlDatabase | Database name |
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlSchema | Schema name |
| SqlHost | Host name |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Advanced Database Objects to EDC

This conversion path extracts database objects definitions from a database, computes lineage, and then generates appropriate objects and flows.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “Database object reader” on page 132
- “Entry Point Detector” on page 166
- “Advanced PL/SQL Converter” on page 73
- “SQL Evaluator” on page 292
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “Graph Property Adder” on page 172
- “HTML Visualizer” on page 178
- “Assets Corrector” on page 78
- “Context Mapper” on page 130
- “Enterprise Data Catalog Writer” on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**Database object reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlDatabase | Database name |
| SqlHost | SqlHost |
| SqlSchema | Schema name |
| SourceModificationDate | Last source modification date |
| SqlExternalTable | Name of External Table |
| SqlTrigger | Name of the SQL Trigger |
| SqlExternalTablePath | File path of External Table |

**Entry Point Detector**

This module does not contain any graph properties.

**Advanced PL/SQL Converter**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedure | SQL Procedure name, if exists |
| SqlPackage | SQL Package name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |
| SqlView | Name of SQL View |
| GraphUri | VOLATILE: Fully qualified URI of Graph |
| SourceLocationFilename | Source file name |
| SourceLocationLineNumber | Position of SourceStatementText according to SourceCompleteText |

| Property name | Description |
|---|---|
| SourceLocationColumnNumber | Position of SourceStatementText according to SourceCompleteText |
| SourceLocation | DEPRECATED: Location of the source |
| SourceStatementText | Source Text |
| SourceCompleteText | Source Complete Text |
| SqlDatabase | Database name |
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlSchema | Schema name |
| SqlHost | Host name |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Advanced PL/SQL to EDC

Generates Enterprise Data Catalog links and objects from PL/SQL scripts.

## List of modules

This chain calls the following modules:

- "General" on page 171
- "Advanced PL/SQL Reader" on page 73
- "SQL Evaluator" on page 292
- "Assets Corrector" on page 78
- "Context Mapper" on page 130
- "Graph Property Adder" on page 172
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**Advanced PL/SQL Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlPackage | SQL Package name, if exists |
| SqlProcedure | SQL Procedure name, if exists |
| SourceLocationFilename | Source file name |
| SourceLocationFilePath | Source file path |
| SourceLocationLineNumber | Position of SourceStatementText according to SourceCompleteText |
| SourceLocation | DEPRECATED: Location of the source |
| ProcCallEntryId | Identifies procedure call. It's logical call's name. |
| ProcCallEntryComment | ProcCallEntryComment |
| SourceCompleteText | Source Complete Text |
| SqlDatabase | Database name |
| SqlHost | Host name |
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |
| SqlSchema | Schema name |
| SqlStatementComment | Comment for SQL statement |
| SqlProcedureComment | Comment for SQL procedure |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Azure Data Factory to EDC

## List of modules

This chain calls the following modules:

- "General" on page 171
- "Azure Data Factory Reader" on page 79
- "SSIS Processor" on page 315
- "Databricks evaluator" on page 137
- "Python Evaluator" on page 256
- "SQL Evaluator" on page 292
- "Assets Corrector" on page 78
- "Context Mapper" on page 130
- "Graph Property Adder" on page 172
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**Azure Data Factory Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceLocationFilePath | Source file path |
| SourceModificationDate | Last source modification date |
| AdfFactory | Factory name |

| Property name | Description |
|---|---|
| AdfPipeline | Pipeline name |
| AdfPipelineActivityName | Pipeline Activity name |
| AdfPipelineActivityDescription | Pipeline Activity description |
| AdfPipelineActivityType | The type of Pipeline Activity |
| PipelineFolder | Pipeline folder |
| PipelineDatasets | Pipeline datasets |
| PipelineDataflows | Pipeline dataflows |
| ActivityDataflow | Activity dataflow |
| AdfParentPipeline | Parent pipeline name |
| AdfParentActivity | Parent Activity name |
| PipelineRun | Pipeline run id |

**SSIS Processor**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocationFilePath | Source file path |
| InputRootPath | Root path containing input files |
| SourceDescription | Description of the source |
| SsisPackage | SSIS Package name |
| SsisDataTask | SSIS Data Task name |
| SsisTaskType | SSIS data task type |
| ParentPackageFiles | List of file names of SSIS Execute Package Task parent packages |
| ParentTasks | List of qualified SSIS Execute Package Task parent tasks |
| SsisScriptFilePath | Path to SSIS script |
| SsisScriptFilename | Name of the script file |

| Property name | Description |
|---|---|
| SsisScriptDirectory | Directory which contains script |
| SsisScriptLineNumber | Line number in SSIS script |
| SsisScriptLineText | Line text in SSIS script |
| SsisFileSystemTaskType | SSIS File System Task type |
| Application host name | Name used as output model name, host name of SSIS files and server name where the Visual Basic scripts are run. |

**Databricks evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Notebook name | Name of called Databricks notebook |

**Python Evaluator**

This module does not contain any graph properties.

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Bteq to EDC

Generates EDC objects and links from BTEQ scripts.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “BTEQ Reader” on page 98
- “SQL Evaluator” on page 292
- “Assets Corrector” on page 78
- “Context Mapper” on page 130
- “Graph Property Adder” on page 172
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “HTML Visualizer” on page 178
- “Enterprise Data Catalog Writer” on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**BTEQ Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocation | DEPRECATED: Location of the source |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# COBOL Parsing Test

Parses given COBOL programs and gathers parsing errors

## List of modules

This chain calls the following modules:

- "General" on page 171
- "COBOL Test Parser" on page 127
- "File Writer" on page 171

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**COBOL Test Parser**

This module does not contain any graph properties.

**File Writer**

This module does not contain any graph properties.

# COBOL to EDC

Generates EDC model from COBOL files.

## List of modules

This chain calls the following modules:

- "General" on page 171
- "COBOL Reader" on page 117
- "SQL Evaluator" on page 292
- "Column Properties PostProcessor" on page 129
- "Assets Corrector" on page 78
- "Context Mapper" on page 130
- "Graph Property Adder" on page 172
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**COBOL Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceLocationFilename | Source file name |
| SourceLocationFilePath | Source file path |
| SourceCompleteText | Source Complete Text |
| CobolProgramId | The PROGRAM-ID of the COBOL program. |
| CallingCobolProgramId | Id of COBOL program that called current program |
| CobolProgramAuthor | Author of the COBOL program. |
| CobolProgramInstallation | INSTALLATION property of COBOL program |
| CobolProgramDateWritten | DATE-WRITTEN property of COBOL program |
| CobolProgramDateCompiled | DATE-COMPILED property of COBOL program |
| CobolProgramSecurity | SECURITY property of COBOL program |
| CobolProgramCallStack | Call stack of COBOL program |
| CobolProgramCallId | Id of COBOL program call. E.g. first call of program 'PROG' will have id: 'PROG', second will have id: 'PROG_1 etc. |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Column Properties PostProcessor**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Databricks to EDC

Generates Enterprise Data Catalog links and objects from Databricks notebooks.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “Databricks Reader” on page 140
- “Unity Lineage” on page 364
- “Databricks Assets” on page 136
- “SQL Evaluator” on page 292
- “Python Evaluator” on page 256
- “Assets Corrector” on page 78
- “Context Mapper” on page 130
- “Graph Property Adder” on page 172
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “HTML Visualizer” on page 178
- “Enterprise Data Catalog Writer” on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**Databricks Reader**

This module does not contain any graph properties.

**Unity Lineage**

This module does not contain any graph properties.

**Databricks Assets**

This module does not contain any graph properties.

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Python Evaluator**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# FastLoad to EDC

Generates EDC objects and links from FastLoad scripts.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “FastLoad Reader” on page 167
- “SQL Evaluator” on page 292
- “Assets Corrector” on page 78
- “Context Mapper” on page 130
- “Graph Property Adder” on page 172
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “HTML Visualizer” on page 178
- “Enterprise Data Catalog Writer” on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

#### General

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

#### FastLoad Reader

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocation | DEPRECATED: Location of the source |
| SourceLocationLineNumber | Position of SourceStatementText according to SourceCompleteText |

#### SQL Evaluator

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

#### Assets Corrector

This module does not contain any graph properties.

#### Context Mapper

This module does not contain any graph properties.

#### Graph Property Adder

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Hive Script to EDC

Generates appropriate files to be loaded into Enterprise Data Catalog from Hive Scripts.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “Hive Script Reader” on page 175
- “SQL Evaluator” on page 292
- “Assets Corrector” on page 78
- “Context Mapper” on page 130
- “Graph Property Adder” on page 172
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “HTML Visualizer” on page 178
- “Enterprise Data Catalog Writer” on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

#### General

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

#### Hive Script Reader

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocationLineNumber | Position of SourceStatementText according to SourceCompleteText |
| SourceLocationColumnNumber | Position of SourceStatementText according to SourceCompleteText |
| SourceStatementText | Source Text |
| SourceLocation | DEPRECATED: Location of the source |

#### SQL Evaluator

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

#### Assets Corrector

This module does not contain any graph properties.

#### Context Mapper

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# IBM DataStage to EDC

Generates EDC objects and links from DataStage ISX files.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “DataStage Reader” on page 148
- “SQL Evaluator” on page 292
- “Assets Corrector” on page 78
- “Context Mapper” on page 130
- “Graph Property Adder” on page 172
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “HTML Visualizer” on page 178
- “Enterprise Data Catalog Writer” on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**DataStage Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceLocationFilename | Source file name |
| SourceLocationFilePath | Source file path |
| SourceModificationDate | Last source modification date |
| DataStageEngine | Engine |
| DataStageProject | Project |
| DataStageFolder | Folder path |
| DataStageJob | Job name |
| DataStageJobType | Job type |
| DataStageParentJob | Parent job name |
| DataStageParentJobFolder | Parent job folder path |
| ParentJobActivityStage | Parent JobActivity stage name |
| DataStageParentSequenceJobShortDescription | DataStage parent sequence job short description |
| DataStageParentSequenceJobLongDescription | DataStage parent sequence job long description |
| ShortDescription | Short job description |
| LongDescription | Full job description |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# IICS to EDC

Generates Enterprise Data Catalog objects and links from Informatica Intelligent Cloud Services (IICS) sessions.

## List of modules

This chain calls the following modules:

- "General" on page 171
- "IICS Reader" on page 181
- "B360 Processor" on page 94
- "SQL Evaluator" on page 292
- "Informatica Post SQL Fixer" on page 200
- "Oracle Synonym Resolver" on page 255
- "Assets Corrector" on page 78
- "Context Mapper" on page 130
- "Graph Property Adder" on page 172
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "Graph Lineage Hash Calculator" on page 171
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

#### IICS Reader

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| IicsProject | IICS project name |
| IicsTask | IICS task name |
| IicsTaskDesc | IICS task description |
| IicsTaskType | IICS task type |

#### B360 Processor

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocationFilePath | Source file path |
| InputRootPath | Root path containing input files |
| SourceDescription | Description of the source |

#### SQL Evaluator

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

#### Informatica Post SQL Fixer

This module does not contain any graph properties.

#### Oracle Synonym Resolver

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**Graph Lineage Hash Calculator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| LineageHash | Lineage hash of the Graph |

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Informatica PowerCenter to EDC

Generates EDC objects and links from Informatica sessions.

## List of modules

This chain calls the following modules:

- "General" on page 171
- "Informatica Reader" on page 200
- "SQL Evaluator" on page 292
- "Informatica Post SQL Fixer" on page 200
- "Oracle Synonym Resolver" on page 255

- "Assets Corrector" on page 78
- "Context Mapper" on page 130
- "Graph Property Adder" on page 172
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**Informatica Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocation | DEPRECATED: Location of the source |
| InformaticaServer | Name of the Informatica Server |
| InformaticaRepository | Name of the Informatica Repository |
| InformaticaWorkflowFolder | Name of the Informatica Workflow Folder (from where mapping is run) |
| InformaticaMappingFolder | Name of the Informatica Mapping Folder (where mapping is placed) |
| InformaticaWorkflow | Name of the Informatica Workflow |
| InformaticaSession | Name of the Informatica Session |
| InformaticaMapping | Name of the informatica Mapping |
| InformaticaWorklets | Names of Informatica Worklets |

| Property name | Description |
|---|---|
| InformaticaTaskDisabled | Names of disabled Informatica tasks |
| InformaticaWorkletInstances | Names of Informatica Worklet instances |
| InformaticaWorkflowInstance | Name of the Workflow Instance |
| explicitWorkflowParameterFile | Name of the explicit parameter file |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Informatica Post SQL Fixer**

This module does not contain any graph properties.

**Oracle Synonym Resolver**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# JCL to EDC

Generates EDC objects and links from JCL files.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “JCL Reader” on page 218
- “SQL Plus Evaluator for JCL” on page 309
- “SQL Loader Evaluator for JCL” on page 305
- “SQL Evaluator” on page 292
- “SAS Evaluator for JCL” on page 273
- “COBOL Evaluator for JCL” on page 107
- “BTEQ Evaluator for JCL” on page 97
- “MultiLoad Evaluator for JCL” on page 240
- “Columns Propagator” on page 129
- “Column Properties PostProcessor” on page 129
- “Assets Corrector” on page 78
- “Context Mapper” on page 130
- “Graph Property Adder” on page 172
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “HTML Visualizer” on page 178
- “Enterprise Data Catalog Writer” on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**JCL Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| JclJob | The name of JCL job |
| JclStep | The name of JCL step |
| JclJobFilename | The absolute location of JCL job file |
| JclProgram | The name of called JCL program or procedure |
| JclStepLineNumber | Line number of JCL step declaration |
| JclStepSourceText | Source text of JCL step |
| JclCallStack | JCL step call stack trace |
| JclProcedure | The name of JCL procedure |
| JclProcedureParameters | Parameters of JCL procedure |

**SQL Plus Evaluator for JCL**

This module does not contain any graph properties.

**SQL Loader Evaluator for JCL**

This module does not contain any graph properties.

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |

| Property name | Description |
|---|---|
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**SAS Evaluator for JCL**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| JclSasInputFile | Name of SAS file that was an input for JCL step |
| SourceLocation | DEPRECATED: Location of the source |
| SourceLocationLineNumber | Position of SourceStatementText according to SourceCompleteText |
| SourceLocationColumnNumber | Position of SourceStatementText according to SourceCompleteText |
| SourceStatementText | Source Text |
| FileUniquePosition | Unique position in file |
| OrderNumber | DEPRECATED: It is usefull in creating distinct name of Extension Mapping Documents e.g. in parsing procedure calls. |

**COBOL Evaluator for JCL**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceLocationFilename | Source file name |
| CobolProgramId | The PROGRAM-ID of the COBOL program. |
| CallingCobolProgramId | Id of COBOL program that called current program |
| CobolProgramAuthor | Author of the COBOL program. |
| CobolProgramInstallation | INSTALLATION property of COBOL program |
| CobolProgramDateWritten | DATE-WRITTEN property of COBOL program |
| CobolProgramDateCompiled | DATE-COMPILED property of COBOL program |
| CobolProgramSecurity | SECURITY property of COBOL program |
| CobolProgramCallStack | Call stack of COBOL program |

**BTEQ Evaluator for JCL**

This module does not contain any graph properties.

**MultiLoad Evaluator for JCL**

This module does not contain any graph properties.

**Columns Propagator**

This module does not contain any graph properties.

**Column Properties PostProcessor**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Microsoft PowerBI to EDC

Generates EDC objects and links from Microsoft PowerBI workspaces.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “Microsoft Power BI Reader” on page 230
- “SQL Evaluator” on page 292
- “Microsoft MDX Query Evaluator” on page 229
- “Assets Corrector” on page 78

- "Context Mapper" on page 130
- "Graph Property Adder" on page 172
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**Microsoft Power BI Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| PowerBiCloudWorkspaceName | PowerBi Cloud Workspace Name |
| PowerBiCloudWorkspaceNameDescription | PowerBi Cloud Workspace Description |
| PowerBiCloudWorkspaceCapacityId | PowerBi Cloud Workspace Capacity ID |
| PowerBiCloudWorkspaceDataRetrievalState | PowerBi Cloud Workspace Data Retrieval State |
| PowerBiCloudWorkspaceId | PowerBi Cloud Workspace ID |
| PowerBiCloudWorkspaceIsOnDedicatedCapacity | PowerBi Cloud Workspace Is On Dedicated Capacity |
| PowerBiCloudWorkspaceState | PowerBi Cloud Workspace State |
| PowerBiCloudWorkspaceType | PowerBi Cloud Workspace Type |
| PowerBiCloudReportId | PowerBi Cloud Report ID |
| PowerBiCloudReportName | PowerBi Cloud Report Name |

| Property name | Description |
|---|---|
| PowerBiCloudReportDescription | PowerBi Cloud Report Description |
| PowerBiCloudReportCreatedBy | PowerBi Cloud Report Created By |
| PowerBiCloudReportCreatedDateTime | PowerBi Cloud Report Created Date Time |
| PowerBiCloudReportEmbedUrl | PowerBi Cloud Report Embed URL |
| PowerBiCloudReportModifiedBy | PowerBi Cloud Report Modified By |
| PowerBiCloudReportModifiedDateTime | PowerBi Cloud Report Modified Date Time |
| PowerBiCloudReportType | PowerBi Cloud Report Type |
| PowerBiCloudReportUrl | PowerBi Cloud Report URL |
| PowerBiCloudDatasetName | PowerBi Cloud Dataset Name |
| PowerBiCloudDatasetDescription | PowerBi Cloud Dataset Description |
| PowerBiCloudDatasetConfiguredBy | PowerBi Cloud Dataset Configured By |
| PowerBiCloudDatasetUrl | PowerBi Cloud Dataset URL |
| PowerBiCloudDatasetContentProviderType | PowerBi Cloud Dataset Content Provider Type |
| PowerBiCloudDatasetCreateReportEmbedUrl | PowerBi Cloud Dataset Create Report Embed URL |
| PowerBiCloudDatasetCreatedDate | PowerBi Cloud Dataset Created Date |
| PowerBiCloudDatasetIsEffectiveIdentityRequired | PowerBi Cloud Dataset Is Effective Identity Required |
| PowerBiCloudDatasetIsEffectiveIdentityRolesRequired | PowerBi Cloud Dataset Is Effective Identity Roles Required |
| PowerBiCloudDatasetIsOnPremGatewayRequired | PowerBi Cloud Dataset Is On Prem Gateway Required |
| PowerBiCloudDatasetQnaEmbedUrl | PowerBi Cloud Dataset Qna Embed Url |
| PowerBiCloudDatasetAddRowsApiEnabled | PowerBi Cloud Dataset Add Rows Api Enabled |
| PowerBiCloudDatasetSchemaMayNotBeUpToDate | PowerBi Cloud Dataset Schema May Not Be Up To Date |
| PowerBiCloudDatasetSchemaRetrievalError | PowerBi Cloud Dataset Schema Retrieval Error |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Microsoft MDX Query Evaluator**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Microsoft SSAS Multidimensional to EDC

Generates EDC objects and links from SSAS project files.

## List of modules

This chain calls the following modules:

- "General" on page 171
- "SSAS Reader" on page 310
- "SQL Evaluator" on page 292
- "Microsoft MDX Query Evaluator" on page 229
- "Assets Corrector" on page 78
- "Graph Property Adder" on page 172
- "Context Mapper" on page 130
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**SSAS Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocationFilePath | Source file path |
| SsasServer | The name of SSAS server |

| Property name | Description |
|---|---|
| SsasProject | The name of SSAS project |
| SsasCube | The name of SSAS cube |
| SsasOutputObjectType | The type of output SSAS object (e.g. data source view table, dimension, measure group, KPI) |
| SsasParentDimension | The name of SSAS parent's dimension name |
| SsasParentType | Ssas parent type name |
| SsasOutputObject | The name of SSAS output object (e.g. dimension name) |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Microsoft MDX Query Evaluator**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Microsoft SSAS Tabular to EDC

Generates EDC objects and links from SSAS Tabular project files.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “SSAS Tabular Reader” on page 313
- “SQL Evaluator” on page 292
- “Assets Corrector” on page 78
- “Graph Property Adder” on page 172
- “Context Mapper” on page 130
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “HTML Visualizer” on page 178
- “Enterprise Data Catalog Writer” on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

#### General

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

#### SSAS Tabular Reader

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocationFilePath | Source file path |
| SsasServer | The name of SSAS server |
| SsasProject | The name of SSAS project |
| SsasTabularSmProjectFilePath | The path to main project file: *.smproj |
| SsasOutputObject | The name of SSAS output object (e.g. dimension name) |
| SsasOutputObjectType | The type of output SSAS object (e.g. data source view table, dimension, measure group, KPI) |

#### SQL Evaluator

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

#### Assets Corrector

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Microsoft SSIS to EDC

Generates EDC model from SSIS files.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “SSIS Reader” on page 330
- “SQL Evaluator” on page 292
- “Temporary Table Names Post Processor” on page 364
- “Oracle Synonym Resolver” on page 255
- “Assets Corrector” on page 78
- “Context Mapper” on page 130
- “Graph Property Adder” on page 172
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “HTML Visualizer” on page 178
- “Enterprise Data Catalog Writer” on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**SSIS Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocationFilePath | Source file path |
| InputRootPath | Root path containing input files |
| SourceDescription | Description of the source |
| SsisPackage | SSIS Package name |
| SsisDataTask | SSIS Data Task name |
| SsisTaskType | SSIS data task type |
| ParentPackageFiles | List of file names of SSIS Execute Package Task parent packages |
| ParentTasks | List of qualified SSIS Execute Package Task parent tasks |
| SsisScriptFilePath | Path to SSIS script |
| SsisScriptFilename | Name of the script file |
| SsisScriptDirectory | Directory which contains script |
| SsisScriptLineNumber | Line number in SSIS script |
| SsisScriptLineText | Line text in SSIS script |

| Property name | Description |
|---|---|
| SsisFileSystemTaskType | SSIS File System Task type |
| Application host name | Name used as output model name, host name of SSIS files and server name where the Visual Basic scripts are run. |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Temporary Table Names Post Processor**

This module does not contain any graph properties.

**Oracle Synonym Resolver**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Microsoft SSRS to EDC

Generates EDC objects and links from SSRS reports.

## List of modules

This chain calls the following modules:

- "General" on page 171
- "SSRS Reader" on page 346
- "SQL Evaluator" on page 292
- "Microsoft MDX Query Evaluator" on page 229
- "Oracle Synonym Resolver" on page 255
- "Assets Corrector" on page 78
- "Context Mapper" on page 130
- "Graph Property Adder" on page 172
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**SSRS Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceLocationFilename | Source file name |
| SsrsObject | SSRS Object name |
| SsrsObjectType | DataSet or Report |
| SsrsServer | SSRS Server name |
| SsrsFolder | SSRS Folder name |
| SsrsReport | Report name that this object belongs to |
| SourceLocation | DEPRECATED: Location of the source |
| SourceModificationDate | Last source modification date |
| GraphUri | VOLATILE: Fully qualified URI of Graph |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Microsoft MDX Query Evaluator**

This module does not contain any graph properties.

**Oracle Synonym Resolver**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# MultiLoad to EDC

Generates EDC objects and links from MultiLoad scripts.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “MultiLoad Reader” on page 241
- “SQL Evaluator” on page 292
- “Assets Corrector” on page 78
- “Context Mapper” on page 130
- “Graph Property Adder” on page 172
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “HTML Visualizer” on page 178
- “Enterprise Data Catalog Writer” on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

#### General

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

#### MultiLoad Reader

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocation | DEPRECATED: Location of the source |

#### SQL Evaluator

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

#### Assets Corrector

This module does not contain any graph properties.

#### Context Mapper

This module does not contain any graph properties.

#### Graph Property Adder

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# ODI to EDC

Generates EDC objects and links from Oracle Data Integrator files.

## List of modules

This chain calls the following modules:

- "General" on page 171
- "ODI Reader" on page 244
- "SQL Evaluator" on page 292
- "Graph Property Adder" on page 172
- "Assets Corrector" on page 78
- "Context Mapper" on page 130
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**ODI Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocationFilePath | Source file path |
| OdiProject | Project name |
| OdiPackageFolder | Name of package folder which executed analyzed object |
| OdiPackage | Name of package which executed analyzed object |
| OdiFolder | Folder path inside a project |
| OdiObjectName | Name of analyzed ODI object |
| OdiObjectType | Type of analyzed ODI object |
| OdiScenario | Name of analyzed scenario |
| OdiScenarioStep | Name of scenario step which defines flow execution |
| OdiScenarioStepTask | Internal ODI task name in current step |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |

| Property name | Description |
|---|---|
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Graph Property Adder**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Salesforce to EDC

## List of modules

This chain calls the following modules:

- “General” on page 171
- “Salesforce Reader” on page 259
- “Lineage Analyzer” on page 226
- “Enterprise Data Catalog Writer” on page 156

### Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**Salesforce Reader**

This module does not contain any graph properties.

**Lineage Analyzer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# SAP BODS to EDC

Generates EDC objects and links from BODS files.

## List of modules

This chain calls the following modules:

- "General" on page 171
- "Business Objects Data Services Reader" on page 100
- "SQL Evaluator" on page 292
- "Assets Corrector" on page 78
- "Context Mapper" on page 130
- "Graph Property Adder" on page 172
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**Business Objects Data Services Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocationFilePath | Source file path |
| SourceModificationDate | Last source modification date |
| BodsRepository | BODS Repository name |
| BodsProject | BODS Project name |
| BodsJob | BODS Job name |
| BodsWorkflow | BODS Workflow name |
| BodsDataflow | BODS Dataflow name |
| BodsScript | BODS Script name |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |

| Property name | Description |
|---|---|
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# SAS EGP to EDC

Generates EDC model from SAS EGP programs.

## List of modules

This chain calls the following modules:

- "General" on page 171
- "SAS EGP Reader" on page 264
- "SQL Evaluator" on page 292
- "Oracle Synonym Resolver" on page 255
- "Column Properties PostProcessor" on page 129

- "Assets Corrector" on page 78
- "Context Mapper" on page 130
- "Graph Property Adder" on page 172
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**SAS EGP Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceLocationFilename | Source file name |
| SourceLocation | DEPRECATED: Location of the source |
| SourceLocationLineNumber | Position of SourceStatementText according to SourceCompleteText |
| SourceLocationColumnNumber | Position of SourceStatementText according to SourceCompleteText |
| SourceStatementText | Source Text |
| SasHost | Host in which sas file is processed |
| SasStepId | Sas step identifier |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Oracle Synonym Resolver**

This module does not contain any graph properties.

**Column Properties PostProcessor**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# SAS to EDC

Generates appropriate files to be loaded into Enterprise Data Catalog from SAS programs.

## List of modules

This chain calls the following modules:

- "General" on page 171
- "SAS Reader" on page 280
- "SQL Evaluator" on page 292
- "Oracle Synonym Resolver" on page 255
- "Assets Corrector" on page 78
- "Column Properties PostProcessor" on page 129
- "Context Mapper" on page 130
- "Graph Property Adder" on page 172
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**SAS Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceLocationFilename | Source file name |
| SourceLocation | DEPRECATED: Location of the source |
| SourceLocationLineNumber | Position of SourceStatementText according to SourceCompleteText |

| Property name | Description |
|---|---|
| SourceLocationColumnNumber | Position of SourceStatementText according to SourceCompleteText |
| SourceStatementText | Source Text |
| FileUniquePosition | Unique position in file |
| SasHost | Host in which sas file is processed |
| SasStepId | Sas step identifier |
| SasIsSelectInto | Determine if statement is proc sql select into |
| SasEgpProject | Sas EGP project identifier |
| SasEgpFlow | Sas EGP flow identifier |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Oracle Synonym Resolver**

This module does not contain any graph properties.

**Assets Corrector**

This module does not contain any graph properties.

**Column Properties PostProcessor**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# SQL*Loader to EDC

Generates EDC objects and links from Oracle SQL*Loader files.

## List of modules

This chain calls the following modules:

- “General” on page 171
- “SQL Loader Reader” on page 305
- “Assets Corrector” on page 78
- “Context Mapper” on page 130
- “Graph Property Adder” on page 172
- “Image Dumper” on page 198
- “Lineage Analyzer” on page 226
- “HTML Visualizer” on page 178
- “Enterprise Data Catalog Writer” on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**SQL Loader Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceType | DEPRECATED: Type of the source |
| SourceLocationFilename | Source file name |
| SourceLocation | DEPRECATED: Location of the source |
| SqlExternalTable | Name of External Table |

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

**Enterprise Data Catalog Writer**

This module does not contain any graph properties.

# Talend Data Integration to EDC

Generates EDC objects and links from Talend files.

## List of modules

This chain calls the following modules:

- "General" on page 171
- "Talend Reader" on page 354
- "SQL Evaluator" on page 292
- "Assets Corrector" on page 78
- "Context Mapper" on page 130
- "Graph Property Adder" on page 172
- "Image Dumper" on page 198
- "Lineage Analyzer" on page 226
- "HTML Visualizer" on page 178
- "Enterprise Data Catalog Writer" on page 156

## Graph properties

Objects that you process are represented as graphs in the MetaDex tool. In rare use cases, you can use the graph properties for advanced modifications.

This chain sets the following graph properties:

**General**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| Version | Product Version |
| GenerationDate | Graph generation date |
| GraphId | VOLATILE: Graph id unique in given job |

**Talend Reader**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SourceLocationFilename | Source file name |
| SourceLocationFilePath | Source file path |
| SourceModificationDate | Last source modification date |
| TalendProject | The name of Talend project |
| TalendFolder | The name of Talend Job folder |

| Property name | Description |
|---|---|
| TalendJobName | The name of Talend Job |
| TalendSubjobName | The name of Talend Subjob |
| TalendStackTrace | Stack trace of Subjob execution |

**SQL Evaluator**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| SqlProcedureCallLocalSignature | Local call id of SQL procedure: 'db.schema.(optionalPackage.)? procedureName(fullArg1, fullArg2)' or ETL process signature executing given code |
| SqlProcedureNameWithArguments | Procedure name with arguments: procedureName(arg1, arg2, ...) |
| SourceCompleteText | Source Complete Text |
| SqlProcedure | SQL Procedure name, if exists |
| SqlQualifiedProcedure | Qualified name for SQL procedure: schema(.optionalPackage).procedureName |

**Assets Corrector**

This module does not contain any graph properties.

**Context Mapper**

This module does not contain any graph properties.

**Graph Property Adder**

This module does not contain any graph properties.

**Image Dumper**

The following table describes the graph properties for the module:

| Property name | Description |
|---|---|
| GraphImagePath | PNG file name |
| GraphImageNotCalculatedCause | This property is set when graph has problems during image calculation - contains the reason of the problem |

**Lineage Analyzer**

This module does not contain any graph properties.

**HTML Visualizer**

This module does not contain any graph properties.

#### Enterprise Data Catalog Writer

This module does not contain any graph properties.

CHAPTER 2

# Module Reference

## Advanced PL/SQL Converter

A module for converting a PL/SQL statement into a data transformation process.

The Advanced PL/SQL Converter includes the following parameters:

### Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Advanced PL/SQL Reader

A module for reading PL/SQL scripts. It can be used to parse various PL/SQL dialects. For example, this module can be used to read and parse scripts intended for use in an Oracle environment.

The Advanced PL/SQL Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Context

Context for database objects (ie. tables). It is used for searching as well as for creating database objects. Database objects are stored and searched by suffix. Use of a context with dot-separated segments allows the specification of default values for host, database and schema. These are only default values and are overridden by explicit qualification of object references in a query. Context can consist of:

- one segment - This segment is treated as the default host name.
- two segments - (e.g., HOST.DBNAME) - these segments are treated as default host and default database names.
- three segments - (e.g., HOST.DBNAME.SCHEMA) - these segments are treated as default host, default database and default schema names.

Specifying more than three segments causes job error.

| Parameter name | Description |
|---|---|
| Prompt | Context |
| Symbol | sql.sqlContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - db<br>- db.sciencedb<br>- db.sciencedb.astronomy |

## Database vendor

The name of the database vendor. The following vendors are supported:

- ORACLE

- TERADATA
- MYSQL
- DB2 (also DB2ZOS, DB2ISERIES)
- MSSQL
- SYBASE
- NETEZZA

The following vendors are not supported:

- H2
- VSAM

| Parameter name | Description |
|---|---|
| Prompt | Database vendor |
| Symbol | sql.vendor |
| Type | MODULE_MODE |
| Required | No |
| Default value | ORACLE |
| Example values | - ORACLE<br>- MSSQL |

## Input scripts

The paths of scripts to be included for processing. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input scripts |
| Symbol | sql.fileIncludeFilter |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input\plsql\224_magic.plsql<br>- /opt/input/plsql/225_more_magic.plsql |

## Input scripts encoding

Configures charset used to read input scripts. Default charset is UTF-8.

| Parameter name | Description |
|---|---|
| Prompt | Input scripts encoding |
| Symbol | sql.scriptsEncoding |
| Type | STRING |
| Required | No |
| Default value | UTF-8 |
| Example values | - UTF-8<br>- CP1250<br>- CP1252 |

## Input scripts to exclude

The paths of scripts to be excluded from processing. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input scripts to exclude |
| Symbol | sql.fileExcludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\plsql\exclude.plsql<br>- /opt/input/plsql/exclude_2.plsql |

## Procedure calls

When it's set to true, it means that input files contain procedure calls rather than PL/SQL itself. If it's set to false, input files contain PL/SQL source code: packages, functions and procedures. Details for procedure calls input file can be found in the user guide in Appendix C. "Procedure Calls Input File Specification"

| Parameter name | Description |
|---|---|
| Prompt | Procedure calls |
| Symbol | sql.procedureCalls |
| Type | BOOLEAN |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Procedure calls parameters

File with bash variable assignments. Each assignment must be in separate line. Each line must be empty or have following assignment: $[VARIABLE_NAME] = [SUBSTITUTION_STRING] VARIABLE_NAME can consist of letters, digits, and the '_' (underscore), and cannot begin with a digit. If set, replacement will be performed on file with procedure calls.

| Parameter name | Description |
|---|---|
| Prompt | Procedure calls parameters |
| Symbol | sql.procedureCallsParams |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\plsql\bash_vars.txt<br>- /opt/input/plsql/bashVariables.in |

## Scripts Root Directory Path

Root directory of input files. Relative paths of input files resolve in the following location: [Input root directory]. Absolute paths from input files must start from the [Input root directory] to be included into processing. [Input root directory] will not be visible on the output objects structure.

| Parameter name | Description |
|---|---|
| Prompt | Scripts Root Directory Path |
| Symbol | sql.inputRoot |
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\inputs\<br>- /opt/feed-etl/ |

### SQL separator

Configures the separator which will be used to split statements. For example Teradata uses ';' as separator, MS SQL can use "GO" and Oracle can use '/'. This option can be used only for SQL input to process as much as possible despite parse errors. You should NEVER choose a separator for PL/SQL input, otherwise this may cause many additional errors.

| Parameter name | Description |
|---|---|
| Prompt | SQL separator |
| Symbol | sql.sqlSeparator |
| Type | STRING |
| Required | No |
| Default value | Undefined |
| Example values | - ;<br>- GO<br>- / |

### Unresolved Params Handling

Defines behavior in case of unresolved parameter in procedureCall file. ERROR - logs an error and skips analysis of current call, UNKNOWN VALUE - treats variable as UNKNOWN value, QUOTE - replaces unknown parameters to its name (without dollar sign), SKIP - continue analysis with unresolved parameters.

| Parameter name | Description |
|---|---|
| Prompt | Unresolved Params Handling |
| Symbol | sql.unresolvedParamsHandling |
| Type | STRING |
| Required | No |
| Default value | ERROR |
| Example values | - UNKNOWN_VALUE<br>- QUOTE<br>- ERROR<br>- SKIP |

# Assets Corrector

This module uses the metadata defined in the Metadata Repository (MDREPO) to correct the case of table names (host, database, schema, table) according to original database, e.g.: host.DATABASE.schema.TABLE -

> Host.Database.Schema.Table . It works without any further configuration - it will use MDREPO/HOST/DB/db.properties file to connect to appropriate databases to find the proper case of each table.

The Assets Corrector includes the following parameters:

### Enable assets correction of graphs

Disables or enables graph correction in this module.

| Parameter name | Description |
|---|---|
| Prompt | Enable assets correction of graphs |
| Symbol | assetCorrector.enableCorrection |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

# Azure Data Factory Reader

Reads Azure Data Factory Pipelines and DataFlows. Use Azure Resource Manager templates as input. They can be exported directly from ADF web editor.

The Azure Data Factory Reader includes the following parameters:

### ADF input filters

Case sensitive include/exclude filters, allowing to filter ADF activities before processing. These filters use the following language: (Include|Exclude) : FactoryPattern : PipelinePattern : ActivityPattern where: Include - it is the include filter Exclude - it is the exclude filter Pattern - it is the filter syntax supported wildcards:

- Asterisk for 0 or more characters
- Question mark for a single character

| Parameter name | Description |
|---|---|
| Prompt | ADF input filters |
| Symbol | adf.inputFilters |
| Type | STRINGLIST |
| Required | Yes |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Include : * : * : * # process all activities -> default Exclude : Factory: x\|y\|pipeline : * # exclude all activities from pipeline pipeline in folder x/y in Factory Include : *prod : * : * # process all activities from factory name ending with "prod" |

## ADF REST API extractor timeout

The ADF REST API extractor timeout for single scan in seconds.

| Parameter name | Description |
|---|---|
| Prompt | ADF REST API extractor timeout |
| Symbol | adf.restApiExtractorTimeoutSeconds |
| Type | STRING |
| Required | No |
| Default value | 300 |
| Example values | - 100<br>- 200 |

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | adf.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Default value file for parameter data flow expressions

Path to the file that contains default values for data flow expressions.

File format consist of [GLOBAL] or/and custom function section.

---

You can define variables definitions: [GLOBAL] VariableName = VariableExpression

VariableName that contains only standard alphanumeric characters does not require any quotes. If it contains any special characters, it requires double-quotes (").

Example of variables definition: [GLOBAL] a = 42 b = 7 c = a < b ? a + 7 : b - 7 // It's 49 d = e(1,2) + 1 // Expression use call to function. "User::table" = 'table' || "User::tableSuffix" // It's string table concatenated with value of User::tableSuffix variable

---

You can define functions definitions: [GLOBAL] FunctionName = FunctionExpression

FunctionName can be unquoted if contains only standard alphanumeric characters but needs to be quoted with double quoted (") if contains any special character. FunctionName must contain list of arguments. List of argument is defined inside "(" ")" characters and consists of "?" sings separated by "," (one "?" sign for each function argument).

Example of function definition: [GLOBAL] a(?) = 1 a(?,?) = 2 b(?) = a(1) + 1 // Expression use call to another function. c(?,?) = d + 2 // Expression use reference to variable.

---

Additional sections can be provided to match only specific functions. To match an overloaded function please provide placeholders for its arguments. Inside the matched section you can also reference matched function arguments. Example of custom function sections:

- [host.db.schema.func(x)]
- z=x
- [host.db.schema.func(x,y)]
- z=x+y

---

To escape special characters such as \n \t use the \ character. For example, to define E:\file \tgtParameterized.csv, enter E:\\file\\tgtParameterized.csv

---

On the right side of assignments you can use simple expression language that can use values of already defined variables and functions or variables from current job context.

Supported expressions (case insensitive) are:

- literals
  - integer: 42
  - string: 'str'
  - string with quoted single quote: 'str\''
  - boolean: true, false
- equals: 42 == 7 -> FALSE
- not equals: 42 != 7 -> TRUE
- concatenations: 'str1' + 'str2' -> 'str1str2'
- function calls: Replace('abc', 'b', 'd') -> 'adc'
- tertiary operator:
  - 42 == 7 ? 'a' : 'b' -> 'b'
  - 42 != 7 ? 'a' : 'b' -> 'a'

Supported functions:

- Hash(str, maxOutputLen) - MD5, maxOutputLen between 1 and 32 inclusive
  - Hash('abcdefgh', 4) -> 'E8DC'
- Replace(str, from, to)
  - Replace('abc', 'b', 'D') -> 'aDc'
- ReplaceRegexp(str, regex, replacement) - replaces with Java regular expressions
  - ReplaceRegexp('abcde', 'b.*', 'f') -> 'af'
  - ReplaceRegexp('graph_id20', 'id(\d+)', '$1') -> 'graph_20'
- StringLengthLimit(str, limit, hashSize) - returns str[0..limit-hashSize] + Hash( str, hashSize )
  - StringLengthLimit('abcc', 3, 2) -> 'a26'
  - StringLengthLimit('abcc', 4, 2) -> 'abcc'
- StringLengthLimit(str, limit) - hashSize has default size 8
  - StringLengthLimit('abcdabcdabcd', 10) -> 'abE340600C'
- RegexpMatch(pattern, str) - tests if str matches pattern
  - pattern uses standard Java 8 regular expressions: https://docs.oracle.com/javase/8/docs/api/java/util/regex/Pattern.html
  - RegexpMatch('[A-Za-z]+','Abcd') -> TRUE
- Upper(str) - returns uppercased str
  - Upper('Abc') -> 'ABC'
- Lower(str) - return lowercased str
  - Lower('aBC') -> 'abc'
- Date(text,format) - returns object representing date with provided format.
  - Formats uses standard Java 8 SimpleDateFormat: https://docs.oracle.com/javase/8/docs/api/java/text/SimpleDateFormat.html
  - It is designed for default values usage
  - ex. Date('2017-10-31', 'yyyy-MM-dd')
- Contains/ContainsIgnoreCase(stack, needle) - tests if stack string contains needle string
  - Contains('abc', 'ab') -> TRUE
  - Contains('abc', 'AB') -> FALSE
  - ContainsIgnoreCase('abc', 'AB') -> TRUE
  - Contains('stack', 'needle') -> FALSE
- RtfToText(str) - return clean text from rtf string
  - RtfToText('{\rtf1\ansi{\fonttbl\f0\fswiss Helvetica;}\f0\pard text\par }') -> text

Supported comments:

- Single line: \ comment

- Multi line: /* comment */

| Parameter name | Description |
|---|---|
| Prompt | Default value file for parameter data flow expressions |
| Symbol | adf.parameterDataFlowExpressionsDefaultValueFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /home/scanners/adf/input/defaultValues.txt |

## Input files - ARM templates

Provide the ARM template ZIP files exported from the Azure Data Factory web user interface or template JSON files.

Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files - ARM templates |
| Symbol | adf.inputFiles |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - /home/scanners/adf/input/*.zip<br>- /home/scanners/adf/input/*.json |

## Integration runtime to SSISDB connection

Enter the details of the integration runtime environment that connects to the Microsoft SQL Server Integration Services database. The **Integration runtime to SSISDB connection** parameter consists of the following parts: - Integration runtime name that you enter in the **Integration runtime name** field - Connection target that you enter in the **Resolved SSISDB MDREPO entry** field

| Parameter name | Description |
|---|---|
| Prompt | Integration runtime to SSISDB connection |
| Symbol | adf.integrationRuntimeToSsisDbConnection |
| Type | PROPERTIES |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Integration runtime 1=MSSQL:user@ssisdbhost.SSISDB.catalog |

## Log warn on missing variable

Decide about logging warn message when there is usage of variable with unknown value

| Parameter name | Description |
|---|---|
| Prompt | Log warn on missing variable |
| Symbol | adf.logWarnOnMissingVariable |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Operational Metadata Time Range

The time range from which pipeline runs are processed.

| Parameter name | Description |
|---|---|
| Prompt | Operational Metadata Time Range |
| Symbol | adf.operationalMetadataTimeRangeMode |
| Type | STRING |
| Required | No |
| Default value | Last 24 hours |
| Example values | - Last 24 hours<br>- Last 7 days<br>- Last 30 days<br>- Custom |

## Operational metadata time range from

Enter the time at or after which the pipeline run event updates in the ISO 8601 format. The custom time range start date can't be older than 45 days from now because Microsoft Azure Data Factory stores pipeline-run data for only 45 days.

| Parameter name | Description |
|---|---|
| Prompt | Operational metadata time range from |
| Symbol | adf.operationalMetadataTimeRangeFrom |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - 2018-06-16T00:36:44.3345758Z |

## Operational metadata time range to

Enter this value for the 'Custom' time range mode. The time at or before which the pipeline run event was updated in 'ISO 8601' format.

| Parameter name | Description |
|---|---|
| Prompt | Operational metadata time range to |
| Symbol | adf.operationalMetadataTimeRangeTo |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - 2018-06-16T00:36:44.3345758Z |

## Parameter default value file

Path to the file which contains default values.

- Defining pipeline parameters and system variables:
  - ["My factory"."My pipeline"]
  - parameters.Pipeline_Param='X' --> default value for pipeline parameter
  - RunId='X' --> default value for pipeline system variable
- Defining default values for Activity items:
  - ["My factory"."My pipeline"."My Activity"]
  - Dataset*Param='X'* --> *value for @dataset().Dataset*Param when it is used in "My Activity" expressions utcnot()='...' --> value for result of executing function utcnow() in "My Activity" expressions

- Defining default values for expression which are in input and output section of Activity:
  - ["My factory"."My pipeline"."My Activity"."input"]
  - Dataset*Param='S'* --> *value for @dataset().Dataset*Param when it is used in "My Activity" input expression
  - ["My factory"."My pipeline"."My Activity"."output"]
  - Dataset*Param='T'* --> *value for @dataset().Dataset*Param when it is used in "My Activity" output expression

| Parameter name | Description |
|---|---|
| Prompt | Parameter default value file |
| Symbol | adf.parameterDefaultValueFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /home/scanners/adf/input/defaultValues.txt |

## PatternReplacer configuration path

Path to a configuration of the PatternReplacer functionality.The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.For this particular reader there are following possible property keys: ActivityDataflowLocationName.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | adf.patternReplacerFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

## Process Latest Runs

Specifies whether to process only the latest pipeline runs or all pipeline runs.

| Parameter name | Description |
|---|---|
| Prompt | Process Latest Runs |
| Symbol | adf.shouldProcessLatestRuns |
| Type | PRIMITIVE_BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Process Operational Metadata

Specifies whether to process operational metadata.

| Parameter name | Description |
|---|---|
| Prompt | Process Operational Metadata |
| Symbol | adf.shouldProcessOperationalMetadata |
| Type | PRIMITIVE_BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Proxy host

Host name or ip address of http(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy host |
| Symbol | adf.restApiProxyHost |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - https://myproxy |

## Proxy password

Password to authenticate to http(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy password |
| Symbol | adf.restApiProxyPassword |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - proxy-password |

## Proxy port

Port a http(s) proxy server is listening on

| Parameter name | Description |
|---|---|
| Prompt | Proxy port |
| Symbol | adf.restApiProxyPort |
| Type | INTEGER |
| Required | No |
| Default value | N/A |
| Example values | - 30456 |

## Proxy username

Username to authenticate to http(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy username |
| Symbol | adf.restApiProxyUsername |

| Parameter name | Description |
|---|---|
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - proxy-user |

## Reader mode

Choose files to read ADF from existing json files or REST API

| Parameter name | Description |
|---|---|
| Prompt | Reader mode |
| Symbol | adf.readerMode |
| Type | MODULE_MODE |
| Required | No |
| Default value | Files |
| Example values | - Files<br>- Admin User<br>- Service Principal |

## REST API client ID

The clientId of user that is used to connect to ADF REST API

| Parameter name | Description |
|---|---|
| Prompt | REST API client ID |
| Symbol | adf.restApiClientId |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - clientId |

## REST API client secret

The client secret that is used to connect to ADF REST API using Service Principal

| Parameter name | Description |
|---|---|
| Prompt | REST API client secret |
| Symbol | adf.restApiClientSecret |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - secret123 |

## REST API password

The password of user that is used to connect to ADF REST API

| Parameter name | Description |
|---|---|
| Prompt | REST API password |
| Symbol | adf.restApiPassword |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - secret123 |

## REST API subscription ID

The subscription ID that is used to get resources from ADF REST API

| Parameter name | Description |
|---|---|
| Prompt | REST API subscription ID |
| Symbol | adf.restApiSubscriptionId |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - subscription_id |

## REST API tenant ID

The tenantId that is used to connect to ADF REST API using Service Principal

| Parameter name | Description |
|---|---|
| Prompt | REST API tenant ID |
| Symbol | adf.restApiTenantId |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - tenantId-2829-2233 |

## REST API user

The user that is used to connect to ADF REST API

| Parameter name | Description |
|---|---|
| Prompt | REST API user |
| Symbol | adf.restApiUser |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - user1 |

## SecureString override values

SecureString entries as keys and encrypted values. See the details of the encryption process in the Help -> Encode section.

| Parameter name | Description |
|---|---|
| Prompt | SecureString override values |
| Symbol | adf.secureStringOverride |
| Type | SENSITIVE_PROPERTIES |
| Required | No |

| Parameter name | Description |
| --- | --- |
| Default value | N/A |
| Example values | - adfscanner<br>- VeryLongBase64 |

## Supported Pipeline Activities, Datasets and Data Flow components

- Copy - Pipeline activity
- ExecuteDataFlow - Pipeline activity. Partial support. It processes only input and outputs providing generic lineage, not analyzing script transformations.
- ExecuteSSISPackage - Pipeline activity. Supported page locations: - File System (Package) - Embedded Package
- GetMetadata - Pipeline activity
- Lookup - Pipeline activity
- SqlServerStoredProcedure - Pipeline activity
- IfCondition - Pipeline activity
- Switch - Pipeline activity
- Until - Pipeline activity
- ForEach - Pipeline activity
- Wait - Pipeline activity
- SetVariable - Pipeline activity
- AppendVariable - Pipeline activity
- ExecutePipeline - Pipeline activity
- WebActivity - Pipeline activity
- DatabricksNotebook - Pipeline activity
- Unsupported - Pipeline activity
- Binary - Dataset
- DelimitedText - Dataset
- Excel - Dataset
- Json - Dataset
- Parquet - Dataset
- FileShare - Dataset
- AmazonRedshiftTable - Dataset
- AzureSqlDWTable - Dataset
- AzureSqlTable - Dataset
- DynamicsCrmEntity - Dataset
- OracleTable - Dataset
- SalesforceObject - Dataset
- SnowflakeTable - Dataset
- SqlServerTable - Dataset

- AzureDatabricksDeltaLakeDataset - Dataset

# B360 Processor

Reads Business 360 entities and their metadata.

The B360 Processor includes the following parameters:

## IICS Cloud Password

The password that you use to log in to Informatica Intelligent Cloud Service.

| Parameter name | Description |
|---|---|
| Prompt | IICS Cloud Password |
| Symbol | iics.password |
| Type | SENSITIVE |
| Required | Yes |
| Default value | N/A |
| Example values | - mySecretPassword123 |

## IICS Cloud Url

The URL to Informatica Intelligent Cloud Service.

| Parameter name | Description |
|---|---|
| Prompt | IICS Cloud Url |
| Symbol | iics.cloudUrl |
| Type | STRING |
| Required | Yes |
| Default value | N/A |
| Example values | - https://infa.com/identity-service/home |

## IICS Cloud User

The username that you use to log in to Informatica Intelligent Cloud Service.

| Parameter name | Description |
|---|---|
| Prompt | IICS Cloud User |
| Symbol | iics.user |
| Type | STRING |
| Required | Yes |
| Default value | N/A |
| Example values | - user@informatica.com |

## IICS Filters List

Case insensitive include/exclude filters that allow to filter tasks The filters use the following syntax: <FilterType> : <Pattern> Where: FilterType: Include - If nonempty, processes only the assets prefixed with at least one of the given qualifications. Exclude - Omits items that match at least one of the given patterns. Pattern: PROJECT*NAME[:DIRECTORY*PATH[:TASK_NAME]] You can use a question mark and an asterisk inside each pattern. Question mark matches any single character, asterisk matches any number of any characters. If a folder contains spaces or other non-standard characters, quote it with double-quotes. For details, see the following examples:

| Parameter name | Description |
|---|---|
| Prompt | IICS Filters List |
| Symbol | iics.inputFilters |
| Type | STRINGLIST |
| Required | Yes |
| Default value | Include : * |
| Example values | - Include : projectN?me:/Tasks/directory:*<br>- Exclude : 'projectName:"/directory":taskNa*'<br>- Include:project:/Tasks/dir:task<br>- Include : project : /Tasks/dir : task<br>- Include : project : "/Tasks/d ir" : task<br>- Include : * : * : *<br>- Exclude : Default<br>- Include : *MyProject : * : * |

## Proxy Host

A hostname or an IP address of the HTTP(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy Host |
| Symbol | iics.proxyHost |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - 88.99.102.102 or https://myproxy |

## Proxy Password

A password for the HTTP(s) proxy server authentication

| Parameter name | Description |
|---|---|
| Prompt | Proxy Password |
| Symbol | iics.proxyPassword |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - abc123 |

## Proxy Port

A port number of the HTTP(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy Port |
| Symbol | iics.proxyPort |
| Type | INTEGER |
| Required | No |
| Default value | 0 |
| Example values | - 30408 |

### Proxy Username

A username for the HTTP(s) proxy server authentication

| Parameter name | Description |
|---|---|
| Prompt | Proxy Username |
| Symbol | iics.proxyUserName |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - abc |

### Supported Components

- Business Entity - Business entity components contain fields, smart fields, and group fields.

# BTEQ Evaluator for JCL

A module for evaluating BTEQ scripts called from the JCL jobs.

The BTEQ Evaluator for JCL includes the following parameters:

### Bteq program context

Default Bteq program context

| Parameter name | Description |
|---|---|
| Prompt | Bteq program context |
| Symbol | bteq.programContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - host.db.schema |

### PatternReplacer configuration path

The path to a configuration of the PatternReplacer functionality.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | bteq.patternReplacerFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

# BTEQ Reader

Reads Teradata BTEQ files.

The BTEQ Reader includes the following parameters:

### Context

Default SQL context

| Parameter name | Description |
|---|---|
| Prompt | Context |
| Symbol | bteq.sqlContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - host.db.db |

### PatternReplacer configuration path

The path to a configuration of the PatternReplacer functionality.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | bteq.patternReplacerFile |

| Parameter name | Description |
|---|---|
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

## Scripts exclude filters

The paths of BTEQ scripts that the job excludes.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Scripts exclude filters |
| Symbol | bteq.fileExcludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\bteq-script.txt<br>- /opt/input/bteq-script.txt<br>- /opt/input/*.txt |

## Scripts Root Directory Path

Root directory of input files. Relative paths of input files resolve in the following location: [Input root directory]. Absolute paths from input files must start from the [Input root directory] to be included into processing. [Input root directory] will not be visible on the output objects structure.

| Parameter name | Description |
|---|---|
| Prompt | Scripts Root Directory Path |
| Symbol | bteq.inputRoot |
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\inputs\<br>- /opt/feed-etl/ |

### Scripts' paths

Bteq scripts' paths

| Parameter name | Description |
|---|---|
| Prompt | Scripts' paths |
| Symbol | bteq.fileIncludeFilter |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input\bteq-script.txt<br>- /opt/input/bteq-script.txt<br>- /opt/input/*.txt |

# Business Objects Data Services Reader

Business Objects Data Services reader module

The Business Objects Data Services Reader includes the following parameters:

### BODS input files

Paths to Business Objects Data Services files. Provided files should be XML files representing repositories exported through BODS.

| Parameter name | Description |
|---|---|
| Prompt | BODS input files |
| Symbol | bods.inputFiles |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\RootDir\input\home\Jobs\admin |

### BODS input filters

Case-sensitive include or exclude filters that allow to filter jobs. These filters use the following syntax: <FilterType>:<Pattern> Where: FilterType: Include - if non-empty only the assets prefixed with at least one of given qualification will be processed Exclude - items matching at least one of given patters will be omitted Pattern: PROJECT_NAME[/JOB_NAME[/WORKFLOW_NAME]*([/DATAFLOW_NAME\]*|[/SCRIPT_NAME])] Use a question mark (?) and an asterisk mark (*) inside each pattern. A question mark (?) matches any single

character, an asterisk mark (*) matches any number of characters in an element. To match multiple segments, use a double asterisk mark (**) instead of multiple single asterisks for each element. If a folder name contains spaces or other non-standard characters, surround the name with double quotation marks (""). See examples for details.

| Parameter name | Description |
|---|---|
| Prompt | BODS input filters |
| Symbol | bods.inputFilters |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - Include:Project/**<br>- Include:Project/Job/Workflow*/**<br>- Include:*/Job<br>- Include:Job/Workflow?/Dataflow<br>- Include:**/Dataflow<br>- Include:*<br>- Exclude:Project/Job/*/Dataflow<br>- Exclude:Project/**/Script<br>- Exclude:Job/Workflow?<br>- Exclude:**/Dataflow<br>- Exclude:* |

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | bods.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Default variables values file

The filepath with default variable values. File format consist of [GLOBAL] section.

---

You can define substitutions parameters definitions: [GLOBAL] SubstitutionParameterName = SubstitutionParameterExpression

SubstitutionParameterName that contains only standard alphanumeric characters does not require any quotes. If it contains any special characters, it requires double-quotes (").

Example of substitution parameter definition:

[GLOBAL] "[$$Subparam1]" = 42 "[$$Subparam2]" = 7 "[$$Subparam3]" = 'example'

---

You can define functions definitions: [GLOBAL] FunctionName = FunctionExpression

FunctionName can be unquoted if contains only standard alphanumeric characters but needs to be quoted with double quoted (") if contains any special character. FunctionName must contain list of arguments. List of argument is defined inside "(" ")" characters and consists of "?" sings separated by "," (one "?" sign for each function argument).

Example of function definition: [GLOBAL] a(?) = 1 a(?,?) = 2 b(?) = a(1) + 1 // Expression use call to another function. c(?,?) = d + 2 // Expression use reference to variable.

---

Additional sections can be provided to match only specific functions. To match an overloaded function please provide placeholders for its arguments. Inside the matched section you can also reference matched function arguments. Example of custom function sections:

- [host.db.schema.func(x)]
- z=x
- [host.db.schema.func(x,y)]
- z=x+y

---

To escape special characters such as \n \t use the \ character. For example, to define E:\file \tgtParameterized.csv, enter E:\\file\\tgtParameterized.csv

---

On the right side of assignments you can use simple expression language that can use values of already defined variables and functions or variables from current job context.

Supported expressions (case insensitive) are:

- literals
  - integer: 42
  - string: 'str'
  - string with quoted single quote: 'str\''
  - boolean: true, false
- equals: 42 == 7 -> FALSE
- not equals: 42 != 7 -> TRUE
- concatenations: 'str1' + 'str2' -> 'str1str2'
- function calls: Replace('abc', 'b', 'd') -> 'adc'
- tertiary operator:
  - 42 == 7 ? 'a' : 'b' -> 'b'
  - 42 != 7 ? 'a' : 'b' -> 'a'

Supported functions:

- Hash(str, maxOutputLen) - MD5, maxOutputLen between 1 and 32 inclusive
  - Hash('abcdefgh', 4) -> 'E8DC'
- Replace(str, from, to)
  - Replace('abc', 'b', 'D') -> 'aDc'
- ReplaceRegexp(str, regex, replacement) - replaces with Java regular expressions
  - ReplaceRegexp('abcde', 'b.*', 'f') -> 'af'
  - ReplaceRegexp('graph_id20', 'id(\d+)', '$1') -> 'graph_20'
- StringLengthLimit(str, limit, hashSize) - returns str[0..limit-hashSize] + Hash( str, hashSize )
  - StringLengthLimit('abcc', 3, 2) -> 'a26'
  - StringLengthLimit('abcc', 4, 2) -> 'abcc'
- StringLengthLimit(str, limit) - hashSize has default size 8
  - StringLengthLimit('abcdabcdabcd', 10) -> 'abE340600C'
- RegexpMatch(pattern, str) - tests if str matches pattern
  - pattern uses standard Java 8 regular expressions: https://docs.oracle.com/javase/8/docs/api/java/util/regex/Pattern.html
  - RegexpMatch('[A-Za-z]+','Abcd') -> TRUE
- Upper(str) - returns uppercased str
  - Upper('Abc') -> 'ABC'
- Lower(str) - return lowercased str
  - Lower('aBC') -> 'abc'
- Date(text,format) - returns object representing date with provided format.
  - Formats uses standard Java 8 SimpleDateFormat: https://docs.oracle.com/javase/8/docs/api/java/text/SimpleDateFormat.html
  - It is designed for default values usage
  - ex. Date('2017-10-31', 'yyyy-MM-dd')
- Contains/ContainsIgnoreCase(stack, needle) - tests if stack string contains needle string
  - Contains('abc', 'ab') -> TRUE
  - Contains('abc', 'AB') -> FALSE
  - ContainsIgnoreCase('abc', 'AB') -> TRUE
  - Contains('stack', 'needle') -> FALSE
- RtfToText(str) - return clean text from rtf string
  - RtfToText('{\rtf1\ansi{\fonttbl\f0\fswiss Helvetica;}\f0\pard text\par }') -> text

Supported comments:

- Single line: \ comment

- Multi line: /* comment */

| Parameter name | Description |
|---|---|
| Prompt | Default variables values file |
| Symbol | bods.defaultValuesFilePath |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\RootDir\defaultVariableValues.txt<br>- /opt/RootDir/defaultVariableValues.txt |

## Host of BODS input files

The host of Business Objects Data Services files

| Parameter name | Description |
|---|---|
| Prompt | Host of BODS input files |
| Symbol | bods.fileHost |
| Type | STRING |
| Required | Yes |
| Default value | BODS_HOST |
| Example values | - localhost<br>- etl01 |

## Log warn on missing variable

Decide about logging warn message when there is usage of variable with unknown value

| Parameter name | Description |
|---|---|
| Prompt | Log warn on missing variable |
| Symbol | bods.logWarnOnMissingVariable |
| Type | BOOLEAN |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | true |
| Example values | - true<br>- false |

## Smart condition handling

If set, condition evaluation is skipped allowing to be processed: both if and else (if exists) flows, while flow exactly once. If not set, logical condition will be evaluated and the decision will be made depending on its result. It may result in unprocessed elements, e.g. dataflows. In case of errors or unsupported expressions processed condition will return false.

| Parameter name | Description |
|---|---|
| Prompt | Smart condition handling |
| Symbol | bods.smartConditionHandling |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## The BODS repository which contains jobs

BODS repository name

| Parameter name | Description |
|---|---|
| Prompt | The BODS repository which contains jobs |
| Symbol | bods.repositoryOverride |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - ORACLE.BODS01 |

## Supported Components

- Embedded Dataflow Call - Dataflow Component
- Excel Source - Dataflow Component
- File Source - Dataflow Component
- File Target - Dataflow Component

- Ohs Table Source - Dataflow Component
- Query - Dataflow Component
- R3 Dataflow Call - Dataflow Component
- Sap Extractor - Dataflow Component
- Table Source - Dataflow Component
- Table Target - Dataflow Component
- Xml Source - Dataflow Component
- Xml Target - Dataflow Component
- Call Step - Flow Control Component
- If Step - Flow Control Component
- Parallel Step - Flow Control Component
- Script - Flow Control Component
- Try Step - Flow Control Component
- While Step - Flow Control Component
- Base_EntityExtraction - Dataflow SDK Transform Component
- Wizard_MatchBatch - Dataflow SDK Transform Component
- Case Operation - Dataflow Transform Component
- Data Transfer - Dataflow Transform Component
- Date Generation - Dataflow Transform Component
- Hierarchy Flattening - Dataflow Transform Component
- History Preserving - Dataflow Transform Component
- Key Generation - Dataflow Transform Component
- Map Operation - Dataflow Transform Component
- Merge - Dataflow Transform Component
- Pivot - Dataflow Transform Component
- Reverse Pivot - Dataflow Transform Component
- Row Generation - Dataflow Transform Component
- Sql - Dataflow Transform Component
- Table Comparison - Dataflow Transform Component
- Xml Map - Dataflow Transform Component
- Validation - Dataflow Transform Component

# COBOL Evaluator for JCL

A module for evaluating COBOL programs called JCL jobs.

The COBOL Evaluator for JCL includes the following parameters:

## Call stack max. size

Maximum number of nested CALL statements to process. Deeper level calls will not be processed (proper WARN message will appear in logs).

| Parameter name | Description |
|---|---|
| Prompt | Call stack max. size |
| Symbol | cobol.callStackMaxSize |
| Type | STRING |
| Required | No |
| Default value | 200 |
| Example values | - 10<br>- 40 |

## Cobol evaluator

Enable the use of this module by selecting "Process". Selecting "Skip" disables the module.

| Parameter name | Description |
|---|---|
| Prompt | Cobol evaluator |
| Symbol | cobol.shouldProcessCobol |
| Type | MODULE_MODE |
| Required | No |
| Default value | Skip |
| Example values | - Skip<br>- Process |

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Copybook files

The paths of copybook files that could be used by COBOL programs. Directories are searched recursively.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Copybook files |
| Symbol | cobol.copybookIncludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\copybook*<br>- /opt/input/*.cpy<br>- sftp://username:password@hostname[:port]/userLocalDir/*.cpy<br>- sftp://username:password@hostname[:port]//rootDir/*.cpy |

## DB bindings

Bindings between COBOL program and DB system

| Parameter name | Description |
|---|---|
| Prompt | DB bindings |
| Symbol | cobol.dbBindings |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - CBLPROG=PWG1 |

## Default values file path

Path to the file which contains default values of variables defined in DATA DIVISION. Each section can contain 0 or more default values definitions. Definitions that should be applied for all processed programs need to be defined in [GLOBAL] section. If you want to define values for single COBOL program, put them in [program name] section. As value you have to provide text literal (e.g. "'value"'). Optionally you can provide condition after the text literal. In such case each condition is evaluated consecutively and the first value for

which condition is fulfilled is used. Eventually, if none of conditions is fulfilled, value without any condition clause will be used. For example, it may have the following content:

- [GLOBAL]
- FIELD1 OF REC1 = 'value1'
- FIELD2 OF REC1 = 'value2'
- FIELD2 OF REC1 = 'value2' if(FIELD3 OF REC1 = 2)
- FIELD2 OF REC1 = 'value2' if(FIELD3 OF REC1 = 4)
- [CBLPROG1]
- FIELD1 OF REC1 = 'other value'

| Parameter name | Description |
|---|---|
| Prompt | Default values file path |
| Symbol | cobol.defaultValuesFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\defaultVariableValues.txt<br>- /opt/etc/defaultVariableValues.txt |

## Do not fail on paragraph error

Enables skipping incorrect paragraphs

| Parameter name | Description |
|---|---|
| Prompt | Do not fail on paragraph error |
| Symbol | cobol.doNotFailOnParagraphError |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Enable programs cache

Enables caching COBOL programs after parsing for further use (it speeds up performance but increase memory usage).

| Parameter name | Description |
|---|---|
| Prompt | Enable programs cache |
| Symbol | cobol.enableProgramsCache |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## External files

The paths of external files (e.g. SQL sources) that could be used by COBOL programs. Directories are searched recursively.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | External files |
| Symbol | cobol.externalFileIncludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\sql*<br>- /opt/input/sql*<br>- sftp://username:password@hostname[:port]/userLocalDir/sql/sql*<br>- sftp://username:password@hostname[:port]//rootDir/sql/sql* |

## IMS Database definitions files

The paths of database definitions files that could be used by COBOL programs to establish connection to IMS database. Directories are searched recursively.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | IMS Database definitions files |
| Symbol | cobol.imsDatabaseDefinitionsFiles |

| Parameter name | Description |
|---|---|
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\DBDs*<br>- /opt/input/*.dbd<br>- sftp://username:password@hostname[:port]/userLocalDir/*.dbd<br>- sftp://username:password@hostname[:port]//rootDir/*.dbd |

## Ims PSB definitions files

Path to files containing PSB definitions. Required for proper execution of IMS Cobol program launched via JCL DFSRC00 utility.

| Parameter name | Description |
|---|---|
| Prompt | Ims PSB definitions files |
| Symbol | cobol.imsPsbFileIncludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\PSB*<br>- /opt/input/*.psb<br>- sftp://username:password@hostname[:port]/input/*.psb |

## Input files

The paths of all COBOL files.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | cobol.fileIncludeFilter |
| Type | FILELIST |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Y:\input*.cbl<br>- /opt/input/*.cbl<br>- sftp://username:password@hostname[:port]/userLocalDir/*.cbl<br>- sftp://username:password@hostname[:port]//rootDir/*.cbl |

## Load all copybooks

Set the parameter to true to load all copybooks or set the parameter to false to load only used copybooks

| Parameter name | Description |
|---|---|
| Prompt | Load all copybooks |
| Symbol | cobol.loadAllCopybooks |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Log additional debug information

Enables logging source text of programs that had errors during processing.

| Parameter name | Description |
|---|---|
| Prompt | Log additional debug information |
| Symbol | cobol.logAdditionalDebugInformation |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Pattern replacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | Pattern replacer configuration path |
| Symbol | cobol.patternReplacerFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\patternReplacer.json<br>- /opt/etc/patternReplacer.json |

## Perform simple processing after error

Enables simplified file assets extraction after error

| Parameter name | Description |
|---|---|
| Prompt | Perform simple processing after error |
| Symbol | cobol.simpleProcessingAfterError |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Program specific parameters file path

Path to the file which contains parameters that can be set per COBOl program. Each section can contain 0 or more parameter definitions. Definitions that should be applied for all processed programs need to be defined in [GLOBAL] section. If you want to define parameters for single COBOL program, put them in [program name] section. In section header ([...]) you can use wildcards to address more that one program. As value you have to provide text literal (e.g. "'value"'). Sections are processed top-down so latter definition override former one. In configuration file, you can define following parameters:

- analyseNestedCalls - enables analysis of programs called from program being processed
- analyseNestedCallsOnlyOnce - if true and analyseNestedCalls is true, enables analysis of program called from program being processed only once, any subsequent call of the specific subprogram will be omitted
- fastAnalysisMode - disables processing paragraphs that were already processed with similar set of variables. Produced lineage may not by 100% accurate in Fast Analysis Mode.

- fullDebugLog - enables verbose logging (information about skipped paragraph processing, etc)
- originalTextMode - enables setting original text for processed cobol items (paragraphs, sections, programs, division).

Possible values:

- nothing - do not set input text for any element (default value),
- paragraphs - set input text only for paragraphs
- all - set input text only all items

maxProcessingTime - maximum number of seconds that single COBOL program processing could last. To set unlimited time, use asterisk: '*'.

smartConditionHandling - if true, conditional statements will not be analysed if condition is evaluated to FALSE

byteLevelLineageOnProgramCall - enables processing different record structures from called and calling programs' USING lists and compute lineage between leafs of those records using their length extracted from PICTURE clauses.

simpleVariableValuesTracing - enables simple variable values tracking. It means that variable values will not be analysed independently in each conditional statement branch. Assignments from ELSE statement will override corresponding assignments from IF statements if the same variable was modified in both branches. In this mode, variables hierarchy will be ignored while tracking variables values. It means that if WS-A is parent of WS-B and value of WS-B will be modified, value of WS-A will remain the same. Note that this will strongly affect smart condition handling so you should not use smart condition handling and simple variable values tracing at the same time.

analyseParagraphsOnceForTheSameStackTrace - disables analysis of paragraphs that were already analysed in context of the same stack trace (sequence of paragraphs analysed before).

maxNumberOfProcessingGoToOrPerform - disables processing GO TO or PERFORM statement when called paragraphs has been already processed max number times.

useWorkingStorageFileLayout - enables usage of assigned structure from WORKING STORAGE SECTION when processing WRITE statement with single data item (without child data items) as an argument.

treatDeclareTableAsCreate - enables handling DECLARE TABLE statements in COBOL program as if they were CREATE TABLE statements (new table with given columns will be created in metadata repository). It might be used to reduce the amount of processing errors when metadata repository is incomplete. By default DECLARE TABLE statements are ignored.

qualifyCobolProgramNameWithStackTrace - enables using whole program call stack separated with dots as a COBOL program name, i.e. JCL_JOB.JCL_STEP.COBOL_PROGRAM. Note: this parameter could be used in GLOBAL section only.

optimizeGoToStatements - enables optimization of GO TO statements through the exclusion of paths that are unlikely to have additional lineage.For example, configuration file may have the following content:

- [GLOBAL]
- analyseNestedCalls = 'true'
- fastAnalysisMode = 'false'
- fullDebugLog = 'false'
- maxProcessingTime = '600'
- maxNumberOfProcessingGoToOrPerform = '100'
- smartConditionHandling = 'false'

- simpleVariableValuesTracing = 'false'
- treatDeclareTableAsCreate = 'true'
- qualifyCobolProgramNameWithStackTrace = 'true'
- [CBL10*]
- smartConditionHandling = 'true'
- useWorkingStorageFileLayout = 'true'
- [*9999]
- maxProcessingTime = '*'
- maxNumberOfProcessingGoToOrPerform = '*'

Values presented in GLOBAL section in example above, represent also values that will be used when no configuration file is provided.

| Parameter name | Description |
|---|---|
| Prompt | Program specific parameters file path |
| Symbol | cobol.programSpecificParametersFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\programSpecificParameters.txt<br>- /opt/etc/programSpecificParameters.txt |

## Programs exclude filters

PROGRAM-IDs of COBOL programs that should not be processed.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Programs exclude filters |
| Symbol | cobol.programExcludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - CBLDEMO1<br>- CBLDEMO*<br>- CBL*1 |

### Programs to inline filters

PROGRAM-IDs of COBOL programs that will be inlined when called

| Parameter name | Description |
|---|---|
| Prompt | Programs to inline filters |
| Symbol | cobol.programsToInlineFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - CBLDEMO1<br>- CBLDEMO*<br>- CBL*1 |

# COBOL Reader

Reads COBOL files and converts them to Mf3 Graphs.

The COBOL Reader includes the following parameters:

### Call stack max. size

Maximum number of nested CALL statements to process. Deeper level calls will not be processed (proper WARN message will appear in logs).

| Parameter name | Description |
|---|---|
| Prompt | Call stack max. size |
| Symbol | cobol.callStackMaxSize |
| Type | STRING |
| Required | No |
| Default value | 200 |
| Example values | - 10<br>- 40 |

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Copybook files

The paths of copybook files that could be used by COBOL programs. Directories are searched recursively.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Copybook files |
| Symbol | cobol.copybookIncludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\copybook*<br>- /opt/input/*.cpy<br>- sftp://username:password@hostname[:port]/userLocalDir/*.cpy<br>- sftp://username:password@hostname[:port]//rootDir/*.cpy |

## DB Contexts

Contexts used during EXEC SQL statements processing specified per COBOL program. Wildcards can be used in program ID.

| Parameter name | Description |
|---|---|
| Prompt | DB Contexts |
| Symbol | cobol.dbContexts |
| Type | PROPERTIES |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | N/A |
| Example values | - CBLPROG=HOST.DB.SCHEMA<br>- CBL*=HOST2.DB2.SCHEMA2 |

## Default DB Context

Default context used during EXEC SQL statements processing (three segmented name - host, db, schema)

| Parameter name | Description |
|---|---|
| Prompt | Default DB Context |
| Symbol | cobol.defaultDbContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - HOST.DB.SCHEMA |

## Default values file path

Path to the file which contains default values of variables defined in DATA DIVISION. Each section can contain 0 or more default values definitions. Definitions that should be applied for all processed programs need to be defined in [GLOBAL] section. If you want to define values for single COBOL program, put them in [program name] section. As value you have to provide text literal (e.g. "'value"'). Optionally you can provide condition after the text literal. In such case each condition is evaluated consecutively and the first value for which condition is fulfilled is used. Eventually, if none of conditions is fulfilled, value without any condition clause will be used. For example, it may have the following content:

- [GLOBAL]
- FIELD1 OF REC1 = 'value1'
- FIELD2 OF REC1 = 'value2'
- FIELD2 OF REC1 = 'value2' if(FIELD3 OF REC1 = 2)
- FIELD2 OF REC1 = 'value2' if(FIELD3 OF REC1 = 4)
- [CBLPROG1]
- FIELD1 OF REC1 = 'other value'

| Parameter name | Description |
|---|---|
| Prompt | Default values file path |
| Symbol | cobol.defaultValuesFile |

| Parameter name | Description |
|---|---|
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\defaultVariableValues.txt<br>- /opt/etc/defaultVariableValues.txt |

## Do not fail on paragraph error

Enables skipping incorrect paragraphs

| Parameter name | Description |
|---|---|
| Prompt | Do not fail on paragraph error |
| Symbol | cobol.doNotFailOnParagraphError |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Enable programs cache

Enables caching COBOL programs after parsing for further use (it speeds up performance but increase memory usage).

| Parameter name | Description |
|---|---|
| Prompt | Enable programs cache |
| Symbol | cobol.enableProgramsCache |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## External files

The paths of external files (e.g. SQL sources) that could be used by COBOL programs. Directories are searched recursively.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | External files |
| Symbol | cobol.externalFileIncludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\sql*<br>- /opt/input/sql*<br>- sftp://username:password@hostname[:port]/userLocalDir/sql/sql*<br>- sftp://username:password@hostname[:port]//rootDir/sql/sql* |

## IMS Database definitions files

The paths of database definitions files that could be used by COBOL programs to establish connection to IMS database. Directories are searched recursively.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | IMS Database definitions files |
| Symbol | cobol.imsDatabaseDefinitionsFiles |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\DBDs*<br>- /opt/input/*.dbd<br>- sftp://username:password@hostname[:port]/userLocalDir/*.dbd<br>- sftp://username:password@hostname[:port]//rootDir/*.dbd |

## Input files

The paths of all COBOL files.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | cobol.fileIncludeFilter |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input*.cbl<br>- /opt/input/*.cbl<br>- sftp://username:password@hostname[:port]/userLocalDir/*.cbl<br>- sftp://username:password@hostname[:port]//rootDir/*.cbl |

## Load all copybooks

Set the parameter to true to load all copybooks or set the parameter to false to load only used copybooks

| Parameter name | Description |
|---|---|
| Prompt | Load all copybooks |
| Symbol | cobol.loadAllCopybooks |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Log additional debug information

Enables logging source text of programs that had errors during processing.

| Parameter name | Description |
|---|---|
| Prompt | Log additional debug information |
| Symbol | cobol.logAdditionalDebugInformation |
| Type | BOOLEAN |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | true |
| Example values | - true<br>- false |

## Pattern replacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | Pattern replacer configuration path |
| Symbol | cobol.patternReplacerFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\patternReplacer.json<br>- /opt/etc/patternReplacer.json |

## Perform simple processing after error

Enables simplified file assets extraction after error

| Parameter name | Description |
|---|---|
| Prompt | Perform simple processing after error |
| Symbol | cobol.simpleProcessingAfterError |
| Type | BOOLEAN |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | false |
| Example values | - true<br>- false |

## Processing entry points

List of COBOL processing entry points. Empty list means that all input files will be processed. There are two types of entry point supported: COBOL and DB2. COBOL entry point is the name of COBOL program. DB2 entry point is the name of DB2 procedure. Syntax:

- <cobol_program_name>
- COBOL: <cobol_program_name>
- DB2: <host>.<db>.<schema>.<procedure_name>

| Parameter name | Description |
|---|---|
| Prompt | Processing entry points |
| Symbol | cobol.programIncludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - CBLDEMO1<br>- CBLDEMO*<br>- CBL*1<br>- COBOL: CBLDEMO1<br>- DB2: HOST.DB.SCHEMA.PROC_NAME |

## Program specific parameters file path

Path to the file which contains parameters that can be set per COBOl program. Each section can contain 0 or more parameter definitions. Definitions that should be applied for all processed programs need to be defined in [GLOBAL] section. If you want to define parameters for single COBOL program, put them in [program name] section. In section header ([...]) you can use wildcards to address more that one program. As value you have to provide text literal (e.g. "'value"'). Sections are processed top-down so latter definition override former one. In configuration file, you can define following parameters:

- analyseNestedCalls - enables analysis of programs called from program being processed
- analyseNestedCallsOnlyOnce - if true and analyseNestedCalls is true, enables analysis of program called from program being processed only once, any subsequent call of the specific subprogram will be omitted
- fastAnalysisMode - disables processing paragraphs that were already processed with similar set of variables. Produced lineage may not by 100% accurate in Fast Analysis Mode.
- fullDebugLog - enables verbose logging (information about skipped paragraph processing, etc)
- originalTextMode - enables setting original text for processed cobol items (paragraphs, sections, programs, division).

Possible values:

- nothing - do not set input text for any element (default value),
- paragraphs - set input text only for paragraphs
- all - set input text only all items

maxProcessingTime - maximum number of seconds that single COBOL program processing could last. To set unlimited time, use asterisk: '*'.

smartConditionHandling - if true, conditional statements will not be analysed if condition is evaluated to FALSE

byteLevelLineageOnProgramCall - enables processing different record structures from called and calling programs' USING lists and compute lineage between leafs of those records using their length extracted from PICTURE clauses.

simpleVariableValuesTracing - enables simple variable values tracking. It means that variable values will not be analysed independently in each conditional statement branch. Assignments from ELSE statement will override corresponding assignments from IF statements if the same variable was modified in both branches. In this mode, variables hierarchy will be ignored while tracking variables values. It means that if WS-A is parent of WS-B and value of WS-B will be modified, value of WS-A will remain the same. Note that this will strongly affect smart condition handling so you should not use smart condition handling and simple variable values tracing at the same time.

analyseParagraphsOnceForTheSameStackTrace - disables analysis of paragraphs that were already analysed in context of the same stack trace (sequence of paragraphs analysed before).

maxNumberOfProcessingGoToOrPerform - disables processing GO TO or PERFORM statement when called paragraphs has been already processed max number times.

useWorkingStorageFileLayout - enables usage of assigned structure from WORKING STORAGE SECTION when processing WRITE statement with single data item (without child data items) as an argument.

treatDeclareTableAsCreate - enables handling DECLARE TABLE statements in COBOL program as if they were CREATE TABLE statements (new table with given columns will be created in metadata repository). It might be used to reduce the amount of processing errors when metadata repository is incomplete. By default DECLARE TABLE statements are ignored.

qualifyCobolProgramNameWithStackTrace - enables using whole program call stack separated with dots as a COBOL program name, i.e. JCL_JOB.JCL_STEP.COBOL_PROGRAM. Note: this parameter could be used in GLOBAL section only.

optimizeGoToStatements - enables optimization of GO TO statements through the exclusion of paths that are unlikely to have additional lineage.For example, configuration file may have the following content:

- [GLOBAL]
- analyseNestedCalls = 'true'
- fastAnalysisMode = 'false'
- fullDebugLog = 'false'
- maxProcessingTime = '600'
- maxNumberOfProcessingGoToOrPerform = '100'
- smartConditionHandling = 'false'
- simpleVariableValuesTracing = 'false'
- treatDeclareTableAsCreate = 'true'
- qualifyCobolProgramNameWithStackTrace = 'true'

- [CBL10*]
- smartConditionHandling = 'true'
- useWorkingStorageFileLayout = 'true'
- [*9999]
- maxProcessingTime = '*'
- maxNumberOfProcessingGoToOrPerform = '*'

Values presented in GLOBAL section in example above, represent also values that will be used when no configuration file is provided.

| Parameter name | Description |
|---|---|
| Prompt | Program specific parameters file path |
| Symbol | cobol.programSpecificParametersFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\programSpecificParameters.txt<br>- /opt/etc/programSpecificParameters.txt |

## Programs exclude filters

PROGRAM-IDs of COBOL programs that should not be processed.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Programs exclude filters |
| Symbol | cobol.programExcludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - CBLDEMO1<br>- CBLDEMO*<br>- CBL*1 |

### Programs to inline filters

PROGRAM-IDs of COBOL programs that will be inlined when called

| Parameter name | Description |
|---|---|
| Prompt | Programs to inline filters |
| Symbol | cobol.programsToInlineFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - CBLDEMO1<br>- CBLDEMO*<br>- CBL*1 |

# COBOL Test Parser

Tries to parse COBOL files to gather parser errors

The COBOL Test Parser includes the following parameters:

### COBOL input files

The paths of all COBOL files.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | COBOL input files |
| Symbol | cobolTest.fileIncludeFilter |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input\cobol*.cbl<br>- /opt/input/cobol/*.cbl<br>- sftp://username:password@hostname[:port]/input/*.cbl |

## Copybooks

The paths of all copybooks used by COBOL files.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Copybooks |
| Symbol | cobolTest.copybookFiles |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input\copybooks*.cpy<br>- /opt/input/copybooks/*<br>- sftp://username:password@hostname[:port]/input/*.cpy |

## Error on missing copybook

If true, COBOL preprocessor will throw exception on missing copybook

| Parameter name | Description |
|---|---|
| Prompt | Error on missing copybook |
| Symbol | cobolTest.errorOnMissingCopybook |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Output directory

Path to directory where parse errors details will be saved

| Parameter name | Description |
|---|---|
| Prompt | Output directory |
| Symbol | cobolTest.parsingTestOutputDirectory |
| Type | FILE |
| Required | Yes |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Y:\output\parse_test\results<br>- /opt/output/parse_test/results |

# Column Properties PostProcessor

GraphPostProcessorModule is used for refining previously created graph, in order to provide proper data lineage.Corrects output fields with "GenericType" or empty Column NativeType. If no valid NativeType is present for output field it will be resolved from this field input field source. This refining will be performed only for output fields which has always only a single source reference all the way to its input field source.

The Column Properties PostProcessor includes the following parameters:

## Enable columns properties correction of output fields

Disables or enables columns properties correction of output fields in graph.

| Parameter name | Description |
|---|---|
| Prompt | Enable columns properties correction of output fields |
| Symbol | columnChecker.enableCorrection |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

# Columns Propagator

A module that propagates column names from known source structure to target with unknown structure. Columns are propagated backward from a step (ie. COBOL) to its predecessor (ie. FTP in JCL). Propagation requires a single default (LINE) column in each of the target structures. Each propagation is performed separately for each file (distinguished by its fully qualified path). The action will be performed only if there is no doubt that the component / step does not modify the format in any way.

The Columns Propagator includes the following parameters:

### Mode

Module is enabled when "Process" is selected. Selecting "Skip" disables this module.

| Parameter name | Description |
|---|---|
| Prompt | Mode |
| Symbol | columnPropagator.mode |
| Type | MODULE_MODE |
| Required | No |
| Default value | Skip |
| Example values | - Process<br>- Skip |

# Context Mapper

A module for renaming contexts in the output. This can be useful when the input metadata has different names than what is required in the target metadata repository.

The Context Mapper includes the following parameters:

### Context Mapper mode

Available modes of the Context Mapper. Basic mode allows only few core functionalities of regular expressions. Advanced mode fully supports Java Regular Expressions allowing you to create complex mappings.

| Parameter name | Description |
|---|---|
| Prompt | Context Mapper mode |
| Symbol | contextMapper.mode |
| Type | MODULE_MODE |
| Required | No |
| Default value | Basic |
| Example values | - Basic<br>- Advanced |

## Name mapping

Context mappings to be applied (1-3 dot-separated segments for Host, DB, Schema) to input or output components. Each successfully matched segment from "from" column will be replaced with corresponding value from "to" column This module has the following modes:

- Advanced - allows to use full Java Regexp syntax. Segments are separated by '.' sign so if you want to use regular expression wildcard sign '.', you must quote entire segment. See examples below. You can use brackets to define group. Group can be used in "to" field with $n where n is number of group(each group number is unique for the whole pattern, counting from the left). $n sign will be replaced with matched value. For more details about Java Regular Expressions see http://docs.oracle.com/javase/7/docs/api/java/util/regex/Pattern.html. By default context mapping is applied to all components. To filter for which types of components mapping should be done write object type before "from" column. Supported object types are: File, Table, BiReport, Object, StoredProcedure, BiCollection, Entity, Generic. Filtering by object type works only in Advanced mode. All I/O components are processed by each mapping sequentially according to order defined in configuration. Examples:
  - F[a-b]*./file(\w)+=HOST./newfile.txt will match F./filelocal and replace it with HOST./newfile.txt
  - \w+.db.schema([0-9])=LOCALHOST.newdb.anotherschema$1 will match HOST.db.schema5 and replace it with LOCALHOST.newdb.anotherschema5
  - "(.)*"./file.txt=/$1.$1.txt will match LOCAL./file.txt and replace it with LOCAL./LOCAL.txt
  - HOST."(.*?).(csv)"."(.*).(csv)"=NEWHOST.$1.$3 will match HOST.c:/a/file1.csv.c:/a/file2.csv and replace it with NEWHOST.c:/a/file1.c:/a/file2 Above we would like to match file name from second and third segment so that we use group number 1 and 3. To clarify, groups are: 1 = "c:/a/file1", 2 = "csv", 3 = "c:/a/file2", 4 = "csv".
  - File:F[a-b]*./file(\w)+=HOST./newfile.txt will match F./filelocal and replace it with HOST./newfile.txt. Above mapping will be done only for files and only in Advanced mode
  - Table:"(.*)"."(.*)"."(.*)"."table"=$1.$2.$3.newtable will replace table with newtable. Above mapping will be done only for tables and only in Advanced mode
- Basic - allows to use wildcard '*' operator and any sign '?' operator. Examples:
  - HOST.*.SCHEMA=HOST.OTHERDB.SCHEMA will match HOST.DB.SCHEMA and replace it with HOST.OTHERDB.SCHEMA
  - HOST*.*.SCHEMA=HOST.OTHERDB.SCHEMA will match HOST100.DB.SCHEMA and replace it with HOST.OTHERDB.SCHEMA
  - HOST.*.SCH?MA=HOST.OTHERDB.SCHEMA will match HOST.DB.SCHEMA and replace it with HOST.OTHERDB.SCHEMA

| Parameter name | Description |
|---|---|
| Prompt | Name mapping |
| Symbol | contextMapper.mapping |
| Type | MULTIMAP |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - A.B.C=D.E.F - Maps the schema identified as A.B.C to D.E.F<br>- A.*.C=D.E.F - All names with host A and schema C will be mapped to D.E.F<br>- A.B=D.E - Maps only the database B on host A to D.E<br>- A=D - Maps host A to D<br>- A.(?i)b=C.D - Advanced - Ignores case sensitivity and maps schema A.b or A.B to C.D |

# Database object reader

A module for reading stored procedure bodies or view definitions from database. It can be used to parse various SQL dialects.

The Database object reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Deprecated Netezza active User name

Unused, if set in Netezza config, please move it to SQLEvaluatorModule.activeUser instead

| Parameter name | Description |
|---|---|
| Prompt | Deprecated Netezza active User name |
| Symbol | sql.defaultUserName |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - admin |

## Enable Teradata view column data type

Enables the job to fetch the Teradata view column data type. Setting the property to true increases processing time.

| Parameter name | Description |
|---|---|
| Prompt | Enable Teradata view column data type |
| Symbol | sql.enableTeradataViewColumnDataType |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Include invalid and encrypted objects

Include invalid and encrypted objects in analysis. Invalid objects are specific for ORACLE. Encrypted objects are specific for SqlServer.

| Parameter name | Description |
|---|---|
| Prompt | Include invalid and encrypted objects |
| Symbol | sql.includeInvalidObjects |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Object exclude filters

List of exclude filters for database objects(view or stored procedures). If there is no include filters every database object except ones matching any of exclude filters will be processed. IF there are specified include filters objects matching any of include filters and not matching any of exclude filters will be processed Each filter is composed as subfilters separated by dots. In each subfilter user can provide special two characters.Question mark matches any single character, asterisk matches any number of any characters. See

examples for details. First two filter segments should contain host and database name. (e.g., HOST.DBNAME) Next segments are database vendor specific.

| Parameter name | Description |
|---|---|
| Prompt | Object exclude filters |
| Symbol | sql.objectExcludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - hostA.sciencedb.astronomy.* |

## Object include filters

List of include filters for database objects (views or stored procedures). It is used for processing limited set of objects. Object is processed when it matches minimum one filter. Each filter is composed as subfilters separated by dots. In each subfilter user can provide special two characters.Question mark matches any single character, asterisk matches any number of any characters. See examples for details. First two filter segments should contain host and database name. (e.g., HOST.DBNAME) Next segments are database vendor specific.

| Parameter name | Description |
|---|---|
| Prompt | Object include filters |
| Symbol | sql.objectIncludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - hostA.sciencedb.astronomy.*<br>- hostA.db.triggerTableOwner.triggerTable.triggerOwner.triggerName |

## PatternReplacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | sql.patternReplacerFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

## Read from

Select at least one of the following modes:

- "Database Assets" - table and view structures with columns (no lineage),
- "Views" - SQLs will be extracted from views definition,
- "Stored procedures" - SQLs will be extracted from stored procedures,
- "Type definitions" - SQLs will be extracted from type definitions (Oracle only),
- "Table-valued functions" - SQLs will be extracted from table-valued functions,
- "External tables" - SQLs will be extracted from External tables,
- "Triggers" - SQLs will be extracted from triggers,
- "Tasks" - SQLs will be extracted from tasks,
- "Packages" - SQLs will be extracted from packages,
- "Pipes" - SQLs will be extracted from pipes,
- "Tags" - Tag and relationships with databases, schemas, tables, stage and columns (no lineage)

| Parameter name | Description |
|---|---|
| Prompt | Read from |
| Symbol | sql.ddlReadSource |
| Type | MODULE_MODES |
| Required | No |
| Default value | Stored procedures |
| Example values | - Views<br>- Stored procedures |

### Skip unmodified objects

Database objects, which did not have their source changed will not be processed.

| Parameter name | Description |
|---|---|
| Prompt | Skip unmodified objects |
| Symbol | sql.skipUnmodified |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

# Databricks Assets

A module for retrieving assets, such as tables and views, from Databricks.

The Databricks Assets includes the following parameters:

### Assets exclude filter

A list of exclude filters for Databricks assets, such as tables or views. If there are no include filters, every Databricks asset, except the ones that match any of the exclude filters, is processed. If you specify any include filters, assets that match any of the include filters and do not match any of the exclude filters are processed. A filter contains four segments separated by a period. In each segment, you can provide two special characters. A question mark (?) matches any single character, and an asterisk (*) matches any number of any characters. For details, see the example values. The first two filter segments should contain the host and database name, for example, HOST.DBNAME. The other two segments are the schema name and asset name.

| Parameter name | Description |
|---|---|
| Prompt | Assets exclude filter |
| Symbol | databricks.assetExcludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - hostA.sciencedb.astronomy.*<br>- hostA.db.schema.tableName |

### Assets include filter

A list of include filters for Databricks assets, such as tables or views. It is used for processing a limited set of assets. An asset is processed when it matches at least one filter. A filter contains four segments separated by a period. In each segment, you can provide two special characters. A question mark (?) matches any single character, and an asterisk (*) matches any number of any characters. For details, see the example values. The first two filter segments should contain the host and database name, for example, HOST.DBNAME. The other two segments are the schema name and asset name.

| Parameter name | Description |
|---|---|
| Prompt | Assets include filter |
| Symbol | databricks.assetIncludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - hostA.sciencedb.astronomy.*<br>- hostA.db.schema.tableName |

# Databricks evaluator

A module for evaluating Databricks notebooks embedded in ETL processes. This prevents the breaking of data lineage at table components in a mapping.

The Databricks evaluator includes the following parameters:

### Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Databricks Catalog Preload exclude filter

A list of exclude filters for Databricks Catalog assets, such as tables or views. If there are no include filters, every internal Databricks asset, except the ones that match any of the exclude filters, is processed. If you specify any include filters, assets that match any of the include filters and do not match any of the exclude filters are processed. A filter contains four segments separated by a period. In each segment, you can provide two special characters. A question mark (?) matches any single character, and an asterisk (*) matches any number of any characters. For details, see the example values. The first two filter segments should contain the host and database name, for example, HOST.DBNAME. The other two segments are the schema name and asset name.

| Parameter name | Description |
|---|---|
| Prompt | Databricks Catalog Preload exclude filter |
| Symbol | databricks.catalogPreloadExcludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - hostA.sciencedb.astronomy.*<br>- hostA.db.schema.tableName |

## Databricks Catalog Preload include filter

A list of include filters for preloading Databricks Catalog assets, such as tables or views. It is used for loading a limited set of assets to optimize processing time. An asset is processed when it matches at least one filter. A filter contains four segments separated by a period. In each segment, you can provide two special characters. A question mark (?) matches any single character, and an asterisk (*) matches any number of any characters. For details, see the example values. The first two filter segments should contain the host and database name, for example, HOST.DBNAME. The other two segments are the schema name and asset name.

| Parameter name | Description |
|---|---|
| Prompt | Databricks Catalog Preload include filter |
| Symbol | databricks.catalogPreloadIncludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - hostA.sciencedb.astronomy.*<br>- hostA.db.schema.tableName |

## Databricks connection

Connections details for Databricks notebooks.

| Parameter name | Description |
|---|---|
| Prompt | Databricks connection |
| Symbol | databricks.connection |
| Type | SENSITIVE_PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - databricksHost.com=token:dapi8be46d02f290489a1defbad48bbbbf77-3<br>- databricksHost.com/User/MyUser=token:dapi8be46d02f290489a1defbad48bbbbf77-3<br>- databricksHost.com=dbcfile:Y:\input\databricks\example.dbc<br>- databricksHost.com/User/MyUser=dbcfile:Y:\input\databricks\example.dbc |

## Databricks environment initialization file

The absolute path to the Python code file that defines the mount points and other environment properties related to the Databricks source. Specify the file path to resolve mount points when you extract metadata from the Databricks source.

| Parameter name | Description |
|---|---|
| Prompt | Databricks environment initialization file |
| Symbol | databricks.environmentInitializationFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\notebookInitialization.py<br>- /opt/etc/notebookInitialization.py |

## Delta Lake SQL Context

Delta Lake SQL context for specific host with optional workspace path

| Parameter name | Description |
|---|---|
| Prompt | Delta Lake SQL Context |
| Symbol | databricks.sqlContext |
| Type | PROPERTIES |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | N/A |
| Example values | - databricksHost.com=DELTALAKE-HOST.DELTALAKE-DATABASE.DEFAULT<br>- databricksHost.com/User/MyUser=DELTALAKE-HOST.DELTALAKE-DATABASE.DEFAULT |

### PatternReplacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | databricks.patternReplacerFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

# Databricks Reader

Reads Databricks notebooks.

The Databricks Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | databricks.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Databricks environment initialization file

The absolute path to the Python code file that defines the mount points and other environment properties related to the Databricks source. Specify the file path to resolve mount points when you extract metadata from the Databricks source.

| Parameter name | Description |
|---|---|
| Prompt | Databricks environment initialization file |
| Symbol | databricks.environmentInitializationFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\notebookInitialization.py<br>- /opt/etc/notebookInitialization.py |

## Databricks input read mode

Choose one of the following modes to read input: Files mode. Reads local files. Workspace mode. Reads remote Databricks workspaces. Fetch Databricks assets mode. Reads only Databricks assets.

| Parameter name | Description |
|---|---|
| Prompt | Databricks input read mode |
| Symbol | databricks.inputReadMode |

| Parameter name | Description |
|---|---|
| Type | MODULE_MODE |
| Required | No |
| Default value | Files |
| Example values | - Files<br>- Workspace<br>- Fetch Databricks assets |

## Databricks notebook exclude filters

A list of exclude filters for Databricks notebooks. Empty list means that there will be no exclusions. Each notebook matching at least one exclude filter will not be processed. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Databricks notebook exclude filters |
| Symbol | databricks.notebookExcludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - /notebook1<br>- */directory/notebook1 |

## Databricks notebook include filters

A list of include filters for Databricks notebooks used for processing a limited set of notebooks. Empty list means that all found notebooks will be processed. Notebook is processed when it matches minimum one filter. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Databricks notebook include filters |
| Symbol | databricks.notebookIncludeFilter |
| Type | STRINGLIST |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - /notebook1<br>- */directory/notebook1 |

## Delta Lake SQL Context

Delta Lake SQL context

| Parameter name | Description |
|---|---|
| Prompt | Delta Lake SQL Context |
| Symbol | databricks.sqlContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - DELTALAKE-HOST.DELTALAKE-DATABASE.DEFAULT |

## Enable entry point detection

Decide whether to detect entry points in Databricks Notebooks or not

| Parameter name | Description |
|---|---|
| Prompt | Enable entry point detection |
| Symbol | databricks.enableEntryPointDetection |
| Type | PRIMITIVE_BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Exclude filters for the IDs of Databricks workflow jobs

A list of exclude filters for Databricks job IDs used for processing. If the list is empty, nothing is excluded. Jobs matching at least one exclude filter are not processed. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Exclude filters for the IDs of Databricks workflow jobs |
| Symbol | databricks.jobsExcludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - job1*<br>- job?0 |

## Exclude filters for the names of Databricks pipelines

A list of exclude filters for Databricks pipelines names used for processing. If the list is empty, nothing is excluded. Pipelines matching at least one exclude filter are not processed. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Exclude filters for the names of Databricks pipelines |
| Symbol | databricks.pipelinesExcludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - pipeline1*<br>- pipeline?0 |

## Include filters for the IDs of Databricks workflow jobs

A list of include filters for Databricks job IDs used for processing. If the list is empty, all jobs are included. Jobs matching at least one include filter are processed. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Include filters for the IDs of Databricks workflow jobs |
| Symbol | databricks.jobsIncludeFilter |

| Parameter name | Description |
|---|---|
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - job1*<br>- job?0 |

## Include filters for the names of Databricks pipelines

A list of include filters for Databricks pipelines names used for processing. If the list is empty, all pipelines are included. Pipelines matching at least one include filter are processed. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Include filters for the names of Databricks pipelines |
| Symbol | databricks.pipelinesIncludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - pipeline1*<br>- pipeline?0 |

## Input files

The paths of files to be processed. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | databricks.fileIncludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\databricks\example.py<br>- /opt/input/jcl/example.sql |

## Notebooks Workspace Path

The path from which scanner will fetch notebooks.

| Parameter name | Description |
|---|---|
| Prompt | Notebooks Workspace Path |
| Symbol | databricks.workspacePath |
| Type | STRING |
| Required | No |
| Default value | /Users |
| Example values | - /Users/MyUser/ |

## PatternReplacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | databricks.patternReplacerFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

### Workspace authentication token

Authentication token for the Databricks workspace access. A random string of 38 symbols generated by Databricks.

| Parameter name | Description |
|---|---|
| Prompt | Workspace authentication token |
| Symbol | databricks.authenticationToken |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - dapi8be46d02f290489a1defbad48bbbbf77-3 For information about generating the authentication token, see: https://docs.databricks.com/dev-tools/api/latest/authentication.html#generate-a-personal-access-token |

### Workspace host

Host name of Databricks workspace.

| Parameter name | Description |
|---|---|
| Prompt | Workspace host |
| Symbol | databricks.workspaceHost |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - databricksHost.com<br>- databricksHost.com:1234 |

### Supported Languages

- Python - Supported language
- SQL - Supported language
- Markdown - Supported language

# DataStage Reader

Reads DataStage files. Requires: - .isx files, which can be exported from DataStage project by IBM Infosphere Information Manager; - access to databases used by .isx file.

The DataStage Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | dataStage.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## File Host name / Output Model Engine name

Host name used for file input/output components and Engine name for output model

| Parameter name | Description |
|---|---|
| Prompt | File Host name / Output Model Engine name |
| Symbol | dataStage.hostName |
| Type | STRING |
| Required | Yes |
| Default value | DATASTAGE_HOST |
| Example values | - servername.com |

## Filename Mapper

DataStage uses the following mappings to:

- resolve paths to user files in stages
- resolve paths for SCH files in the Sequential File stage with the Schema File property. Upload the Schema File SCH files to the server where you installed the advanced scanner. The part you provided in "From" section is replaced with the part you provided in "To" section.

| Parameter name | Description |
|---|---|
| Prompt | Filename Mapper |
| Symbol | dataStage.filesMap |

| Parameter name | Description |
|---|---|
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - /export/home/RootDir/datastage=C:\RootDir\datastage<br>- C:\RootDir\datastage=/export/home/RootDir/datastage |

## Include/Exclude job filters

Case sensitive include/exclude filters, allowing to filter jobs to be processed. These filters use the following syntax: <FilterType> : <Pattern> Where: FilterType: Include - if nonempty only the assets prefixed with at least one of given qualification will be processed Exclude - items matching at least one of given patters will be omitted Pattern: PROJECT_NAME[:DIRECTORY_PATH[:JOB_NAME]] Question mark and asterisk mark can be used inside each pattern. Question mark matches any single character, asterisk matches any number of any characters. If folder contains spaces or other non-standard characters - quote it with double quotes. See examples for details.

| Parameter name | Description |
|---|---|
| Prompt | Include/Exclude job filters |
| Symbol | dataStage.jobFilters |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - Include : projectN?me:/Jobs/directory:*<br>- Exclude : projectName:"/directory":jobNa*<br>- Include:project:/Jobs/dir:job<br>- Include : project : /Jobs/dir : job<br>- Include : project : "/Jobs/d ir" : job |

## Input File Encoding

Input File Encoding

| Parameter name | Description |
|---|---|
| Prompt | Input File Encoding |
| Symbol | dataStage.inputFileEncoding |
| Type | CHARSET |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | UTF-8 |
| Example values | - UTF-8<br>- UTF-16<br>- UTF-16BE<br>- UTF-16LE<br>- ISO-8859-1<br>- US-ASCII |

## Input files

The paths of ISX files to be processed

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | dataStage.inputs |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\RootDir\input\datastage\sample.isx |

## Job parameter override

These mappings can be used to override job parameters or specify the parameter set values file. Specify the parameter name using the job name, the parameter set name and the parameter name. Separate the components with periods (.), as shown in the following examples: Parameter name: jobName.parameterName Value: parameterValue Parameter name: jobName.parameterSetName.parameterName Value: value Wildcards can also be used e.g: Parameter name: *.parameterName Value: value Specify the parameter set values file using the job name and the parameter set name. Separate the components with a period (.). The Value File corresponds to the DataStage Parameter Set object in the Value tab, …on the Value tab, as shown in the following example: Parameter name: jobName.parameterSetName Value: valuesFileName You can also use a wildcard as a job name. For more information, see MetaDex documentation.

| Parameter name | Description |
|---|---|
| Prompt | Job parameter override |
| Symbol | dataStage.parameterOverrideMap |
| Type | PROPERTIES |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - jobName.parameterName=parameterValue<br>- *.parameterSetName.parameterName=value<br>- jobName.parameterSetName=valuesFileName |

## Maximum variable nesting depth

Maximum variable nesting depth

| Parameter name | Description |
|---|---|
| Prompt | Maximum variable nesting depth |
| Symbol | dataStage.maxVariableNestingDepth |
| Type | STRING |
| Required | No |
| Default value | 50000 |
| Example values | - 100<br>- 500<br>- 1000<br>- 2000 |

## Operational metadata directory

A directory with operational metadata XML files, such as the *.xml.flow file format. If you specify an operation metadata directory parameter, the MetaDex scanner job searches for metadata for every processed job and parameter values. If found, the job is processed as many times as there are runs available for the metadata, with parameter values specified for every run. Overwrites of parameters provided in the configuration do not work.

| Parameter name | Description |
|---|---|
| Prompt | Operational metadata directory |
| Symbol | dataStage.operationalMetadataDir |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\RootDir\input\datastage\SampleFolder |

## Override DataStage host name

Override DataStage components server name specified in ISX file

| Parameter name | Description |
|---|---|
| Prompt | Override DataStage host name |
| Symbol | dataStage.serverOverride |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - myhost.com |

## Override DataStage project name

Override DataStage components project name specified in ISX file

| Parameter name | Description |
|---|---|
| Prompt | Override DataStage project name |
| Symbol | dataStage.projectOverride |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - myProject |

## Process sequence jobs

Process sequence jobs contained in the project

| Parameter name | Description |
|---|---|
| Prompt | Process sequence jobs |
| Symbol | dataStage.processSequenceJobs |
| Type | BOOLEAN |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | false |
| Example values | - true<br>- false |

## Supported Stages

- Aggregator - Parallel
- Aggregator - Server
- CODBCStage - Server
- Change Capture - Parallel
- Checksum - Parallel
- Complex Flat File - Parallel
- Copy - Parallel
- DB2 Connector - Parallel and Server
- DRS Connector - Parallel
- Data Set - Parallel
- FTP - Parallel
- FTP - Server
- FTP Plug In - Parallel
- File Connector - Parallel
- File Set - Parallel
- Filter - Parallel
- Funnel - Parallel
- Hashed File - Server
- Hierarchical Data - Parallel - partial support - only as internal component not being input nor output for Job
- Hive Connector - Parallel
- JDBC Connector - Parallel
- Join - Parallel
- Lookup - Parallel
- Merge - Parallel
- Modify - Parallel
- Netezza Connector - Parallel and Server
- ODBC Connector - Parallel and Server
- Oracle Connector - Parallel and Server
- OracleOCI - Server
- Peek - Parallel
- Pivot - Parallel
- Pivot Enterprise - Parallel

- Remove Duplicates - Parallel
- Row Generator - Parallel
- Sequential File - Parallel
- Sequential File - Server
- Slowly Changing Dimensions - Parallel
- Sort - Parallel
- Sort - Server
- Stored Procedure Connector - Parallel and Server
- Surrogate Key Generator - Parallel
- Switch - Parallel
- Teradata Connector - Parallel and Server
- Transformer - Parallel
- Transformer - Server
- Unstructured Data - Parallel
- Xml Output - Parallel
- Local Container - Parallel and Server
- Shared Container - Parallel and Server
- End Loop Activity - Sequence - partial support. Parent sequence Job is not shown as independent object in output model, but target jobs are processed.
- Execute Command - Sequence - partial support. Parent sequence Job is not shown as independent object in output model, but target jobs are processed.
- Job Activity - Sequence - partial support. Parent sequence Job is not shown as independent object in output model, but target jobs are processed.
- Start Loop Activity - Sequence - partial support. Parent sequence Job is not shown as independent object in output model, but target jobs are processed.
- User Variables - Sequence - partial support. Parent sequence Job is not shown as independent object in output model, but target jobs are processed.

# Enterprise Data Catalog Writer

Imports EDC objects using native format

The Enterprise Data Catalog Writer includes the following parameters:

## Auto assign connections

Sets option "Auto Assign Connections" on EDC resource.

| Parameter name | Description |
|---|---|
| Prompt | Auto assign connections |
| Symbol | edc.autoAssignConnections |

| Parameter name | Description |
|---|---|
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Case Sensitive

Sets Case Sensitive option of the created EDC resource.Impacts endpoint (database, file) objects and linking from other resources to those objects.

| Parameter name | Description |
|---|---|
| Prompt | Case Sensitive |
| Symbol | edc.caseSensitive |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Connection naming pattern

The naming convention pattern for connections. By default you use the following pattern to name connections: "DB/SCHEMA". You can change the pattern to use a source system connection name followed by a schema: "CONNECTION_SCHEMA".

| Parameter name | Description |
|---|---|
| Prompt | Connection naming pattern |
| Symbol | edc.connectionNamingPattern |
| Type | STRING |
| Required | No |
| Default value | Database and schema |
| Example values | - Connection and schema<br>- Database and schema |

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Control conditions maximum length

Maximum length of "Control conditions" attribute. If "Control conditions" will be longer than defined value, they will be cut off and "..." will be added at the end. Value of this parameter must be >= 3 and <= 100000. If value out of the allowed scope will be provided, this parameter will be set to nearest correct value.

| Parameter name | Description |
|---|---|
| Prompt | Control conditions maximum length |
| Symbol | edc.controlConditionMaxLength |
| Type | PRIMITIVE_INTEGER |
| Required | No |
| Default value | 5000 |
| Example values | - 6000 |

## EDC address

Address of Enterprise Data Catalog.

| Parameter name | Description |
|---|---|
| Prompt | EDC address |
| Symbol | edc.address |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | ${EDC_URL} |
| Example values | - infa11:9085 |

## EDC password

EDC password.

| Parameter name | Description |
|---|---|
| Prompt | EDC password |
| Symbol | edc.password |
| Type | SENSITIVE |
| Required | No |
| Default value | ${EDC_PASSWORD} |
| Example values | - password |

## EDC resource name

Defines the name of the EDC resource. This value will be used to:

- name a zip file containing resource objects,
- refer to this resource when generating external links (if any exists).

| Parameter name | Description |
|---|---|
| Prompt | EDC resource name |
| Symbol | edc.resourceName |
| Type | STRING |
| Required | Yes |
| Default value | N/A |
| Example values | - EDC_RESOURCE |

## EDC Resource Reload Timeout

After triggering resource reload in EDC,EDC is queried periodically (every 10 seconds) for the status of the reload job . This parameter defines the maximum amount of time in seconds that Scanner will wait for job completion. If after this time job is still running, an ERROR will be logged and scanner job will be stopped.

| Parameter name | Description |
|---|---|
| Prompt | EDC Resource Reload Timeout |
| Symbol | edc.resourceReloadTimeout |
| Type | INTEGER |
| Required | No |
| Default value | 1800 |
| Example values | - 1000 |

## EDC REST API request timeout

Defines timeout (in seconds) for a single HTTP request while communicating with EDC using REST API.

| Parameter name | Description |
|---|---|
| Prompt | EDC REST API request timeout |
| Symbol | edc.restApiRequestTimeout |
| Type | INTEGER |
| Required | No |
| Default value | 180 |
| Example values | - 120 |

## EDC user

Name of the EDC user. Username should be prefixed with domain name followed by a backslash. In case of using NATIVE domain, no prefix should be defined.

| Parameter name | Description |
|---|---|
| Prompt | EDC user |
| Symbol | edc.user |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | ${EDC_USER} |
| Example values | - DOMAIN\user<br>- Admin |

## EDC writer mode

EDC writer mode.

- Load using REST API - generates output files and saves them in the output directory. Generated data is loaded to EDC via REST API,
- Generate output files - generates output files and saves them in the output directory.

| Parameter name | Description |
|---|---|
| Prompt | EDC writer mode |
| Symbol | edc.mode |
| Type | MODULE_MODE |
| Required | No |
| Default value | Generate output files |
| Example values | - Load using REST API<br>- Generate output files |

## Enable detailed lineage. It is set automatically

Lineage between tables/columns will be compacted with the possibility to extend it to show all intermediate expressions.

| Parameter name | Description |
|---|---|
| Prompt | Enable detailed lineage. It is set automatically |
| Symbol | edc.enableDetailedLineage |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - It is set automatically |

## Enable reference resources

Sets option "Enable Reference Resources" on EDC resource.

| Parameter name | Description |
|---|---|
| Prompt | Enable reference resources |
| Symbol | edc.enableReferenceResources |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Keep unresolved reference objects

Sets option "Retain Unresolved Reference Assets" on EDC resource. Has effect only if "Enable Reference Resources" is set to true.

| Parameter name | Description |
|---|---|
| Prompt | Keep unresolved reference objects |
| Symbol | edc.keepUnresolvedReferenceObjects |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Load skipping mode

Mode that determines whether loading job results to EDC with REST API should be skipped or not. Possible values are:

- Don't skip - The result always loads to EDC,
- Skip on errors
- Loading skips if any ERROR occurred earlier,
- Skip on warns or errors

- Loading skips if any ERROR or WARN occurred earlier.

| Parameter name | Description |
|---|---|
| Prompt | Load skipping mode |
| Symbol | edc.loadSkippingMode |
| Type | STRING |
| Required | No |
| Default value | Don't skip |
| Example values | - Don't skip<br>- Skip on errors<br>- Skip on warns or errors |

## Output directory

Path to the output directory. Under this location following generated files will be stored:

- xml files containing model definitions,
- a zip file containing resource objects,
- a csv file containing external link definitions (if any exists).

| Parameter name | Description |
|---|---|
| Prompt | Output directory |
| Symbol | edc.outputDir |
| Type | DIR |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\metadata\edc\loadable\<br>- /usr/local/share/metadata/edc/loadable/ |

## Override class output type

Comma separated list of classes that should be present in output. Those classes will be created as reference resources, other classes will be ignored

| Parameter name | Description |
|---|---|
| Prompt | Override class output type |
| Symbol | edc.overrideOutputTypeForClass |
| Type | STRING |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | NONE |
| Example values | - primaryKey,foreignKey |

## Process control type lineage as value lineage

Regardless of this option, control type lineage will always be processed using appropriate control type associations. If this option is set to "true", control type lineage will be additionally processed as value type lineage.

| Parameter name | Description |
|---|---|
| Prompt | Process control type lineage as value lineage |
| Symbol | edc.processControlLineageAsValueLineage |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Resource name map

Defines mapping between object context and resource name. When an external link (for example, between table and objects from generated resource) is generated, , the name of the resource in which the external object is located must be known. Object context (id) provided in "object context" section will be matched with resource name provided in "resource name" section. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Resource name map |
| Symbol | edc.resourceNameMap |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - HOST/DB/SCHEMA/TABLE/COL1=ORACLE_RESOURCE<br>- HOST/DB/*=ORACLE_RESOURCE |

## Resource type

This option is responsible for setting "Resource Type" in EDC System Attributes. It is set automatically

| Parameter name | Description |
|---|---|
| Prompt | Resource type |
| Symbol | edc.modelName |
| Type | STRING |
| Required | Yes |
| Default value | N/A |
| Example values | - It is set automatically |

## Scanner memory usage

Memory option value used to configure scanner when loading resource data

| Parameter name | Description |
|---|---|
| Prompt | Scanner memory usage |
| Symbol | edc.scannerMemoryUsage |
| Type | STRING |
| Required | No |
| Default value | Low |
| Example values | - Low |

## Track data source changes

Sets option "Track Data Source Changes" on EDC resource.

| Parameter name | Description |
|---|---|
| Prompt | Track data source changes |
| Symbol | edc.trackDataSourceChanges |
| Type | BOOLEAN |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | false |
| Example values | - true<br>- false |

### Use in-app visualizations

Use the dynamic graph visualizations and link them in EDC.

| Parameter name | Description |
|---|---|
| Prompt | Use in-app visualizations |
| Symbol | edc.useInAppVisualizations |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

# Entry Point Detector

A module that detects entry points in provided database objects

The Entry Point Detector includes the following parameters:

### Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

### Enable entry point detection

Decide whether to detect entry points in provided database objects

| Parameter name | Description |
|---|---|
| Prompt | Enable entry point detection |
| Symbol | entryPointDetector.enableDetection |
| Type | PRIMITIVE_BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

# FastLoad Reader

Reads Teradata Fast Load files.

The FastLoad Reader includes the following parameters:

### Context

Default SQL context

| Parameter name | Description |
|---|---|
| Prompt | Context |
| Symbol | fastLoad.sqlContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - host.db.db |

## Data Files Working Directory

Path for Data Files Working Directory

| Parameter name | Description |
|---|---|
| Prompt | Data Files Working Directory |
| Symbol | fastLoad.dataFilesWorkingDirectory |
| Type | PATH |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\<br>- /root/dir1/ |

## Data host name

Name of the host where FastLoad executes

| Parameter name | Description |
|---|---|
| Prompt | Data host name |
| Symbol | fastLoad.dataHostName |
| Type | STRING |
| Required | Yes |
| Default value | N/A |
| Example values | - fastload-z4032<br>- fl.company.com |

## DEPRECATED: DB host name

Database host qualification. Use sqlContext 1st segment instead

| Parameter name | Description |
|---|---|
| Prompt | DEPRECATED: DB host name |
| Symbol | fastLoad.teradataHostName |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - fastload-z4032<br>- fl.company.com |

## DEPRECATED: Default database name

Name of the default database. Use sqlContext 3rd segment instead

| Parameter name | Description |
|---|---|
| Prompt | DEPRECATED: Default database name |
| Symbol | fastLoad.teradataDbName |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - salesdb<br>- hr |

## Repository path

Path to table metadata repository

| Parameter name | Description |
|---|---|
| Prompt | Repository path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input\MDREPO\<br>- /opt/input/MDREPO/ |

## Scripts exclude filters

The paths of FastLoad scripts that the job excludes.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Scripts exclude filters |
| Symbol | fastLoad.fileExcludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\fastload-script.fl<br>- /opt/input/fastload-script.fl<br>- /opt/input/*.fl |

## Scripts Root Directory Path

Root directory of input files. Relative paths of input files resolve in the following location: [Input root directory]. Absolute paths from input files must start from the [Input root directory] to be included into processing. [Input root directory] will not be visible on the output objects structure.

| Parameter name | Description |
|---|---|
| Prompt | Scripts Root Directory Path |
| Symbol | fastLoad.inputRoot |
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\inputs\<br>- /opt/feed-etl/ |

## Scripts' paths

Paths to input scripts. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Scripts' paths |
| Symbol | fastLoad.fileIncludeFilter |
| Type | FILELIST |

| Parameter name | Description |
|---|---|
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input\fastload\224_magic.fl<br>- /opt/input/fastload/225_more_magic.fl |

# File Writer

Writes data to file.

The File Writer includes the following parameters:

## Output file

Output file to be written

| Parameter name | Description |
|---|---|
| Prompt | Output file |
| Symbol | test.outputFile |
| Type | PATH |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\etl\input\expenses.dat<br>- /etl/input/expenses.dat |

# General

Properties available on all Mf3 Graphs.

The General has no configurable parameters defined.

# Graph Lineage Hash Calculator

LineageHashCalculator generates a lineage hash of a graph and adds it as a graph property only if the job requires it. A job might require a lineage hash to deduplicate graphs with the same lineage. Lineage hash is

based on the following lineage data of all output fields of a graph: names, expressions, and their dependencies.

The Graph Lineage Hash Calculator includes the following parameters:

### Ignore output field filters

The list of fields to ignore during calculating a lineage hash. These are usually technical fields, not meaningful to business lineage. You may want to exclude them when you want to deduplicate graphs that contain the same lineage based on their lineage hash, but these technical fields constantly change their expressions. Each element of the list contains a segmented identifier separated by a period. Each segment can contain the following wildcard characters: * Question mark. Represents a single character. * Asterisk. Represents multiple characters or empty text. Each segment represents one part of a fully qualified identifier. For example, databases can have a fully qualified identifier in the host.database.schema.table.column format but you don't have to use all elements. Since elements match by prefix, from the most specific to the least specific, if you want to exclude column1 from table1, use: "table1.column1"

| Parameter name | Description |
|---|---|
| Prompt | Ignore output field filters |
| Symbol | lineageHash.ignoreOutputFieldFilters |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - host.db.schema.tab.col1<br>- tab2.col*,table1.col* |

# Graph Property Adder

GraphPropertyAdderModule is used for adding new Graph properties conditionally. Conditions are based on existing Graph properties. It is recommended to not override following graph properties: SqlHost, SqlDatabase, SqlSchema. These properties may be mapped during context mapping process.

The Graph Property Adder includes the following parameters:

### Configuration file

When a property adder rule condition for an entry is fulfilled, the rule adds the configured graph property. The property has the following functions:

- Overrides the existing property of the same name if the property type is single value.
- Appends to the existing property of the same name if the property type multi-value.
- Overrides the existing property of the same name if the property type is multi-value and additional flag `overrideExistingValues` is set to `true` at the end of the configuration for the given condition.

Supported expressions (case insensitive) are:

- literals
  - integer: 42
  - string: 'str'
  - string with quoted single quote: 'str\''
  - boolean: true, false
- equals: 42 == 7 -> FALSE
- not equals: 42 != 7 -> TRUE
- concatenations: 'str1' + 'str2' -> 'str1str2'
- function calls: Replace('abc', 'b', 'd') -> 'adc'
- tertiary operator:
  - 42 == 7 ? 'a' : 'b' -> 'b'
  - 42 != 7 ? 'a' : 'b' -> 'a'

Supported functions:

- Hash(str, maxOutputLen) - MD5, maxOutputLen between 1 and 32 inclusive
  - Hash('abcdefgh', 4) -> 'E8DC'
- Replace(str, from, to)
  - Replace('abc', 'b', 'D') -> 'aDc'
- ReplaceRegexp(str, regex, replacement) - replaces with Java regular expressions
  - ReplaceRegexp('abcde', 'b.*', 'f') -> 'af'
  - ReplaceRegexp('graph_id20', 'id(\d+)', '$1') -> 'graph_20'
- StringLengthLimit(str, limit, hashSize) - returns str[0..limit-hashSize] + Hash( str, hashSize )
  - StringLengthLimit('abcc', 3, 2) -> 'a26'
  - StringLengthLimit('abcc', 4, 2) -> 'abcc'
- StringLengthLimit(str, limit) - hashSize has default size 8
  - StringLengthLimit('abcdabcdabcd', 10) -> 'abE340600C'
- RegexpMatch(pattern, str) - tests if str matches pattern
  - pattern uses standard Java 8 regular expressions: https://docs.oracle.com/javase/8/docs/api/java/util/regex/Pattern.html
  - RegexpMatch('[A-Za-z]+','Abcd') -> TRUE
- Upper(str) - returns uppercased str
  - Upper('Abc') -> 'ABC'
- Lower(str) - return lowercased str
  - Lower('aBC') -> 'abc'
- Date(text,format) - returns object representing date with provided format.
  - Formats uses standard Java 8 SimpleDateFormat: https://docs.oracle.com/javase/8/docs/api/java/text/SimpleDateFormat.html
  - It is designed for default values usage
  - ex. Date('2017-10-31', 'yyyy-MM-dd')

- Contains/ContainsIgnoreCase(stack, needle) - tests if stack string contains needle string
  - Contains('abc', 'ab') -> TRUE
  - Contains('abc', 'AB') -> FALSE
  - ContainsIgnoreCase('abc', 'AB') -> TRUE
  - Contains('stack', 'needle') -> FALSE
- RtfToText(str) - return clean text from rtf string
  - RtfToText('{\rtf1\ansi{\fonttbl\f0\fswiss Helvetica;}\f0\pard text\par }') -> text

Supported comments:

- Single line: \ comment
- Multi line: /* comment */

You may use property names inside String literals. Condition examples:

- 'abc' == 'a%SourceFile%'
- true
- 'abc' == 'a' + '%SourceFile%'
- RegexpMatch('[A-Za-z]', '%Property%')
- '%?potentiallyExistingProperty%' == ''

To set property or override existing consider following set property examples:

- AbcProperty := 'a%SourceFile%'
- x := 'a' + 'b'
- y := 'HASH-' + Hash('%SourceFile%', 3)

Rules file must be consistent with following schema:

```
{
 "propertyAdderRulesList":     [
  {
   "condition": "CONDITION",
   "newPropertyName": "PROPERTY_NAME",
   "newPropertyValue": "PROPERTY_VALUE"
  }
 ]
}
```

Example:

```
{
 "propertyAdderRulesList":
 [
  {
   "condition": "true",
   "newPropertyName": "ViewSubFolder",
   "newPropertyValue": "'HASH-' + Hash('%SqlView%',2)"
  }
 ]
}
```

| Parameter name | Description |
|---|---|
| Prompt | Configuration file |
| Symbol | propertyAdder.conditionsToPropertiesFile |

| Parameter name | Description |
|---|---|
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\conditionsToProperties.json<br>- /opt/input/conditionsToProperties.json |

### Update graph components

- False. Updates graph properties.
- True. Updates graph properties and graph component properties.

| Parameter name | Description |
|---|---|
| Prompt | Update graph components |
| Symbol | propertyAdder.updateGraphComponents |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

# Hive Script Reader

Reads Hive scripts.

The Hive Script Reader includes the following parameters:

### Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Context

Default SQL context

| Parameter name | Description |
|---|---|
| Prompt | Context |
| Symbol | hive.sqlContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - host.db.db |

## Default variables values file

| Parameter name | Description |
|---|---|
| Prompt | Default variables values file |
| Symbol | hive.defaultValuesFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\defaultVariableValues.txt<br>- /opt/etc/defaultVariableValues.txt |

## Scripts paths

Hive scripts paths

| Parameter name | Description |
|---|---|
| Prompt | Scripts paths |
| Symbol | hive.fileIncludeFilter |

| Parameter name | Description |
|---|---|
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\hive-script.txt<br>- /opt/input/hive-script.txt<br>- /opt/input/*.txt |

## Scripts Root Directory Path

Root directory of input files. Relative paths of input files resolve in the following location: [Input root directory]. Absolute paths from input files must start from the [Input root directory] to be included into processing. [Input root directory] will not be visible on the output objects structure.

| Parameter name | Description |
|---|---|
| Prompt | Scripts Root Directory Path |
| Symbol | hive.inputRoot |
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\inputs\<br>- /opt/feed-etl/ |

## Tool type

Hive tool type

| Parameter name | Description |
|---|---|
| Prompt | Tool type |
| Symbol | hive.toolType |
| Type | STRING |
| Required | No |
| Default value | Hive |
| Example values | - Hive<br>- Impala |

# HTML Visualizer

Writes visualisation in html format.

The HTML Visualizer includes the following parameters:

## Filename pattern

Filename pattern for the generated extension mapping documents. You may use Graph Properties here, check documentation for details.

| Parameter name | Description |
|---|---|
| Prompt | Filename pattern |
| Symbol | visualizer.filenamePattern |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - SQL.%SourceProject%.%SourceFile%:%SourceFileLineNumber%<br>- %SourceFileLocation%-%SourceFile% |

## Generate control components

If it's set to true, control components will be generated.

| Parameter name | Description |
|---|---|
| Prompt | Generate control components |
| Symbol | visualizer.withControlComponents |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - True<br>- False |

## Graph property names filter

Identifies properties to keep in the document (especially for Metadata Manager and html visualisation). You can use regular wildcards: * (0 or more characters) and ? (0 or 1 character). Default: * .

| Parameter name | Description |
|---|---|
| Prompt | Graph property names filter |
| Symbol | visualizer.graphPropertyFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - Informatica*<br>- SourceType<br>- Source?ption |

## HTML Output Directory

If specified, this is the directory for HTML visualization. The value must be an absolute path.full html visualization path is generated by extending "HTML Output Directory" by system name and application name.

| Parameter name | Description |
|---|---|
| Prompt | HTML Output Directory |
| Symbol | visualizer.htmlOutputDir |
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\processing\output\html\<br>- /usr/local/share/processing/output/html/ |

## HTML Output Http address

If specified, new property will be added to graph: GraphVisualizationUrl, which will contain address of HTML output with visualization generated by MetaDex, which will be accessible from Metadata Workbench. Address has to begin with http:// and end with /.

| Parameter name | Description |
|---|---|
| Prompt | HTML Output Http address |
| Symbol | visualizer.httpVisualizationLocation |
| Type | STRING |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | N/A |
| Example values | - http://example.com/edcAdvancedScannersReports/ |

## Max expression length

Max length of expression which will be imported to Metadata Workbench. Expression will be cut off and "..." will be added. Max length must be >= 3.

| Parameter name | Description |
|---|---|
| Prompt | Max expression length |
| Symbol | visualizer.maxExpressionLength |
| Type | STRING |
| Required | No |
| Default value | 5000 |
| Example values | - 5<br>- 79<br>- 101 |

## Warn if max expression length reached

If max expression is reached message is logged. If it's set to true, it's logged as WARN. If it's false, it's logged as TRACE.

| Parameter name | Description |
|---|---|
| Prompt | Warn if max expression length reached |
| Symbol | visualizer.warnExpressionsExceedingMaxLength |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - True<br>- False |

# IICS Reader

A module that extracts XML files from Informatica Intelligent Cloud Services (IICS), parses Informatica ETL jobs, and generates data lineage with other types of processes.

The IICS Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema

- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | iics.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL<br>- MQ_CON=QUEUE:host.queue-manager.queue-name - Similar to above but define connection to message queue |

## Enable indirect source files support

Whether indirect source file should be supported. If it is set to false source files from indirect file will be not included in lineage

| Parameter name | Description |
|---|---|
| Prompt | Enable indirect source files support |
| Symbol | iics.enableIndirectSourceFileSupport |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | No example is available for this parameter. |

## Explicit parameters file

Additional Informatica parameter file with overridden or missing parameter definitions.

| Parameter name | Description |
|---|---|
| Prompt | Explicit parameters file |
| Symbol | iics.overrideParamsFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /export/home/RootDir/infa/param-files/bikeshop_overrides.prm<br>- Y:\RootDir\infa\param-files\bikeshop_overrides.prm |

## IICS Cloud Password

The password that you use to log in to Informatica Intelligent Cloud Service.

| Parameter name | Description |
|---|---|
| Prompt | IICS Cloud Password |
| Symbol | iics.password |
| Type | SENSITIVE |
| Required | Yes |
| Default value | N/A |
| Example values | - mySecretPassword123 |

## IICS Cloud Url

The URL to Informatica Intelligent Cloud Service.

| Parameter name | Description |
|---|---|
| Prompt | IICS Cloud Url |
| Symbol | iics.cloudUrl |
| Type | STRING |
| Required | Yes |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - https://infa.com/identity-service/home |

## IICS Cloud User

The username that you use to log in to Informatica Intelligent Cloud Service.

| Parameter name | Description |
|---|---|
| Prompt | IICS Cloud User |
| Symbol | iics.user |
| Type | STRING |
| Required | Yes |
| Default value | N/A |
| Example values | - user@informatica.com |

## IICS Filters List

Case insensitive include/exclude filters that allow to filter tasks The filters use the following syntax: <FilterType> : <Pattern> Where: FilterType: Include - If nonempty, processes only the assets prefixed with at least one of the given qualifications. Exclude - Omits items that match at least one of the given patterns. Pattern: PROJECT*NAME[:DIRECTORY*PATH[:TASK_NAME]] You can use a question mark and an asterisk inside each pattern. Question mark matches any single character, asterisk matches any number of any characters. If a folder contains spaces or other non-standard characters, quote it with double-quotes. For details, see the following examples:

| Parameter name | Description |
|---|---|
| Prompt | IICS Filters List |
| Symbol | iics.inputFilters |
| Type | STRINGLIST |
| Required | Yes |

| Parameter name | Description |
|---|---|
| Default value | Include : * |
| Example values | - Include : projectN?me:/Tasks/directory:*<br>- Exclude : 'projectName:"/directory":taskNa*'<br>- Include:project:/Tasks/dir:task<br>- Include : project : /Tasks/dir : task<br>- Include : project : "/Tasks/d ir" : task<br>- Include : * : * : *<br>- Exclude : Default<br>- Include : *MyProject : * : * |

## Informatica file variables

List of Informatica files mapped to local files that allow MetaDex to use local copies of files instead of connecting to an Informatica Intelligent Cloud Services installation.This mapping is used with Informatica parameter files and with indirect source files. Values in the "From" column are substituted by values in the "To" column, in order to resolve workflow parameter file references.

| Parameter name | Description |
|---|---|
| Prompt | Informatica file variables |
| Symbol | iics.paramFilesMap |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - /export/home/RootDir/infa/param-files=C:\PrmFiles<br>- Y:\RootDir\infa\param-files=C:\PrmFiles |

## Informatica MDREPO database (HOST.DATABASE.SCHEMA)

Host, db name and schema in MDREPO under which Informatica internal db connection details are being held

| Parameter name | Description |
|---|---|
| Prompt | Informatica MDREPO database (HOST.DATABASE.SCHEMA) |
| Symbol | iics.database |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - host01.db.schema |

## Informatica Web Services Hubs

List of Web Services Hubs mapped to corresponding URLs. These values will be used when processing Web Service Providers.

| Parameter name | Description |
|---|---|
| Prompt | Informatica Web Services Hubs |
| Symbol | iics.webServicesHubMap |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - hub1=http://infa06:7333/wsh/<br>- hub2=http://localhost:3333/wsh/ |

## Integration service name

Name of the integration service on which the workflows run. The name is used when searching parameter file sections for parameter value

| Parameter name | Description |
|---|---|
| Prompt | Integration service name |
| Symbol | iics.integrationServiceName |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | No example is available for this parameter. |

## Log warnings on empty variable or parameter

Whether warning message should be logged on empty variable or parameter. If false the message will be logged on info level.

| Parameter name | Description |
|---|---|
| Prompt | Log warnings on empty variable or parameter |
| Symbol | iics.warnOnEmptyVariable |
| Type | BOOLEAN |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | true |
| Example values | No example is available for this parameter. |

## Override output model repository

The name for repository appears in output model. It's picked from an input Informatica Xml unless it's overridden here.

| Parameter name | Description |
|---|---|
| Prompt | Override output model repository |
| Symbol | iics.overrideRepository |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - Test_Repository |

## Process workflow link conditions

Whether link conditions should affect tasks processing, when true link conditions will be evaluated.

| Parameter name | Description |
|---|---|
| Prompt | Process workflow link conditions |
| Symbol | iics.processTasksConditionally |
| Type | PRIMITIVE_BOOLEAN |
| Required | No |
| Default value | false |
| Example values | No example is available for this parameter. |

## Proxy Host

A hostname or an IP address of the HTTP(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy Host |
| Symbol | iics.proxyHost |

| Parameter name | Description |
|---|---|
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - 88.99.102.102 or https://myproxy |

## Proxy Password

A password for the HTTP(s) proxy server authentication

| Parameter name | Description |
|---|---|
| Prompt | Proxy Password |
| Symbol | iics.proxyPassword |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - abc123 |

## Proxy Port

A port number of the HTTP(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy Port |
| Symbol | iics.proxyPort |
| Type | INTEGER |
| Required | No |
| Default value | 0 |
| Example values | - 30408 |

## Proxy Username

A username for the HTTP(s) proxy server authentication

| Parameter name | Description |
|---|---|
| Prompt | Proxy Username |
| Symbol | iics.proxyUserName |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - abc |

## Server in output model and files host name

Host name used for the data files read or written by the mappings and Server name for output model

| Parameter name | Description |
|---|---|
| Prompt | Server in output model and files host name |
| Symbol | iics.filesHost |
| Type | STRING |
| Required | Yes |
| Default value | IICS_HOST |
| Example values | - DATAF<br>- File Storage |

## Session instances to be excluded during processing

Expression which matches full path to session instance to be excluded during processing. Full path is (without worklets): FL:[InformaticaFolderName].WF:[WorkflowName].SI:[SessionInstanceName].MP:[MappingName] or (with worklet): FL:[InformaticaFolderName].WF:[WorkflowName].WL:[Worklet1InstanceName].WL:[Worklet2InstanceName].SI:[SessionInstanceName].MP:[MappingName]. You can add zero, one or more worklets definitions. Question mark and asterisk mark can be used inside folder, workflow, worklet instance, session instance or mapping name. Question mark matches any single character, asterisk matches any number of any characters. See examples for details.

| Parameter name | Description |
|---|---|
| Prompt | Session instances to be excluded during processing |
| Symbol | iics.excludeSessionInstances |
| Type | STRINGLIST |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | N/A |
| Example values | - FL:temp*.WF:*.SI:*.MP:*- exclude folders starting with 'temp'<br>- FL:*.WF:*.SI:*.MP:mapping1? - exclude mappings starting with 'mapping1' and having one additional characters at the end of the name |

## Session instances to be included during processing

Expression which matches full path to session instance to be included during processing. Full path is (without worklets): FL:[InformaticaFolderName].WF:[WorkflowName].SI:[SessionInstanceName].MP:[MappingName] or (with worklet): FL:[InformaticaFolderName].WF:[WorkflowName].WL:[Worklet1InstanceName].WL:[Worklet2InstanceName].SI:[SessionInstanceName].MP:[MappingName]. You can add zero, one or more worklets definitions. Question mark and asterisk mark can be used inside folder, workflow, worklet instance, session instance or mapping name. Question mark matches any single character, asterisk matches any number of any characters. See examples for details.

Include patterns will also define the order in which workflows (and only workflows) will be processed. For example the following patterns:

- FL:.*WF:Workflow1*.*SI:*.MP:*
- FL:.*WF:Workflow2*.*SI:*.MP:*
- FL:.*WF:*.SI:.*MP:*

will result in the following workflow processing order: Workflow1, Workflow2, any other workflow.

| Parameter name | Description |
|---|---|
| Prompt | Session instances to be included during processing |
| Symbol | iics.includeSessionInstances |
| Type | STRINGLIST |
| Required | No |
| Default value | FL:*.WF:*.SI:*.MP:* |
| Example values | - FL:*.WF:*.SI:*.MP:*- process everything<br>- FL:*.WF:*.SI:*.MP:mapping1? - process only mappings starting with 'mapping1' and having one additional characters at the end of the name |

## Skip disabled tasks

Do not process any workflows, worklets, sessions, or other tasks marked as "Disabled" by Informatica Intelligent Cloud Services.

| Parameter name | Description |
|---|---|
| Prompt | Skip disabled tasks |
| Symbol | iics.skipDisabled |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Source Qualifier command file list

List of source files for the Source Definition of the component of Informatica Intelligent Cloud Services. In this component list of files to process can be defined as a result of command line command. For lineage purposes, the user can define list of input files for this particular component. The field "Qualified file input component name" should contain a qualified name to user's choice Source Definition Informatica component. A Qualified name means the "Informatica component path" and the string "FileList". An Informatica component path means the Informatica folder, workflow, worklets (if any), session instance name and component name. Each part of the qualification is prefixed by two letter acronym FL- folder, WF- workflow, WL- worklet, SI- session instance, CP- component. A colon separates each prefix from its value, and a dot separates each segment. The field "List of file paths" can contain comma separated list of file paths that will be used as input files processed by this component.

| Parameter name | Description |
|---|---|
| Prompt | Source Qualifier command file list |
| Symbol | iics.sourceQualifierCommandFileList |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - FL:INFA_01.WF:wf_DAILY_TRANS.WL:wl_REGISTER_INVOICE.SI:sm_so_st_REGISTER_INVOICE.CP:inFiles.FileList=path/to/file1.dat;path/to/file2.dat --- Here, "INFA_01" is the name of the folder, "wf_DAILY_TRNS" is the name of the workflow, "wl_REGISTER_INVOICE" is the worklet, "sm_so_st_REGISTER_INVOICE" is the session instance, "inFiles" is the component and "FileList" is the required constant string. Next is the comma separated list of file paths. |

## Sql Transform components excluded from logging warns

Expression which matches full path to Sql Transform component in script mode for which WARN messages will NOT be logged. Full path is (without worklets): FL:[InformaticaFolderName].WF:[WorkflowName].SI:[SessionInstanceName].MP:[MappingName].CP:[ComponentName] or (with worklet): FL:[InformaticaFolderName].WF:[WorkflowName].WL:[Worklet1InstanceName].WL:[Worklet2InstanceName].SI:[SessionInstanceName].MP:[MappingName].CP:[ComponentName]. You can add zero, one or more worklets definitions. Question mark and asterisk mark can be used inside folder, workflow, worklet instance, session instance, mapping or component name. Question mark matches any single character, asterisk matches any number of any characters. See examples for details. Example patterns:

- FL:.*WF:*.SI:.*MP:*.CP. - Do not log WARN for all sql transform component in script mode
- FL:.*WF:*.SI:.*MP:*.CP:component - Do not log WARN for sql transform component in script mode that name is "component"

| Parameter name | Description |
|---|---|
| Prompt | Sql Transform components excluded from logging warns |
| Symbol | iics.sqlTransformComponentExcludedInWarnMessageFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - FL:*.WF:*.SI:*.MP:*.CP:* - Log info instead of warn for all sql transform component in script mode<br>- FL:*.WF:*.SI:*.MP:*.CP:filterSql? - Log info instead of warn for transform component in script mode that name starts with filterSql and log warn for others |

## SQLTransformer SQL resolution

Map for replacing the entire SQL query or parts of it in the SQL Transform component of Informatica Intelligent Cloud Services. This component builds a dynamic SQL query for every record using values from input ports as part of the query. For lineage purposes, the user can define a single value for each input port or override original component's SQL query or do not provide any additional information. The "Qualified SQL part to replace" field should contain a qualified name to user's choice input field in particular SQL Transform component or qualified name of that component in case of sql query override. A Qualified field name means the "Informatica component path" and the string "SQLReplace", and the input port name or "Informatica component path" and the string "SQLReplaceAll". An Informatica component path means the Informatica folder, workflow, worklets (if any), session instance name and component name. Each part of the qualification is prefixed by two letter acronym FL- folder, WF-workflow, WL- worklet, SI- session instance, CP-component. A colon separates each prefix from its value, and a dot separates each segment. The field

"Replacement" can contain parameters (for example: $Param1, $$Param2) that will be resolved according to their current value.

| Parameter name | Description |
|---|---|
| Prompt | SQLTransformer SQL resolution |
| Symbol | iics.sqlStringReplacers |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - FL:INFA_01.WF:wf_DAILY_TRANS.WL:wl_REGISTER_INVOICE.SI:sm_so_st_REGISTER_INVOICE.CP:SQL.SQLReplace.~o_SID~ --- Here, "INFA_01" is the name of the folder, "wf_DAILY_TRNS" is the name of the workflow, "wl_REGISTER_INVOICE" is the worklet, "sm_so_st_REGISTER_INVOICE" is the session instance, "SQL" is the component, "SQLReplace" is the required constant string, and "~o_SID~" is the port.<br>- FL:INFA_01.WF:wf_DAILY_TRANS.WL:wl_REGISTER_INVOICE.SI:sm_so_st_REGISTER_INVOICE.CP:SQL.SQLReplaceAll --- Here, "INFA_01" is the name of the folder, "wf_DAILY_TRNS" is the name of the workflow, "wl_REGISTER_INVOICE" is the worklet, "sm_so_st_REGISTER_INVOICE" is the session instance, "SQL" is the component and "SQLReplaceAll" is the required constant string. |

## Taskflow workaround cache fetch file age in days

The age of files in the workaround cache folder for Informatica Intelligent Cloud Services taskflows to filter.Default is -1. Ignores the value and includes all files in the scan. If you enter an invalid value or a negative value, the property uses the default value and includes all files in the scan. If you enter 0, the job fails.Ensure that you set the Taskflow workaround cache fetch folder path parameter to use the Taskflow workaround cache fetch file age in days parameter.

| Parameter name | Description |
|---|---|
| Prompt | Taskflow workaround cache fetch file age in days |
| Symbol | iics.cdiTasksCacheFetchFileAgeInDays |
| Type | STRING |
| Required | No |
| Default value | -1 |
| Example values | - 1<br>- 7<br>- 10 |

## Taskflow workaround cache fetch folder path

Complete path to the IICS Taskflow workaround cache folder. Determines whether to read the Taskflow workaround cache fetch mode and Taskflow workaround cache fetch file age in days parameters. Ensure that you set this before you use these parameters.

| Parameter name | Description |
|---|---|
| Prompt | Taskflow workaround cache fetch folder path |
| Symbol | iics.cdiTasksCacheFetchFolder |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - /root/taskflowFiles/ |

## Taskflow workaround cache fetch mode

One of the three modes for fetching IICS mapping task XMLs. Ensure that you have configured the Taskflow workaround cache fetch folder path parameter before you enable cache fetch mode.

| Parameter name | Description |
|---|---|
| Prompt | Taskflow workaround cache fetch mode |
| Symbol | iics.cdiTasksCacheFetchMode |
| Type | STRING |
| Required | No |
| Default value | files |
| Example values | - Files<br>- API<br>- Both |

## workflow with assigned parameter file

When workflow is run with different parameter file than defined inside workflow add this workflow and parameter file as MetaDex parameter. Use folder qualification for workflow and full parameter file path

| Parameter name | Description |
|---|---|
| Prompt | workflow with assigned parameter file |
| Symbol | iics.workflowAndParameter |
| Type | MULTIMAP |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | N/A |
| Example values | - FL:PRD_CREDIT.WF.wf_LoadCredit=$PMParamFileDir/loadCredit.prm |

## Supported Components

- AddressValidation Transformation - Custom
- Aggregator - Transformation
- Amazon Athena (AmazonAthena_amazonathena Reader) - Reader
- Amazon Redshift (AmazonRSCloudAdapter Reader) - Reader
- Amazon Redshift (AmazonRSCloudAdapter Writer) - Writer
- Amazon Redshift (AmazonRedshiftV2_amazonredshift Reader) - Reader
- Amazon Redshift (AmazonRedshiftV2_amazonredshift Writer) - Writer
- Amazon S3 (AmazonS3 Reader) - Reader
- Amazon S3 (AmazonS3 Writer) - Writer
- Amazon S3 (AmazonS3V2_amazons3 Reader) - Reader
- Amazon S3 (AmazonS3V2_amazons3 Writer) - Writer
- App Multi-Group Source Qualifier - Transformation
- Application Source Qualifier - Transformation
- Azure Blob (AzureBlobV3_blob Reader) - Reader
- Azure Blob (AzureBlobV3_blob Writer) - Writer
- Azure Data Lake Storage Gen2 (adlsgen2_adlsg2 Reader) - Reader
- Azure Data Lake Storage Gen2 (adlsgen2_adlsg2 Writer) - Writer
- Azure Data Warehouse (AzureDWV3_table Reader) - Reader
- Azure Data Warehouse (AzureDWV3_table Writer) - Writer
- Business 360 Flat End Point (mdmflat_flatmdmentity Writer) - Writer
- Business Entity (mdmV1_businessentity Reader) - Reader
- Business Entity (mdmV1_businessentity Writer) - Writer
- Cluster Transformation - Custom
- Comparison Transformation - Custom
- Consolidation Transformation - Custom
- Custom Transformation - Transformation
- DB2 Warehouse (Db2Warehouse_table Reader) - Reader
- Databricks Delta (databricksdelta_databricksdelta Reader) - Reader
- Databricks Delta (databricksdelta_databricksdelta Writer) - Writer
- Db2 Warehouse (Db2Warehouse_table Writer) - Writer
- Decision Transformation - Custom
- Expression - Transformation

- File (File Reader) - Reader
- File (File Writer) - Writer
- Filter - Transformation
- Generic JDBC (GenericJDBCAdapter Reader) - Reader
- Generic JDBC (GenericJDBCAdapter Writer) - Writer
- Google Big Query (GoogleBigQueryV2_table Reader) - Reader
- Google Big Query (GoogleBigQueryV2_table Writer) - Writer
- Google Cloud Storage (GoogleCloudStorageV2_storage Reader) - Reader
- Google Cloud Storage (GoogleCloudStorageV2_storage Writer) - Writer
- Google Storage (googlestoragev2_storage Reader) - Reader
- Google Storage (googlestoragev2_storage Writer) - Writer
- HDFS Flat File (HDFS Flat File Reader) - Reader
- HDFS Flat File (HDFS Flat File Writer) - Writer
- Hive (hiveadapter_table Reader) - Reader
- Hive (hiveadapter_table Writer) - Writer
- IDQParserZTransform - IDQ Custom Transformation
- Input Transformation - Transformation
- JDBC (JDBC_V2_table Reader) - Reader
- JDBC Writer (JDBC_V2_table Writer) - Writer
- Java Transformation - Custom
- Joiner - Transformation
- KeyGenerator Transformation - Custom
- Labeler Transformation - Custom
- Lookup Procedure - Transformation
- MQ (MQ Reader) - Reader
- MQ (MQ Writer) - Writer
- MQ Source Qualifier - Transformation
- Mapplet - Transformation
- Microsoft Fabric One Lake (onelake_onelake Reader) - Reader
- Microsoft Fabric One Lake (onelake_onelake Writer) - Writer
- MicrosoftExcel (MicrosoftExcel Reader) - Reader
- MongoDB (mongodbadapter_table Reader) - Reader
- MongoDB (mongodbadapter_table Writer) - Writer
- Netezza Bulk (Netezza Bulk Reader) - Reader
- Netezza Bulk (Netezza Bulk Writer) - Writer
- Normalizer - Transformation
- Operational Data Provisioning Browser (ODP_browser Reader) - Reader
- Output Transformation - Transformation
- PairGenerator Transformation - Custom

- Parser Transformation - Custom
- Postgres (PostgreSQLadapter_table Reader) - Reader
- Postgres (PostgreSQLadapter_table Writer) - Writer
- PowerExchange Batch for VSAM files (PowerExchange Batch Reader for VSAM files) - Reader
- PowerExchange CDC for DB2zOS (PowerExchange CDC Reader for DB2zOS) - Reader
- PowerExchange CDC for ORACLE (PowerExchange CDC Reader for ORACLE) - Reader
- PowerExchange for Sequential files (PowerExchange Reader for Sequential files) - Reader
- PowerExchange for Sequential files (PowerExchange Writer for Sequential files) - Writer
- RESTv2 (RESTv2 Reader) - Reader
- RESTv2 (RESTv2 Writer) - Writer
- Rank - Transformation
- Relational Table (Relational Reader) - Reader
- Relational Table (Relational Writer) - Writer
- Router - Transformation
- SAP Business Explorer (SAPBExQuery Reader) - Reader
- SAP Business Warehouse (SAPBWREADER Reader) - Reader
- SAP Tables (SAPTables Reader) - Reader
- SAP Tables (SAP_Table_Writer) - Writer
- SPI (SPI_READER) - Reader
- SQL Transform - Custom
- Salesforce (SalesForce Reader_New) - Reader
- Salesforce (SalesForce Writer_New) - Writer
- Salesforce (Salesforce Reader) - Reader
- Salesforce (Salesforce Writer) - Writer
- Sequence - Transformation
- SharePoint Online (SharepointOnline Reader) - Reader
- Snowflake Data Warehouse (SnowflakeCloudDataWarehouse Reader) - Reader
- Snowflake Data Warehouse (SnowflakeCloudDataWarehouse Writer) - Writer
- Snowflake Data Warehouse (SnowflakeCloudDataWarehouseV2_table Reader) - Reader
- Snowflake Data Warehouse (SnowflakeCloudDataWarehouseV2_table Writer) - Writer
- Sorter - Transformation
- Source Definition - Transformation
- Source Qualifier - Transformation
- Stored Procedure - Transformation
- Target Definition - Transformation
- Teradata FastExport (Teradata FastExport Reader) - Reader
- Teradata Parallel Transporter (Teradata Parallel Transporter Reader) - Reader
- Teradata Parallel Transporter (Teradata Parallel Transporter Writer) - Writer
- TokenStandardiser Transformation - Custom

- Transaction Control - Transformation
- Union Transformation - Custom
- Unstructured Data Transformation - Custom
- Update Strategy - Transformation
- WebServices Consumer - Custom
- WebServices Provider (WebServices Provider Reader) - Reader
- WebServices Provider (WebServices Provider Writer) - Writer
- WeightBasedAnalyser Transformation - Custom
- WorkdayH (WorkdayH Reader) - Reader
- WorkdayH (WorkdayH Writer) - Writer
- XML (XML Reader) - Reader
- XML (XML Writer) - Writer
- XML GENERATOR - Custom
- XML PARSER - Custom
- XML Source Qualifier - Transformation

# Image Dumper

A module for generating visualizations of data lineage generated by MetaDex.

The Image Dumper includes the following parameters:

## Generate component-level visualization

Enable visual generations of data lineage at the component level.Only PNG files will be linked to Workbench.

| Parameter name | Description |
|---|---|
| Prompt | Generate component-level visualization |
| Symbol | imageDumper.componentsImageType |
| Type | STRING |
| Required | No |
| Default value | NONE |
| Example values | - NONE<br>- PNG |

## Generate field-level visualization

Enable visual generations of data lineage at the field level.

| Parameter name | Description |
|---|---|
| Prompt | Generate field-level visualization |
| Symbol | imageDumper.fieldLevelImageType |
| Type | STRING |
| Required | No |
| Default value | NONE |
| Example values | - NONE<br>- SVG |

## Max components count

Specifies max components count per image. Graphs with more components will not be converted to image

| Parameter name | Description |
|---|---|
| Prompt | Max components count |
| Symbol | imageDumper.maxComponentsCount |
| Type | STRING |
| Required | No |
| Default value | 600 |
| Example values | - 100<br>- 500 |

## Visualizations output directory

Specifies location where images will be stored

| Parameter name | Description |
|---|---|
| Prompt | Visualizations output directory |
| Symbol | imageDumper.outputDir |
| Type | DIR_PATH |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - /home/processing/output/image/<br>- Y:\processing\output\image\ |

# Informatica Post SQL Fixer

A module for fix Informatica specific issues with SQL queries on MF3 graphs.

The Informatica Post SQL Fixer has no configurable parameters defined.

# Informatica Reader

A module for parsing Informatica ETL jobs and generating data lineage with other types of processes. To export Informatica folder please use following commands (example for version 9.6.1- should work for latest version as well): cd C:\informatica\9.6.1\clients\PowerCenterClient\client\bin pmrep.exe connect -r REPOSITORY_NAME -d INFORMATICA_REPO_HOST -n USERNAME -s DOMAIN_NAME -x PASSWD pmrep.exe ObjectExport -f FOLDER_TO_BE_EXPORTED -u C:\OUTPUT_FOLDER\OUTPUT_FILE.xml Exported OUTPUT_FILE will include all needed information.

The Informatica Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field

- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | powerCenter.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL<br>- MQ_CON=QUEUE:host.queue-manager.queue-name - Similar to above but define connection to message queue |

## Enable indirect source files support

Whether indirect source file should be supported. If it is set to false source files from indirect file will be not included in lineage

| Parameter name | Description |
|---|---|
| Prompt | Enable indirect source files support |
| Symbol | powerCenter.enableIndirectSourceFileSupport |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | No example is available for this parameter. |

## Explicit parameters file

Additional Informatica parameter file with overridden or missing parameter definitions.

| Parameter name | Description |
|---|---|
| Prompt | Explicit parameters file |
| Symbol | powerCenter.overrideParamsFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /export/home/RootDir/infa/param-files/bikeshop_overrides.prm<br>- Y:\RootDir\infa\param-files\bikeshop_overrides.prm |

## Informatica file variables

List of Informatica files mapped to local files that allow MetaDex to use local copies of files instead of connecting to an Informatica PowerCenter installation.This mapping is used with Informatica parameter files and with indirect source files. Values in the "From" column are substituted by values in the "To" column, in order to resolve workflow parameter file references.

| Parameter name | Description |
|---|---|
| Prompt | Informatica file variables |
| Symbol | powerCenter.paramFilesMap |
| Type | PROPERTIES |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - /export/home/RootDir/infa/param-files=C:\PrmFiles<br>- Y:\RootDir\infa\param-files=C:\PrmFiles |

## Informatica MDREPO database (HOST.DATABASE.SCHEMA)

Host, db name and schema in MDREPO under which Informatica internal db connection details are being held

| Parameter name | Description |
|---|---|
| Prompt | Informatica MDREPO database (HOST.DATABASE.SCHEMA) |
| Symbol | powerCenter.database |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - host01.db.schema |

## Informatica Web Services Hubs

List of Web Services Hubs mapped to corresponding URLs. These values will be used when processing Web Service Providers.

| Parameter name | Description |
|---|---|
| Prompt | Informatica Web Services Hubs |
| Symbol | powerCenter.webServicesHubMap |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - hub1=http://infa06:7333/wsh/<br>- hub2=http://localhost:3333/wsh/ |

## Input files

List of XML files exported from Informatica PowerCenter. Each file should contains at least one workflow. Job starts from workflows and processes only sessions and mappings inside those workflows. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | powerCenter.sourceFile |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - /export/home/RootDir/etl-exports/20130524-extract.XML<br>- Y:\RootDir\etl-exports\20130524-extract.XML |

## Integration service name

Name of the integration service on which the workflows run. The name is used when searching parameter file sections for parameter value

| Parameter name | Description |
|---|---|
| Prompt | Integration service name |
| Symbol | powerCenter.integrationServiceName |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | No example is available for this parameter. |

## Log warnings on empty variable or parameter

Whether warning message should be logged on empty variable or parameter. If false the message will be logged on info level.

| Parameter name | Description |
|---|---|
| Prompt | Log warnings on empty variable or parameter |
| Symbol | powerCenter.warnOnEmptyVariable |
| Type | BOOLEAN |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | true |
| Example values | No example is available for this parameter. |

## Object filter

This parameter allows to select appropriate objects from the repository to be processed. By default every object is included, but you can narrow them by "Include" and "Exclude" filters. The general syntax looks like this: Include|Exclude:FL:<folder>.WF:<workflow>.SI:<session>.MP:<mapping>

In every part you can use wildcards: * to accept any number of any characters or ? to accept 1 character.

To process everything you can leave a default value: Include:FL:.*WF:*.SI:.*MP:* To narrow filter, you can either add include or exclude filters of specific objects, like this: Include:FL:MyFolder.WF:MyWorkflow.SI:.*MP:* Exclude:FL:.*WF:*.SI:*.MP:BadMapping The above will process only MyWorkflow from MyFolder, but will skip BadMapping.

If mode "Pmrep command" is used, only matching folders will be exported (all not matched will be skipped from export).

| Parameter name | Description |
|---|---|
| Prompt | Object filter |
| Symbol | powerCenter.inputFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | Include:FL:*.WF:*.SI:*.MP:* |
| Example values | - Include:FL:*.WF:*.SI:*.MP:* - process everything<br>- Include:FL:*.WF:*.SI:*.MP:mapping1? - process only mappings starting with 'mapping1' and having one additional characters at the end of the name<br>- Exclude:FL:*.WF:*.SI:*.MP:mapping1 - exclude mapping1 from processing |

## Override output model repository

The name for repository appears in output model. It's picked from an input Informatica Xml unless it's overridden here.

| Parameter name | Description |
|---|---|
| Prompt | Override output model repository |
| Symbol | powerCenter.overrideRepository |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Test_Repository |

## Pmrep host:port

The host and port used in pmrep connect command (-h, -o).

| Parameter name | Description |
|---|---|
| Prompt | Pmrep host:port |
| Symbol | powerCenter.pmrepPortalHostAndPort |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - infa-host:6005 |

## Pmrep password

The password used in pmrep connect command (-x).

| Parameter name | Description |
|---|---|
| Prompt | Pmrep password |
| Symbol | powerCenter.pmrepPassword |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - user |

## Pmrep path

The path to pmrep utility

| Parameter name | Description |
|---|---|
| Prompt | Pmrep path |
| Symbol | powerCenter.pmrepPath |

| Parameter name | Description |
|---|---|
| Type | STRING |
| Required | No |
| Default value | pmrep |
| Example values | - pmrep<br>- /opt/informatica_1041_cmd_utilities_linux-x64/PowerCenter/server/bin |

## Pmrep repository

The repository name used in pmrep connect command (-r).

| Parameter name | Description |
|---|---|
| Prompt | Pmrep repository |
| Symbol | powerCenter.pmrepRepository |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - PC_repository.informatica.com |

## Pmrep security domain

The security domain name used in pmrep connect command (-s).

| Parameter name | Description |
|---|---|
| Prompt | Pmrep security domain |
| Symbol | powerCenter.pmrepSecurityDomain |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - infa-security-domain |

## Pmrep timeout in seconds

The timeout in seconds used in each command independently. If it is exceeded, processing of given command is stopped. ERROR log will show up if it happens.

| Parameter name | Description |
|---|---|
| Prompt | Pmrep timeout in seconds |
| Symbol | powerCenter.pmrepTimeoutInSeconds |
| Type | INTEGER |
| Required | No |
| Default value | 600 |
| Example values | - 300<br>- 600<br>- 900 |

## Pmrep user

The user name used in pmrep connect command (-n).

| Parameter name | Description |
|---|---|
| Prompt | Pmrep user |
| Symbol | powerCenter.pmrepUser |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - user |

## Process only active used workflows

When it is set to true, then only workflows executed over "the last few days" are processed. "The last few days" is determined by another parameter: activeFilterWorkflowInformaticaDaysNumber=<number of days>. So, when it is set to true - the activeFilterWorkflowInformaticaDaysNumber has to be set, as well. Other accompanying parameters that must be set (when it is set to true):

- Point to the Informatica database connection defined in MDREPO: informaticaDatabase=<HOST>.<SCHEMA>.<DATABASE>
- Path to Informatica connection registry repositoryPath=<repository path>

Example: activeFilterWorkflowEnabled=true informaticaDatabase=prodHost50.InfaRepoSchema.InfaDBProd activeFilterWorkflowInformaticaDaysNumber=30 with the configuration as above, only the workflows that

were executed over the last 30 days are taken into account for processing. The information about workflow executions is found in the Informatica repository database, determined by dbConnection.

| Parameter name | Description |
|---|---|
| Prompt | Process only active used workflows |
| Symbol | powerCenter.activeFilterWorkflowEnabled |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | No example is available for this parameter. |

## Process workflow link conditions

Whether link conditions should affect tasks processing, when true link conditions will be evaluated.

| Parameter name | Description |
|---|---|
| Prompt | Process workflow link conditions |
| Symbol | powerCenter.processTasksConditionally |
| Type | PRIMITIVE_BOOLEAN |
| Required | No |
| Default value | false |
| Example values | No example is available for this parameter. |

## Process workflows run during last days

Include only the workflows that were run by Informatica Intelligent Cloud Services during the days that you last declared

| Parameter name | Description |
|---|---|
| Prompt | Process workflows run during last days |
| Symbol | powerCenter.activeFilterWorkflowInformaticaDaysNumber |
| Type | STRING |
| Required | No |
| Default value | 7 |
| Example values | No example is available for this parameter. |

## Read source

Choose whether to read from existing XML repository exported files or via pmrep command. If you choose pmrep command, you will need to provide additional connection parameters and set environment settings to allow pmrep command to work. The connection command will look like this: pmrep connect -r <repository> -h <portal*host*> -*o* <*portal*port> -s <security domain> -n <user> -x <password>

| Parameter name | Description |
|---|---|
| Prompt | Read source |
| Symbol | powerCenter.readSource |
| Type | MODULE_MODE |
| Required | No |
| Default value | Files |
| Example values | - Files<br>- Pmrep command |

## Server in output model and files host name

Host name used for the data files read or written by the mappings and Server name for output model

| Parameter name | Description |
|---|---|
| Prompt | Server in output model and files host name |
| Symbol | powerCenter.filesHost |
| Type | STRING |
| Required | Yes |
| Default value | POWERCENTER_HOST |
| Example values | - DATAF<br>- File Storage |

## Session instances to be excluded during processing

Expression which matches full path to session instance to be excluded during processing. Full path is (without worklets): FL:[InformaticaFolderName].WF:[WorkflowName].SI:[SessionInstanceName].MP:[MappingName] or (with worklet): FL:[InformaticaFolderName].WF:[WorkflowName].WL:[Worklet1InstanceName].WL:[Worklet2InstanceName].SI:[SessionInstanceName].MP:[MappingName]. You can add zero, one or more worklets definitions. Question mark and asterisk mark can be used inside folder,

workflow, worklet instance, session instance or mapping name. Question mark matches any single character, asterisk matches any number of any characters. See examples for details.

| Parameter name | Description |
|---|---|
| Prompt | Session instances to be excluded during processing |
| Symbol | powerCenter.excludeSessionInstances |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - FL:temp*.WF:*.SI:*.MP:*- exclude folders starting with 'temp'<br>- FL:*.WF:*.SI:*.MP:mapping1? - exclude mappings starting with 'mapping1' and having one additional characters at the end of the name |

## Session instances to be included during processing

Expression which matches full path to session instance to be included during processing. Full path is (without worklets): FL:[InformaticaFolderName].WF:[WorkflowName].SI:[SessionInstanceName].MP:[MappingName] or (with worklet): FL:[InformaticaFolderName].WF:[WorkflowName].WL:[Worklet1InstanceName].WL:[Worklet2InstanceName].SI:[SessionInstanceName].MP:[MappingName]. You can add zero, one or more worklets definitions. Question mark and asterisk mark can be used inside folder, workflow, worklet instance, session instance or mapping name. Question mark matches any single character, asterisk matches any number of any characters. See examples for details.

Include patterns will also define the order in which workflows (and only workflows) will be processed. For example the following patterns:

- FL:.*WF:Workflow1.SI:*.MP:*
- FL:.*WF:Workflow2.SI:*.MP:*
- FL:.*WF:*.SI:.*MP:*

will result in the following workflow processing order: Workflow1, Workflow2, any other workflow.

| Parameter name | Description |
|---|---|
| Prompt | Session instances to be included during processing |
| Symbol | powerCenter.includeSessionInstances |
| Type | STRINGLIST |
| Required | No |
| Default value | FL:*.WF:*.SI:*.MP:* |
| Example values | - FL:*.WF:*.SI:*.MP:*- process everything<br>- FL:*.WF:*.SI:*.MP:mapping1? - process only mappings starting with 'mapping1' and having one additional characters at the end of the name |

## Skip disabled tasks

Do not process any workflows, worklets, sessions, or other tasks marked as "Disabled" by Informatica Intelligent Cloud Services.

| Parameter name | Description |
|---|---|
| Prompt | Skip disabled tasks |
| Symbol | powerCenter.skipDisabled |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Source Qualifier command file list

List of source files for the Source Definition of the component of Informatica PowerCenter. In this component list of files to process can be defined as a result of command line command. For lineage purposes, the user can define list of input files for this particular component. The field "Qualified file input component name" should contain a qualified name to user's choice Source Definition Informatica component. A Qualified name means the "Informatica component path" and the string "FileList". An Informatica component path means the Informatica folder, workflow, worklets (if any), session instance name and component name. Each part of the qualification is prefixed by two letter acronym FL- folder, WF- workflow, WL- worklet, SI- session instance, CP- component. A colon separates each prefix from its value, and a dot separates each segment. The field "List of file paths" can contain comma separated list of file paths that will be used as input files processed by this component.

| Parameter name | Description |
|---|---|
| Prompt | Source Qualifier command file list |
| Symbol | powerCenter.sourceQualifierCommandFileList |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - FL:INFA_01.WF:wf_DAILY_TRANS.WL:wl_REGISTER_INVOICE.SI:sm_so_st_REGISTER_INVOICE.CP:inFiles.FileList=path/to/file1.dat;path/to/file2.dat --- Here, "INFA_01" is the name of the folder, "wf_DAILY_TRNS" is the name of the workflow, "wl_REGISTER_INVOICE" is the worklet, "sm_so_st_REGISTER_INVOICE" is the session instance, "inFiles" is the component and "FileList" is the required constant string. Next is the comma separated list of file paths. |

## Sql Transform components excluded from logging warns

Expression which matches full path to Sql Transform component in script mode for which WARN messages will NOT be logged. Full path is (without worklets): FL:[InformaticaFolderName].WF:[WorkflowName].SI:[SessionInstanceName].MP:[MappingName].CP:[ComponentName] or (with worklet): FL:[InformaticaFolderName].WF:[WorkflowName].WL:[Worklet1InstanceName].WL:[Worklet2InstanceName].SI:[SessionInstanceName].MP:[MappingName].CP:[ComponentName]. You can add zero, one or more worklets definitions. Question mark and asterisk mark can be used inside folder, workflow, worklet instance, session instance, mapping or component name. Question mark matches any single character, asterisk matches any number of any characters. See examples for details. Example patterns:

- FL:.*WF:*.SI:.*MP:*.CP. - Do not log WARN for all sql transform component in script mode
- FL:.*WF:*.SI:.*MP:*.CP:component - Do not log WARN for sql transform component in script mode that name is "component"

| Parameter name | Description |
|---|---|
| Prompt | Sql Transform components excluded from logging warns |
| Symbol | powerCenter.sqlTransformComponentExcludedInWarnMessageFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - FL:*.WF:*.SI:*.MP:*.CP:* - Log info instead of warn for all sql transform component in script mode<br>- FL:*.WF:*.SI:*.MP:*.CP:filterSql? - Log info instead of warn for transform component in script mode that name starts with filterSql and log warn for others |

## SQLTransformer SQL resolution

Map for replacing the entire SQL query or parts of it in the SQL Transform component of Informatica PowerCenter. This component builds a dynamic SQL query for every record using values from input ports as part of the query. For lineage purposes, the user can define a single value for each input port or override original component's SQL query or do not provide any additional information. The "Qualified SQL part to replace" field should contain a qualified name to user's choice input field in particular SQL Transform component or qualified name of that component in case of sql query override. A Qualified field name means the "Informatica component path" and the string "SQLReplace", and the input port name or "Informatica component path" and the string "SQLReplaceAll". An Informatica component path means the Informatica folder, workflow, worklets (if any), session instance name and component name. Each part of the qualification is prefixed by two letter acronym FL- folder, WF-workflow, WL- worklet, SI- session instance, CP-component. A colon separates each prefix from its value, and a dot separates each segment. The field

"Replacement" can contain parameters (for example: $Param1, $$Param2) that will be resolved according to their current value.

| Parameter name | Description |
|---|---|
| Prompt | SQLTransformer SQL resolution |
| Symbol | powerCenter.sqlStringReplacers |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - FL:INFA_01.WF:wf_DAILY_TRANS.WL:wl_REGISTER_INVOICE.SI:sm_so_st_REGISTER_INVOICE.CP:SQL.SQLReplace.~o_SID~ --- Here, "INFA_01" is the name of the folder, "wf_DAILY_TRNS" is the name of the workflow, "wl_REGISTER_INVOICE" is the worklet, "sm_so_st_REGISTER_INVOICE" is the session instance, "SQL" is the component, "SQLReplace" is the required constant string, and "~o_SID~" is the port.<br>- FL:INFA_01.WF:wf_DAILY_TRANS.WL:wl_REGISTER_INVOICE.SI:sm_so_st_REGISTER_INVOICE.CP:SQL.SQLReplaceAll --- Here, "INFA_01" is the name of the folder, "wf_DAILY_TRNS" is the name of the workflow, "wl_REGISTER_INVOICE" is the worklet, "sm_so_st_REGISTER_INVOICE" is the session instance, "SQL" is the component and "SQLReplaceAll" is the required constant string. |

## workflow with assigned parameter file

When workflow is run with different parameter file than defined inside workflow add this workflow and parameter file as MetaDex parameter. Use folder qualification for workflow and full parameter file path

| Parameter name | Description |
|---|---|
| Prompt | workflow with assigned parameter file |
| Symbol | powerCenter.workflowAndParameter |
| Type | MULTIMAP |
| Required | No |
| Default value | N/A |
| Example values | - FL:PRD_CREDIT.WF.wf_LoadCredit=$PMParamFileDir/loadCredit.prm |

## Supported Components

- AddressValidation Transformation - Custom
- Aggregator - Transformation
- Amazon Athena (AmazonAthena_amazonathena Reader) - Reader
- Amazon Redshift (AmazonRSCloudAdapter Reader) - Reader
- Amazon Redshift (AmazonRSCloudAdapter Writer) - Writer
- Amazon Redshift (AmazonRedshiftV2_amazonredshift Reader) - Reader
- Amazon Redshift (AmazonRedshiftV2_amazonredshift Writer) - Writer

- Amazon S3 (AmazonS3 Reader) - Reader
- Amazon S3 (AmazonS3 Writer) - Writer
- Amazon S3 (AmazonS3V2_amazons3 Reader) - Reader
- Amazon S3 (AmazonS3V2_amazons3 Writer) - Writer
- App Multi-Group Source Qualifier - Transformation
- Application Source Qualifier - Transformation
- Azure Blob (AzureBlobV3_blob Reader) - Reader
- Azure Blob (AzureBlobV3_blob Writer) - Writer
- Azure Data Lake Storage Gen2 (adlsgen2_adlsg2 Reader) - Reader
- Azure Data Lake Storage Gen2 (adlsgen2_adlsg2 Writer) - Writer
- Azure Data Warehouse (AzureDWV3_table Reader) - Reader
- Azure Data Warehouse (AzureDWV3_table Writer) - Writer
- Business 360 Flat End Point (mdmflat_flatmdmentity Writer) - Writer
- Business Entity (mdmV1_businessentity Reader) - Reader
- Business Entity (mdmV1_businessentity Writer) - Writer
- Cluster Transformation - Custom
- Comparison Transformation - Custom
- Consolidation Transformation - Custom
- Custom Transformation - Transformation
- DB2 Warehouse (Db2Warehouse_table Reader) - Reader
- Databricks Delta (databricksdelta_databricksdelta Reader) - Reader
- Databricks Delta (databricksdelta_databricksdelta Writer) - Writer
- Db2 Warehouse (Db2Warehouse_table Writer) - Writer
- Decision Transformation - Custom
- Expression - Transformation
- File (File Reader) - Reader
- File (File Writer) - Writer
- Filter - Transformation
- Generic JDBC (GenericJDBCAdapter Reader) - Reader
- Generic JDBC (GenericJDBCAdapter Writer) - Writer
- Google Big Query (GoogleBigQueryV2_table Reader) - Reader
- Google Big Query (GoogleBigQueryV2_table Writer) - Writer
- Google Cloud Storage (GoogleCloudStorageV2_storage Reader) - Reader
- Google Cloud Storage (GoogleCloudStorageV2_storage Writer) - Writer
- Google Storage (googlestoragev2_storage Reader) - Reader
- Google Storage (googlestoragev2_storage Writer) - Writer
- HDFS Flat File (HDFS Flat File Reader) - Reader
- HDFS Flat File (HDFS Flat File Writer) - Writer
- Hive (hiveadapter_table Reader) - Reader

- Hive (hiveadapter_table Writer) - Writer
- IDQParserZTransform - IDQ Custom Transformation
- Input Transformation - Transformation
- JDBC (JDBC_V2_table Reader) - Reader
- JDBC Writer (JDBC_V2_table Writer) - Writer
- Java Transformation - Custom
- Joiner - Transformation
- KeyGenerator Transformation - Custom
- Labeler Transformation - Custom
- Lookup Procedure - Transformation
- MQ (MQ Reader) - Reader
- MQ (MQ Writer) - Writer
- MQ Source Qualifier - Transformation
- Mapplet - Transformation
- Microsoft Fabric One Lake (onelake_onelake Reader) - Reader
- Microsoft Fabric One Lake (onelake_onelake Writer) - Writer
- MicrosoftExcel (MicrosoftExcel Reader) - Reader
- MongoDB (mongodbadapter_table Reader) - Reader
- MongoDB (mongodbadapter_table Writer) - Writer
- Netezza Bulk (Netezza Bulk Reader) - Reader
- Netezza Bulk (Netezza Bulk Writer) - Writer
- Normalizer - Transformation
- Operational Data Provisioning Browser (ODP_browser Reader) - Reader
- Output Transformation - Transformation
- PairGenerator Transformation - Custom
- Parser Transformation - Custom
- Postgres (PostgreSQLadapter_table Reader) - Reader
- Postgres (PostgreSQLadapter_table Writer) - Writer
- PowerExchange Batch for VSAM files (PowerExchange Batch Reader for VSAM files) - Reader
- PowerExchange CDC for DB2zOS (PowerExchange CDC Reader for DB2zOS) - Reader
- PowerExchange CDC for ORACLE (PowerExchange CDC Reader for ORACLE) - Reader
- PowerExchange for Sequential files (PowerExchange Reader for Sequential files) - Reader
- PowerExchange for Sequential files (PowerExchange Writer for Sequential files) - Writer
- RESTv2 (RESTv2 Reader) - Reader
- RESTv2 (RESTv2 Writer) - Writer
- Rank - Transformation
- Relational Table (Relational Reader) - Reader
- Relational Table (Relational Writer) - Writer
- Router - Transformation

- SAP Business Explorer (SAPBExQuery Reader) - Reader
- SAP Business Warehouse (SAPBWREADER Reader) - Reader
- SAP Tables (SAPTables Reader) - Reader
- SAP Tables (SAP_Table_Writer) - Writer
- SPI (SPI_READER) - Reader
- SQL Transform - Custom
- Salesforce (SalesForce Reader_New) - Reader
- Salesforce (SalesForce Writer_New) - Writer
- Salesforce (Salesforce Reader) - Reader
- Salesforce (Salesforce Writer) - Writer
- Sequence - Transformation
- SharePoint Online (SharepointOnline Reader) - Reader
- Snowflake Data Warehouse (SnowflakeCloudDataWarehouse Reader) - Reader
- Snowflake Data Warehouse (SnowflakeCloudDataWarehouse Writer) - Writer
- Snowflake Data Warehouse (SnowflakeCloudDataWarehouseV2_table Reader) - Reader
- Snowflake Data Warehouse (SnowflakeCloudDataWarehouseV2_table Writer) - Writer
- Sorter - Transformation
- Source Definition - Transformation
- Source Qualifier - Transformation
- Stored Procedure - Transformation
- Target Definition - Transformation
- Teradata FastExport (Teradata FastExport Reader) - Reader
- Teradata Parallel Transporter (Teradata Parallel Transporter Reader) - Reader
- Teradata Parallel Transporter (Teradata Parallel Transporter Writer) - Writer
- TokenStandardiser Transformation - Custom
- Transaction Control - Transformation
- Union Transformation - Custom
- Unstructured Data Transformation - Custom
- Update Strategy - Transformation
- WebServices Consumer - Custom
- WebServices Provider (WebServices Provider Reader) - Reader
- WebServices Provider (WebServices Provider Writer) - Writer
- WeightBasedAnalyser Transformation - Custom
- WorkdayH (WorkdayH Reader) - Reader
- WorkdayH (WorkdayH Writer) - Writer
- XML (XML Reader) - Reader
- XML (XML Writer) - Writer
- XML GENERATOR - Custom
- XML PARSER - Custom

- XML Source Qualifier - Transformation

# JCL Reader

A module for reading JCL files.

The JCL Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Content of missing files or datasets

Missing files or datasets mapped to their content.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Content of missing files or datasets |
| Symbol | jcl.missingFilesOrDatasetsContent |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - DATASET="SOME CONTENT"<br>- IGNORED.DATASET.*="" |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field

- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | jcl.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Default variables values file

Path to the file with local variables values.

Such variable could be one of following variable types:

- simple: %VARIABLE
- symbolic: &VARIABLE
- scheduler: %%VARIABLE

Definitions are global for each script and they are represented by key-value pairs (the same syntax as a java .properties file).

Example:

- VARIABLE=A

| Parameter name | Description |
|---|---|
| Prompt | Default variables values file |
| Symbol | jcl.defaultValuesFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\defaultVariableValues.txt<br>- /opt/etc/defaultVariableValues.txt |

## Exclude programs called from JCL script

Programs called from JCL script to be excluded. Each entry has to contain 1 segment: the name of the program. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Exclude programs called from JCL script |
| Symbol | jcl.excludePrograms |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - SKIPPED_PROGRAM<br>- PROG??<br>- PROG1* |

## Input files

The paths of files to be processed. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | jcl.fileIncludeFilter |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input\jcl\224_super_old.jcl<br>- /opt/input/jcl/225_even_older.jcl<br>- sftp://username:password@hostname[:port]/input/*.jcl |

## JCL files host name

Host name for JCL files

| Parameter name | Description |
|---|---|
| Prompt | JCL files host name |
| Symbol | jcl.fileHost |
| Type | STRING |
| Required | Yes |
| Default value | N/A |
| Example values | - JCL_FILES_HOST |

## JCL preprocessor default values file path

A path to the file that contains JCL preprocessor default values which are used to resolve preprocessing directives

| Parameter name | Description |
|---|---|
| Prompt | JCL preprocessor default values file path |
| Symbol | jcl.preprocessorDefaultValuesFile |
| Type | FILE |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Y:\etc\preprocessorDefaultValuesFile.txt<br>- /opt/etc/preprocessorDefaultValuesFile.txt |

## Jobs to be excluded

JCL jobs that should be excluded.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Jobs to be excluded |
| Symbol | jcl.jobExcludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - JCLJOB<br>- JOB1 |

## Jobs to be processed

JCL jobs that should be processed.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Jobs to be processed |
| Symbol | jcl.jobIncludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - JCLJOB<br>- JOB1 |

## Path mapping

Path mapping

| Parameter name | Description |
|---|---|
| Prompt | Path mapping |
| Symbol | jcl.ftpPathMapping |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - HOST:one/path=OTHER_HOST:other/path |

## PatternReplacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | jcl.patternReplacerFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

## Raise error when JCL files or datasets are missing

This option indicates if JCL files or datasets are missing warning should be logged and file treated as empty or error should be raised.

| Parameter name | Description |
|---|---|
| Prompt | Raise error when JCL files or datasets are missing |
| Symbol | jcl.logErrorOnJclMissingFilesOrDatasets |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Recognize JCL Job files automatically

This option indicates if JCL Job files should be recognized automatically or not.

| Parameter name | Description |
|---|---|
| Prompt | Recognize JCL Job files automatically |
| Symbol | jcl.recognizeJclFiles |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Supported JCL Programs

- IEBGENER
- ICEGENER
- ICETOOL
- ORASQL
- ORALDR
- BTEQ
- BTQMAIN
- MLOAD
- SAS
- SASHOST

- SASB
- SAS92
- SAS94
- IEFBR14
- DSNUTILB
- SORT
- SORTD
- ICEMAN
- IERRCO00
- IGHRCO00
- IKJEFT01
- IKJEFT1A
- IKJEFT1B
- DSNTIAUL
- DSNTIAD
- DSNTEP2
- DSNTEP4
- COBOL
- ORAPGM
- IDCAMS
- PKZIP
- PKUNZIP
- FTBMCPY
- SELCOPY
- IEBCOPY
- INFOMOVR
- FTP
- FILEMGR
- TRSMAIN
- ISRSUPC
- DUMMY
- FTBABEND
- IEBPTPCH
- IEBUPDTE
- ROSCOPY
- EQQYCAIN
- EQQEVPGM
- BPXBATCH
- MISSING

- ABEND
- EZTPA00
- PTLDRIVM
- OCOPY
- OGETX
- UT997
- TXT2PDF
- EDCICONV
- INZUTILB
- DSNUPROC
- CDEXECPA
- DFSRRC00
- DMBATCH
- ZIP390
- XMITIP

# Lineage Analyzer

A module for calculating and filtering data lineage generated by MetaDex.

The Lineage Analyzer includes the following parameters:

## Analyse calculation complexity

For every extension mapping, number of sources referenced in mapping's function will be analyzed. This information will be stored as additional property of extension mapping. There are following complexities recognized:

- Constant - expression has not any sources, eg: 123
- Copy - expression is direct column/variable reference, eg: COL1
- Single Source Calculation - expression has only one source, eg: COL1 + 123
- Multi Source Calculation - expression has more than one source, eg: COL1 + COL2 + 123
- Undefined - unable to determine the number of sources

| Parameter name | Description |
|---|---|
| Prompt | Analyse calculation complexity |
| Symbol | lineageAnalyzer.calculationComplexity |
| Type | PRIMITIVE_BOOLEAN |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | true |
| Example values | - true<br>- false |

## Generate control dependencies

Control dependencies are data not impacting values, but modifies result indirectly. Examples of control dependencies:

- AGGREGATION keys,
- FILTER expressions,
- JOIN conditions,
- IF conditions.

Generate control dependencies will append them to the output.Example:

- Update T set T1=S1 from S where S2=T2;
- Value dependencies for T.T1: S.S1.
- Control dependencies for T.T1: S.S2, T.T2.

| Parameter name | Description |
|---|---|
| Prompt | Generate control dependencies |
| Symbol | lineageAnalyzer.generateControlDependencies |
| Type | PRIMITIVE_BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Graph Transformation: Move filter condition to join

Graph can be transformed before lineage calculation, to move condition from filter(s) after join to join condition (sometimes it can provide better readability).

This effectively transforms a graph for SQL code like:

```
INSERT INTO T3
 SELECT *
 FROM T1
 CROSS JOIN T2
 WHERE T1.ID = T2.ID
```

to be the same as a graph which is created for following SQL code:

```
INSERT INTO T3
 SELECT *
 FROM T1
 JOIN T2 ON T1.ID=T2.ID
```

For input already in second form this option has no effect (unless there is a where condition after the join which can be still moved to join as additional condition).

| Parameter name | Description |
|---|---|
| Prompt | Graph Transformation: Move filter condition to join |
| Symbol | lineageAnalyzer.moveFilterConditionToJoin |
| Type | PRIMITIVE_BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Include/Exclude object filters

Case sensitive include/exclude filters, allowing to filter table and file assets before processing. These filters use the following syntax: <FilterType> : <AssetType> : <FilterText> Where: FilterType: Include - if nonempty only the assets prefixed with at least one of given qualification will be processed Exclude - items matching at least one of given patters will be omitted IncludeSource - ff nonempty only the source assets prefixed with at least one of given qualification will be processed IncludeTarget - if nonempty only the target assets prefixed with at least one of given qualification will be processed ExcludeSource - source items matching at least one of given patters will be omitted ExcludeTarget - target items matching at least one of given patters will be omitted AssetType: Table File FilterText: Full patterns are: For table : HOST_PATTERN[.DB_PATTERN[.SCHEMA_PATTERN[.TABLE_PATTERN[.COLUMN_PATTERN]]]] For file : HOST_PATTERN[.PATH_PATTERN[.COLUMN_PATTERN]] Question mark and asterisk mark can be used inside each pattern Question mark matches any single character, asterisk matches any number of any characters. See examples for details. To transfer from old filter to the new one replace e.g: dlFilterTableIncludes = host1.db* --> objectFilters = Include : Table : host1.db* dlFilterTableExcludes = File:HOST."/home/dummy.csv".column --> objectFilters = Exclude : File : HOST."/home/dummy.csv".column dlFilterTableTargetExcludes = Table:*.*.*.*.dummy --> objectFilters = ExcludeTarget : Table : *.*.*.*.dummy

| Parameter name | Description |
|---|---|
| Prompt | Include/Exclude object filters |
| Symbol | lineageAnalyzer.objectFilters |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - Include : Table : .*.*.*.*<br>- ExcludeSource : File : 'HOST.*.*.txt' |

### Only control conditions

A parameter that shows lineage and control conditions in Enterprise Data Catalog and control conditions in Cloud Data Governance and Catalog

| Parameter name | Description |
|---|---|
| Prompt | Only control conditions |
| Symbol | lineageAnalyzer.onlyControlConditionsInModifiedBy |
| Type | PRIMITIVE_BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

### Remove from job if has no lineage

This parameter could be used to remove some graphs from job if they have no lineage. Thanks to that the number of graphs to process could be reduced. The graph has no lineage if after all filters (dlFilter* parameters) applied, the graph has no single source to target mapping. E.g. Consider SQL "INSERT INTO T1 SELECT * FROM T2", which has lineage T2 -> T1.If filters are set to remove T2, the graph will not have lineage T2 -> T1 = will have no source to target mappings. Set this parameter to true and if graph from job has no lineage, it will be removed from processing. Set to false all graphs will be process regardless if they have lineage or not.

| Parameter name | Description |
|---|---|
| Prompt | Remove from job if has no lineage |
| Symbol | lineageAnalyzer.removeFromProcessingIfHasNoLineage |
| Type | PRIMITIVE_BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

# Microsoft MDX Query Evaluator

A module for evaluating MDX statements embedded in SSRS Reports.

The Microsoft MDX Query Evaluator has no configurable parameters defined.

# Microsoft Power BI Reader

Reads Microsoft Power BI files

The Microsoft Power BI Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema

- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | powerBiCloud.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Force dataset field forwarding to reports

Enable mandatory dataset field forwards into report components.

| Parameter name | Description |
|---|---|
| Prompt | Force dataset field forwarding to reports |
| Symbol | powerBiCloud.forceFieldForwardingToReports |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Include personal workspaces

Specify whether to include personal workspaces

| Parameter name | Description |
|---|---|
| Prompt | Include personal workspaces |
| Symbol | powerBiCloud.includePersonalWorkspaces |

| Parameter name | Description |
|---|---|
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Input files

Files extracted using PowerBi Cloud API

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | powerBiCloud.inputFiles |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - /home/powerbi/*.json |

## PatternReplacer configuration path

Path to a configuration of the PatternReplacer functionality.The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | powerBiCloud.patternReplacerFile |
| Type | FILE |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

## Power BI Cloud REST API extractor filters

Power BI Cloud REST API extractor include/exclude filters used to select reports that have to be downloaded. You can use asterisk '*' to match any number of characters

| Parameter name | Description |
|---|---|
| Prompt | Power BI Cloud REST API extractor filters |
| Symbol | powerBiCloud.inputFilters |
| Type | STRINGLIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Include:Workspace 1:Report 1<br>- Exclude:Workspace 2:* |

## Power BI Cloud REST API extractor timeout

The time after which the Microsoft Power BI scan completes for a single job in seconds.

| Parameter name | Description |
|---|---|
| Prompt | Power BI Cloud REST API extractor timeout |
| Symbol | powerBiCloud.restApiExtractorTimeoutSeconds |
| Type | STRING |
| Required | No |
| Default value | 300 |
| Example values | - 100<br>- 200 |

## Power BI Cloud REST API workspaces scan timeout

The Power BI Cloud REST API workspaces scan timeout for single scan in seconds. If the scan does not finish during that time with the "Succeeded" status, the job aborts.

| Parameter name | Description |
|---|---|
| Prompt | Power BI Cloud REST API workspaces scan timeout |
| Symbol | powerBiCloud.restApiWorkspacesScanTimeoutSeconds |
| Type | STRING |
| Required | No |
| Default value | 30 |
| Example values | - 10 |

## Power BI Cloud server name

The Power BI Cloud server in which the files are located

| Parameter name | Description |
|---|---|
| Prompt | Power BI Cloud server name |
| Symbol | powerBiCloud.server |
| Type | STRING |
| Required | Yes |
| Default value | POWERBI_SERVER |
| Example values | - server01 |

## Proxy host

Host name or ip address of http(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy host |
| Symbol | powerBiCloud.restApiProxyHost |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - https://myproxy |

## Proxy password

Password to authenticate to http(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy password |
| Symbol | powerBiCloud.restApiProxyPassword |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - proxy-password |

## Proxy port

Port a http(s) proxy server is listening on

| Parameter name | Description |
|---|---|
| Prompt | Proxy port |
| Symbol | powerBiCloud.restApiProxyPort |
| Type | INTEGER |
| Required | No |
| Default value | 0 |
| Example values | - 30456 |

## Proxy username

Username to authenticate to http(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy username |
| Symbol | powerBiCloud.restApiProxyUsername |

| Parameter name | Description |
|---|---|
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - proxy-user |

## Reader mode

Choose files to read Power BI Cloud from existing json files or REST API

| Parameter name | Description |
|---|---|
| Prompt | Reader mode |
| Symbol | powerBiCloud.readerMode |
| Type | MODULE_MODE |
| Required | No |
| Default value | Files |
| Example values | - Files<br>- Admin User |

## REST API Authority URL

The authority URL that you use to receive the access token.

| Parameter name | Description |
|---|---|
| Prompt | REST API Authority URL |
| Symbol | powerBiCloud.restApiAuthorityUrl |
| Type | STRING |
| Required | No |
| Default value | https://login.microsoftonline.com |
| Example values | - https://login.microsoftonline.us |

## REST API client ID

The clientId of user that is used to connect to Power BI Cloud REST API

| Parameter name | Description |
|---|---|
| Prompt | REST API client ID |
| Symbol | powerBiCloud.restApiClientId |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - clientId |

## REST API client secret

The client secret that is used to connect to Power BI Cloud REST API using Service Principal

| Parameter name | Description |
|---|---|
| Prompt | REST API client secret |
| Symbol | powerBiCloud.restApiClientSecret |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - secret123 |

## REST API password

The password of user that is used to connect to Power BI Cloud REST API

| Parameter name | Description |
|---|---|
| Prompt | REST API password |
| Symbol | powerBiCloud.restApiPassword |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - secret123 |

## REST API scope

The scope that you use to connect to the Microsoft Power BI National Cloud REST API with the service principal authentication method.

| Parameter name | Description |
|---|---|
| Prompt | REST API scope |
| Symbol | powerBiCloud.restApiScope |
| Type | STRING |
| Required | No |
| Default value | https://analysis.windows.net/powerbi/api/.default |
| Example values | - https://analysis.usgovcloudapi.net/powerbi/api/.default |

## REST API tenant ID

The tenantId that is used to connect to Power BI Cloud REST API using Service Principal

| Parameter name | Description |
|---|---|
| Prompt | REST API tenant ID |
| Symbol | powerBiCloud.restApiTenantId |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - tenantId-2829-2233 |

## REST API URL

The url that is used to connect to Power BI Cloud REST API

| Parameter name | Description |
|---|---|
| Prompt | REST API URL |
| Symbol | powerBiCloud.restApiUrl |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | https://api.powerbi.com |
| Example values | - https://api.my.url |

## REST API user

The user that is used to connect to Power BI Cloud REST API

| Parameter name | Description |
|---|---|
| Prompt | REST API user |
| Symbol | powerBiCloud.restApiUser |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - user1 |

## Workspace type

Workspace type that should be processed

| Parameter name | Description |
|---|---|
| Prompt | Workspace type |
| Symbol | powerBiCloud.workspaceType |
| Type | STRINGLIST |
| Required | Yes |
| Default value | Workspace |
| Example values | - Workspace<br>- Group |

### Workspaces modified since

Include workspaces modified in the last specified number of days. Specify the number of days between 1 and 30.

| Parameter name | Description |
|---|---|
| Prompt | Workspaces modified since |
| Symbol | powerBiCloud.workspacesModifiedSince |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - 1<br>- 30 |

### Supported Dashboard, Dataset and Report components

- Dashboard - Dashboard component
- Dataset - Dataset component
- Report - Report component

# MultiLoad Evaluator for JCL

A module for evaluating MULTILOAD scripts called from the JCL jobs.

The MultiLoad Evaluator for JCL includes the following parameters:

### MultiLoad program context

Default MultiLoad program context

| Parameter name | Description |
|---|---|
| Prompt | MultiLoad program context |
| Symbol | multiLoad.sqlContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - host.db.schema |

### PatternReplacer configuration path

The path to a configuration of the PatternReplacer functionality.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | multiLoad.patternReplacerFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

# MultiLoad Reader

Reads Teradata MultiLoad files.

The MultiLoad Reader includes the following parameters:

### Context

Default SQL context

| Parameter name | Description |
|---|---|
| Prompt | Context |
| Symbol | multiLoad.sqlContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - host.db.db |

### Data Files Working Directory

Path for Data Files Working Directory

| Parameter name | Description |
|---|---|
| Prompt | Data Files Working Directory |
| Symbol | multiLoad.dataFilesWorkingDirectory |

| Parameter name | Description |
|---|---|
| Type | PATH |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\<br>- /root/dir1/ |

## Default variables values file

Default values path

| Parameter name | Description |
|---|---|
| Prompt | Default variables values file |
| Symbol | multiLoad.defaultValuesFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\defaultVariableValues.txt<br>- /opt/etc/defaultVariableValues.txt |

## PatternReplacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | multiLoad.patternReplacerFile |
| Type | FILE |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

## Scripts exclude filters

The paths of MultiLoad scripts that the job excludes.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Scripts exclude filters |
| Symbol | multiLoad.fileExcludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\sample_file.ml<br>- /opt/input/sample_file.ml<br>- /opt/input/*.ml |

## Scripts Root Directory Path

Root directory of input files. Relative paths of input files resolve in the following location: [Input root directory]. Absolute paths from input files must start from the [Input root directory] to be included into processing. [Input root directory] will not be visible on the output objects structure.

| Parameter name | Description |
|---|---|
| Prompt | Scripts Root Directory Path |
| Symbol | multiLoad.inputRoot |
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\inputs\<br>- /opt/feed-etl/ |

### Scripts' paths

Path to input script. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Scripts' paths |
| Symbol | multiLoad.fileIncludeFilter |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input\multiload\sample_file.ml<br>- /opt/input/multiload/sample_file.ml |

# ODI Reader

Converts ODI mappings to Graphs

The ODI Reader includes the following parameters:

### Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

### Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | odi.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Default variables values file

Path to the file with default variables values. File format consist of [GLOBAL] or/and custom function section.

---

You can define variables definitions: [GLOBAL] VariableName = VariableExpression

VariableName that contains only standard alphanumeric characters does not require any quotes. If it contains any special characters, it requires double-quotes (").

Example of variables definition: [GLOBAL] a = 42 b = 7 c = a < b ? a + 7 : b - 7 // It's 49 d = e(1,2) + 1 // Expression use call to function. "User::table" = 'table' || "User::tableSuffix" // It's string table concatenated with value of User::tableSuffix variable

---

You can define functions definitions: [GLOBAL] FunctionName = FunctionExpression

FunctionName can be unquoted if contains only standard alphanumeric characters but needs to be quoted with double quoted (") if contains any special character. FunctionName must contain list of arguments. List of argument is defined inside "(" ")" characters and consists of "?" sings separated by "," (one "?" sign for each function argument).

Example of function definition: [GLOBAL] a(?) = 1 a(?,?) = 2 b(?) = a(1) + 1 // Expression use call to another function. c(?,?) = d + 2 // Expression use reference to variable.

---

Additional sections can be provided to match only specific functions. To match an overloaded function please provide placeholders for its arguments. Inside the matched section you can also reference matched function arguments. Example of custom function sections:

- [host.db.schema.func(x)]
- z=x
- [host.db.schema.func(x,y)]
- z=x+y

---

To escape special characters such as \n \t use the \ character. For example, to define E:\file \tgtParameterized.csv, enter E:\\file\\tgtParameterized.csv

---

On the right side of assignments you can use simple expression language that can use values of already defined variables and functions or variables from current job context.

Supported expressions (case insensitive) are:

- literals
  - integer: 42
  - string: 'str'
  - string with quoted single quote: 'str\''
  - boolean: true, false
- equals: 42 == 7 -> FALSE
- not equals: 42 != 7 -> TRUE
- concatenations: 'str1' + 'str2' -> 'str1str2'
- function calls: Replace('abc', 'b', 'd') -> 'adc'
- tertiary operator:
  - 42 == 7 ? 'a' : 'b' -> 'b'
  - 42 != 7 ? 'a' : 'b' -> 'a'

Supported functions:

- Hash(str, maxOutputLen) - MD5, maxOutputLen between 1 and 32 inclusive
  - Hash('abcdefgh', 4) -> 'E8DC'
- Replace(str, from, to)
  - Replace('abc', 'b', 'D') -> 'aDc'
- ReplaceRegexp(str, regex, replacement) - replaces with Java regular expressions
  - ReplaceRegexp('abcde', 'b.*', 'f') -> 'af'
  - ReplaceRegexp('graph_id20', 'id(\d+)', '$1') -> 'graph_20'
- StringLengthLimit(str, limit, hashSize) - returns str[0..limit-hashSize] + Hash( str, hashSize )
  - StringLengthLimit('abcc', 3, 2) -> 'a26'
  - StringLengthLimit('abcc', 4, 2) -> 'abcc'
- StringLengthLimit(str, limit) - hashSize has default size 8
  - StringLengthLimit('abcdabcdabcd', 10) -> 'abE340600C'
- RegexpMatch(pattern, str) - tests if str matches pattern
  - pattern uses standard Java 8 regular expressions: https://docs.oracle.com/javase/8/docs/api/java/util/regex/Pattern.html
  - RegexpMatch('[A-Za-z]+','Abcd') -> TRUE
- Upper(str) - returns uppercased str
  - Upper('Abc') -> 'ABC'
- Lower(str) - return lowercased str
  - Lower('aBC') -> 'abc'
- Date(text,format) - returns object representing date with provided format.
  - Formats uses standard Java 8 SimpleDateFormat: https://docs.oracle.com/javase/8/docs/api/java/text/SimpleDateFormat.html
  - It is designed for default values usage
  - ex. Date('2017-10-31', 'yyyy-MM-dd')
- Contains/ContainsIgnoreCase(stack, needle) - tests if stack string contains needle string
  - Contains('abc', 'ab') -> TRUE
  - Contains('abc', 'AB') -> FALSE
  - ContainsIgnoreCase('abc', 'AB') -> TRUE
  - Contains('stack', 'needle') -> FALSE
- RtfToText(str) - return clean text from rtf string
  - RtfToText('{\rtf1\ansi{\fonttbl\f0\fswiss Helvetica;}\f0\pard text\par }') -> text

Supported comments:

- Single line: \ comment

- Multi line: /* comment */

| Parameter name | Description |
|---|---|
| Prompt | Default variables values file |
| Symbol | odi.defaultValuesFilePath |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /opt/RootDir/defaultVariableValues.txt |

## Export directory

The directory where the export file will be saved

| Parameter name | Description |
|---|---|
| Prompt | Export directory |
| Symbol | odi.exporterExportDir |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - /home/user/odi-export |

## Exporter timeout in minutes

The timeout for Odi Exporter to prepare input files

| Parameter name | Description |
|---|---|
| Prompt | Exporter timeout in minutes |
| Symbol | odi.exporterTimeoutInMinutes |
| Type | STRING |
| Required | No |
| Default value | 10 |
| Example values | - 5<br>- 10<br>- 30 |

## Host for files processed by ODI

Host name where the files processed by ODI are stored.

| Parameter name | Description |
|---|---|
| Prompt | Host for files processed by ODI |
| Symbol | odi.fileHost |
| Type | STRING |
| Required | No |
| Default value | ODI_HOST |
| Example values | - DATAF<br>- File Storage |

## Input files

The paths of files that should be available during job. It could be either XML files or ZIP files containing XML files or another ZIPs. ZIP files are expected to have '.zip' extension. Work repository files and master repository files must be present.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | odi.inputFiles |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - /opt/RootDir/input/*.zip |

## Log warn on missing variable

Decide about logging warn message when there is usage of variable with unknown value

| Parameter name | Description |
|---|---|
| Prompt | Log warn on missing variable |
| Symbol | odi.logWarnOnMissingVariable |
| Type | BOOLEAN |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | true |
| Example values | - true<br>- false |

## Method for providing input files

How input files are provided to the reader

| Parameter name | Description |
|---|---|
| Prompt | Method for providing input files |
| Symbol | odi.inputMode |
| Type | MODULE_MODE |
| Required | No |
| Default value | Files |
| Example values | - Files<br>- Exporter |

## Objects to be excluded during job

Expression which matches full path to object to be excluded during job. Objects to match are represented by the following filter:

Project/FolderPath/ObjectTypeAndName

Where: - Project is a project name in ODI Studio - FolderPath is a folder in ODI Studio which contains a given object - ObjectTypeAndName is an object type and a name you want to process. You define it in the following way:

ObjectType:ObjectName

For example, Scen:Scenario Name

Supported objects types are: - Scen - Scenario - Pack - Package - Proc - Procedure - Map - Mapping

If you want to select a scenario from a package, you can nest your filter in the following way:

Project/FolderPath/Pack:pack1/Scen:Scen1

When you want to process all scenarios from a package, you can use the following filter:

Project/FolderPath/Pack:pack1

It is treated in the same way as specified in the following path that contains a wildcard:

Project/FolderPath/Pack:pack1/Scen:*

If you include a Project name in a filter, the package part (Pack:) is obligatory for all scenarios that are in packages.

Question mark (?) and asterisk (*) *can be used inside a project, a folder or an object name. Double asterisk mark* (**) can be used inside a folder path.

Question mark matches any single character, asterisk matches any number of any characters, double asterisk mark matches all possible folders depths.

| Parameter name | Description |
|---|---|
| Prompt | Objects to be excluded during job |
| Symbol | odi.excludeObjectsFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - */temp*/**/* - Exclude scenarios for any object at first level folders starting with 'temp'<br>- */**/Pack:mapping1? - Exclude scenarios for packages that start with 'mapping1' and that have one additional character at the end of the name |

## Objects to be included during job

Expression which matches full path to object to be included during job. Objects to match are represented by the following filter:

Project/FolderPath/ObjectTypeAndName

Where: - Project is a project name in ODI Studio - FolderPath is a folder in ODI Studio which contains a given object - ObjectTypeAndName is an object type and a name you want to process. You define it in the following way:

ObjectType:ObjectName

For example, Scen:Scenario Name

Supported objects types are: - Scen - Scenario - Pack - Package - Proc - Procedure - Map - Mapping

If you want to select a scenario from a package, you can nest your filter in the following way:

Project/FolderPath/Pack:pack1/Scen:Scen1

When you want to process all scenarios from a package, you can use the following filter:

Project/FolderPath/Pack:pack1

It is treated in the same way as specified in the following path that contains a wildcard:

Project/FolderPath/Pack:pack1/Scen:*

If you include a Project name in a filter, the package part (Pack:) is obligatory for all scenarios that are in packages.

Question mark (?) and asterisk (*) *can be used inside a project, a folder or an object name. Double asterisk mark* (**) can be used inside a folder path.

Question mark matches any single character, asterisk matches any number of any characters, double asterisk mark matches all possible folders depths.

| Parameter name | Description |
|---|---|
| Prompt | Objects to be included during job |
| Symbol | odi.includeObjectsFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - **/Scen:* - A default filter that matches all scenarios from the input files<br>- scenario_folder/Scen:table_to* - Allows the scanner to process only the scenarios with names that start with 'table_to' and are in the scenario_folder<br>- */**/Pack:* - Process scenarios for every package in every project (skips scenarios for mappings, procedures and so on)<br>- PROJ01/My Folder/*/Pack:* - Process scenarios for packages inside child folders of 'My Folder' in PROJ01 |

## ODI context name

Name of ODI context used to resolve physical locations (eg. database schemas) used in job

| Parameter name | Description |
|---|---|
| Prompt | ODI context name |
| Symbol | odi.contextName |
| Type | STRING |
| Required | Yes |
| Default value | Global |
| Example values | - Global<br>- Production<br>- Development<br>- ABC |

## ODI repository Connection registry entry

The 2 segment entry pointing to Connection registry. In this location you should have CONNECTION_REGISTRY/HOST/DB/db.properties file containing:

- jdbc url
- user

- password

| Parameter name | Description |
|---|---|
| Prompt | ODI repository Connection registry entry |
| Symbol | odi.exporterRepositoryConnectionRegistryEntry |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - HOST.DB |

## ODI Studio password

The user password to Odi Studio installation

| Parameter name | Description |
|---|---|
| Prompt | ODI Studio password |
| Symbol | odi.exporterOdiPassword |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - myOdiStudioPassword |

## ODI Studio user

The name of the Odi Studio user

| Parameter name | Description |
|---|---|
| Prompt | ODI Studio user |
| Symbol | odi.exporterOdiUser |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - data_manager<br>- steward |

## Oracle_Home folder

Path to the Oracle_Home folder of the ODI Client installation from Oracle

| Parameter name | Description |
|---|---|
| Prompt | Oracle_Home folder |
| Symbol | odi.exporterOracleHome |
| Type | STRING |
| Required | No |
| Default value | /opt/Oracle/Middleware/Oracle_Home |
| Example values | - /opt/Oracle/Middleware/Oracle_Home<br>- /home/user/Oracle/Middleware/Oracle_Home |

## WORK REPO name

The name of the WORK_REPO with user projects

| Parameter name | Description |
|---|---|
| Prompt | WORK REPO name |
| Symbol | odi.exporterWorkRepo |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - WORKREP |

## Supported Knowledge Modules, Package Steps and Technologies

- BMS CKM Oracle - Knowledge Module
- BMS IKM SQL Control Append - Knowledge Module
- BMS IKM SQL Control Append L0 [LAST IMAGE RECYCLE] - Knowledge Module
- CKM Oracle - Knowledge Module
- CKM SQL - Knowledge Module
- IKM Oracle Control Append - Knowledge Module
- IKM Oracle Incremental Update - Knowledge Module
- IKM Oracle Incremental Update (MERGE) - Knowledge Module
- IKM Oracle Insert - Knowledge Module
- IKM Oracle Merge - Knowledge Module
- IKM Oracle Slowly Changing Dimension - Knowledge Module
- IKM SQL Control Append - Knowledge Module

- IKM SQL Insert - Knowledge Module
- IKM SQL Load - Knowledge Module
- LKM File to Oracle (Built-In) - Knowledge Module
- LKM File to SQL - Knowledge Module
- LKM Oracle to File (datapump) - Knowledge Module
- LKM Oracle to Oracle (Built-In) - Knowledge Module
- LKM Oracle to Oracle (DBLINK) - Knowledge Module
- LKM Oracle to Oracle Pull (DB Link) - Knowledge Module
- LKM SQL to File - Knowledge Module
- LKM SQL to Oracle - Knowledge Module
- LKM SQL to Oracle (Built-In) - Knowledge Module
- LKM SQL to SQL - Knowledge Module
- LKM SQL to SQL (Built-In) - Knowledge Module
- OdiEnterpriseDataQuality - Package Step
- OdiInvokeWebService - Package Step
- OdiSendMail - Package Step
- OdiSqlUnload - Package Step
- OdiStartScen - Package Step
- OdiWaitForChildSession - Package Step
- OdiWaitForData - Package Step
- OdiWaitForLogData - Package Step
- FILE (FTP) - Technology
- GENERIC_SQL (Oracle) - Technology
- ORACLE (Oracle) - Technology
- SNOWFLAKE (Snowflake) - Technology
- UNDEFINED (Generic) - Technology
- UNKNOWN (Generic) - Technology

# Oracle Synonym Resolver

A module for resolve table synonyms in Oracle. A synonym itself has no table definition, so this module is used to match synonyms with their corresponding tables.

The Oracle Synonym Resolver has no configurable parameters defined.

# Python Evaluator

A module for evaluating Python statements.

The Python Evaluator includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Notebooks Python Modules Path

The path to the directory containing custom Python user modules.

| Parameter name | Description |
|---|---|
| Prompt | Notebooks Python Modules Path |
| Symbol | python.userModulesDirectory |
| Type | DIR |
| Required | No |
| Default value | /ext/python/modules |
| Example values | - Y:\etc\user_modules<br>- /opt/etc/user_modules |

## Python default variables values file

File format consist of [GLOBAL] or/and custom function section.

---

You can define variables definitions: [GLOBAL] VariableName = VariableExpression

VariableName that contains only standard alphanumeric characters does not require any quotes. If it contains any special characters, it requires double-quotes (").

Example of variables definition: [GLOBAL] a = 42 b = 7 c = a < b ? a + 7 : b - 7 // It's 49 d = e(1,2) + 1 // Expression use call to function. "User::table" = 'table' || "User::tableSuffix" // It's string table concatenated with value of User::tableSuffix variable

---

You can define functions definitions: [GLOBAL] FunctionName = FunctionExpression

FunctionName can be unquoted if contains only standard alphanumeric characters but needs to be quoted with double quoted (") if contains any special character. FunctionName must contain list of arguments. List of argument is defined inside "(" ")" characters and consists of "?" sings separated by "," (one "?" sign for each function argument).

Example of function definition: [GLOBAL] a(?) = 1 a(?,?) = 2 b(?) = a(1) + 1 // Expression use call to another function. c(?,?) = d + 2 // Expression use reference to variable.

---

Additional sections can be provided to match only specific functions. To match an overloaded function please provide placeholders for its arguments. Inside the matched section you can also reference matched function arguments. Example of custom function sections:

- [host.db.schema.func(x)]
- z=x
- [host.db.schema.func(x,y)]
- z=x+y

---

To escape special characters such as \n \t use the \ character. For example, to define E:\file \tgtParameterized.csv, enter E:\\file\\tgtParameterized.csv

---

On the right side of assignments you can use simple expression language that can use values of already defined variables and functions or variables from current job context.

Supported expressions (case insensitive) are:

- literals
  - integer: 42
  - string: 'str'
  - string with quoted single quote: 'str\''
  - boolean: true, false
- equals: 42 == 7 -> FALSE
- not equals: 42 != 7 -> TRUE
- concatenations: 'str1' + 'str2' -> 'str1str2'
- function calls: Replace('abc', 'b', 'd') -> 'adc'
- tertiary operator:
  - 42 == 7 ? 'a' : 'b' -> 'b'
  - 42 != 7 ? 'a' : 'b' -> 'a'

Supported functions:

- Hash(str, maxOutputLen) - MD5, maxOutputLen between 1 and 32 inclusive
  - Hash('abcdefgh', 4) -> 'E8DC'
- Replace(str, from, to)
  - Replace('abc', 'b', 'D') -> 'aDc'

- ReplaceRegexp(str, regex, replacement) - replaces with Java regular expressions
  - ReplaceRegexp('abcde', 'b.*', 'f') -> 'af'
  - ReplaceRegexp('graph_id20', 'id(\d+)', '$1') -> 'graph_20'
- StringLengthLimit(str, limit, hashSize) - returns str[0..limit-hashSize] + Hash( str, hashSize )
  - StringLengthLimit('abcc', 3, 2) -> 'a26'
  - StringLengthLimit('abcc', 4, 2) -> 'abcc'
- StringLengthLimit(str, limit) - hashSize has default size 8
  - StringLengthLimit('abcdabcdabcd', 10) -> 'abE340600C'
- RegexpMatch(pattern, str) - tests if str matches pattern
  - pattern uses standard Java 8 regular expressions: https://docs.oracle.com/javase/8/docs/api/java/util/regex/Pattern.html
  - RegexpMatch('[A-Za-z]+','Abcd') -> TRUE
- Upper(str) - returns uppercased str
  - Upper('Abc') -> 'ABC'
- Lower(str) - return lowercased str
  - Lower('aBC') -> 'abc'
- Date(text,format) - returns object representing date with provided format.
  - Formats uses standard Java 8 SimpleDateFormat: https://docs.oracle.com/javase/8/docs/api/java/text/SimpleDateFormat.html
  - It is designed for default values usage
  - ex. Date('2017-10-31', 'yyyy-MM-dd')
- Contains/ContainsIgnoreCase(stack, needle) - tests if stack string contains needle string
  - Contains('abc', 'ab') -> TRUE
  - Contains('abc', 'AB') -> FALSE
  - ContainsIgnoreCase('abc', 'AB') -> TRUE
  - Contains('stack', 'needle') -> FALSE
- RtfToText(str) - return clean text from rtf string
  - RtfToText('{\rtf1\ansi{\fonttbl\f0\fswiss Helvetica;}\f0\pard text\par }') -> text

Supported comments:

- Single line: \ comment
- Multi line: /* comment */

| Parameter name | Description |
|---|---|
| Prompt | Python default variables values file |
| Symbol | python.defaultValuesFilePath |
| Type | FILE |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Y:\etc\pythonDefaultVariableValues.txt<br>- /opt/etc/pythonDefaultVariableValues.txt |

# Salesforce Reader

A module for extracting Salesforce metadata.

The Salesforce Reader includes the following parameters:

## Authentication mode

Choose the authentication mode to read metadata from Salesforce.

| Parameter name | Description |
|---|---|
| Prompt | Authentication mode |
| Symbol | salesforce.authMode |
| Type | MODULE_MODE |
| Required | No |
| Default value | OAuth 2.0 Refresh Token Flow |
| Example values | - OAuth 2.0 Username-Password Flow<br>- OAuth 2.0 Refresh Token Flow<br>- Standard Password Flow |

## Proxy Host

A hostname or an IP address of the HTTP(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy Host |
| Symbol | salesforce.proxyHost |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - 88.99.102.102 or https://myproxy |

## Proxy Password

A password for the HTTP(s) proxy server authentication

| Parameter name | Description |
|---|---|
| Prompt | Proxy Password |
| Symbol | salesforce.proxyPassword |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - abc123 |

## Proxy Port

A port number of the HTTP(s) proxy server

| Parameter name | Description |
|---|---|
| Prompt | Proxy Port |
| Symbol | salesforce.proxyPort |
| Type | INTEGER |
| Required | No |
| Default value | 0 |
| Example values | - 30408 |

## Proxy Username

A username for the HTTP(s) proxy server authentication

| Parameter name | Description |
|---|---|
| Prompt | Proxy Username |
| Symbol | salesforce.proxyUserName |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - abc |

## REST API base URI

Use the baseURI to connect to the Salesforce REST API.

| Parameter name | Description |
|---|---|
| Prompt | REST API base URI |
| Symbol | salesforce.restApiBaseURI |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - https://infa8-dev-ed.my.salesforce.com/ |

## REST API Consumer Key

The ConsumerKey of the user that you connect to the Salesforce REST API with.

| Parameter name | Description |
|---|---|
| Prompt | REST API Consumer Key |
| Symbol | salesforce.restApiConsumerKey |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - restApiConsumerKey |

## REST API Consumer Secret

The ConsumerSecret of the user that you use to connect to the Salesforce REST API with.

| Parameter name | Description |
|---|---|
| Prompt | REST API Consumer Secret |
| Symbol | salesforce.restApiConsumerSecret |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - restApiConsumerSecret |

## REST API password

The password of the user that you use to connect to the Salesforce REST API with.

| Parameter name | Description |
|---|---|
| Prompt | REST API password |
| Symbol | salesforce.restApiPassword |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - secret123 |

## REST API Refresh Token

A refresh token that you generate in the Salesforce resource. The token uses the consumer key and the consumer secret

| Parameter name | Description |
|---|---|
| Prompt | REST API Refresh Token |
| Symbol | salesforce.refreshToken |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - eyJhbGciOiJSUzI1NiIsInR5cCI6IkpXVCJ9 |

## REST API Security Token

The Security token that you use to connect to the Salesforce REST API with.

| Parameter name | Description |
|---|---|
| Prompt | REST API Security Token |
| Symbol | salesforce.restApiSecurityToken |
| Type | SENSITIVE |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - restApiSecurityToken |

## REST API user

The user that you use to connect to the Salesforce REST API with.

| Parameter name | Description |
|---|---|
| Prompt | REST API user |
| Symbol | salesforce.restApiUser |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - user1 |

## Salesforce Application Filters

Salesforce application filters that you use to apply filter to Salesforce organizations.As input, you should have a single module parameter with syntax such as: (Include|Exclude) : ObjectType (: ObjectPattern)? The pattern can include the following wildcard characters: "*" - asterisk for 0 or more characters (any characters) "?" - question mark for exactly 1 character (any character) As a result, the Salesforce parameter definition in the property file looks as follows: salesforceFilter.0=Include : Object : * salesforceFilter.1=Exclude : Object : test* Meaning: - Take everything from Type Object, but skip all Objects the name of which begins with "test". As default, you can set the filter as: salesforceFilter.0=Include : * Meaning: - Take everything from all ObjectTypes that the scanner has access to. Other example: salesforceFilter.1=Include : Object : Finance* salesforceFilter.2=Include : ListView : Finance* salesforceFilter.3=Exclude : Object : FinanceInPoland salesforceFilter.4=Exclude : ListView : FinanceInPoland salesforceFilter.5=Include : Trigger : *Schedule* salesforceFilter.6=Exclude : Trigger : Tes? Meaning: - Take All Objects that begin with "Finance" and exclude the Object named FinanceInPoland from them. - Take All ListView that begin with "Finance" and exclude the ListView named FinanceInPoland from them. - Take All Trigger that contain the name "Schedule" and exclude the Trigger name which starts with Tes+Any one character from them

| Parameter name | Description |
|---|---|
| Prompt | Salesforce Application Filters |
| Symbol | salesforce.inputFilters |
| Type | STRINGLIST |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | Include : * |
| Example values | - Include : Object : test* Exclude : ListView : prod* Include : * : ?rod Exclude: Trigger |

### Supported Organization is root asset, contains objects, list views, trigger assets

- Organization - An Organization asset is a root asset of Salesforce.
- Object - An Object asset contains fields and their properties.
- List View - A List View asset contains list view columns and their properties.
- Trigger - A Trigger asset is a single object that contains code.

# SAS EGP Reader

Reads EGP archives with SAS programs.

The SAS EGP Reader includes the following parameters:

### Allow Sas Annotations

Allow Sas comment /@ @/ to be interpreted as annotation.

| Parameter name | Description |
|---|---|
| Prompt | Allow Sas Annotations |
| Symbol | sasEgp.allowAnnotations |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

### Auto Default Values

For MACRO variables with unknown value use its name by default value and log INFO message instead of WARN.

| Parameter name | Description |
|---|---|
| Prompt | Auto Default Values |
| Symbol | sasEgp.autoDefaultValues |

| Parameter name | Description |
|---|---|
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Context

Context for sas objects (ie. datasets, views). It is used for searching as well as for creating database objects. Database objects are stored and searched by suffix. Use of a context with dot-separated segments allows the specification of default values for host, database and schema. These are only default values and are overridden by explicit qualification of object references in a query. Context must consist of three segments - (e.g., HOST.DBNAME.LIBNAME) - these segments are treated as default host and default database. Specifying other number of segments causes processing error.

| Parameter name | Description |
|---|---|
| Prompt | Context |
| Symbol | sasEgp.sqlContext |
| Type | STRING |
| Required | Yes |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Server01.internal.WORK<br>- Server01.external.USERLIB |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | sasEgp.connections |
| Type | PROPERTIES |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Default remote library name

Remote library is used as a default library name for assets created within RSUBMIT block.

| Parameter name | Description |
|---|---|
| Prompt | Default remote library name |
| Symbol | sasEgp.defaultRemoteLibrary |
| Type | STRING |
| Required | No |
| Default value | RWORK |
| Example values | - RWORK<br>- REMOTEWORK |

## Default variables values file

Path to the file with Macro variables values. File may start with [GLOBAL] section. Then, file must be split into FILE-sections. Each file section can contain ANY number of lines with default values like in example below. [FILE: 'filename'] VAR &A='MMDDYY' VAR &B='MMYY'

| Parameter name | Description |
|---|---|
| Prompt | Default variables values file |
| Symbol | sasEgp.defaultValuesFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\defaultVariableValues.txt<br>- /opt/etc/defaultVariableValues.txt |

## Input files

The paths of all EGP files that should be processed.Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | sasEgp.inputFiles |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input*.egp<br>- /opt/input/*.egp |

## Libraries to exclude

Four- to two-segment library patterns to exclude from the current resource. If the host or database is not present in the filter, their names are obtained from the SAS context. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Libraries to exclude |
| Symbol | sasEgp.libraryExcludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - Lib1.Exclude<br>- Exclude*.T?<br>- Db.Lib1.Exclude<br>- H.Db.Lib1.Exclude |

## Libraries to include

Four- to two-segment library patterns to include in the current resource. If the host or database is not present in the filter, their names are obtained from the SAS context. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Libraries to include |
| Symbol | sasEgp.libraryIncludeFilter |

| Parameter name | Description |
|---|---|
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - Lib1.S<br>- Lib*.T?<br>- Db.Lib1.S<br>- H.Db.Lib1.S |

## List of temporary library patterns

The patterns of libnames where datasets are stored temporarily. In this case dataset WORK.D will have a qualification: "WORK-filename".#TMP*[UNIQUE*INTEGER]_DEach entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | List of temporary library patterns |
| Symbol | sasEgp.temporaryLibrariesPatterns |
| Type | STRINGLIST |
| Required | No |
| Default value | WORK |
| Example values | - WORK-*<br>- TEMP-LIBRARY |

## Log each step in debug log file

Indicates if Sas Steps may be added to debug log

| Parameter name | Description |
|---|---|
| Prompt | Log each step in debug log file |
| Symbol | sasEgp.fullDebug |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Macro definitions files

The paths to files with macro definitions. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Macro definitions files |
| Symbol | sasEgp.mautosFiles |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\mautos\MyMacros.sas |

## Maximum loop counter

Value to define maximum number of times every loop can be executed (to prevent hanging Scanner on infinite loops).

| Parameter name | Description |
|---|---|
| Prompt | Maximum loop counter |
| Symbol | sasEgp.maxLoopCounter |
| Type | STRING |
| Required | No |
| Default value | 100 |
| Example values | - 10<br>- 20 |

## Maximum macro execution depth

Value to define maximum size of macro call stack trace.

| Parameter name | Description |
|---|---|
| Prompt | Maximum macro execution depth |
| Symbol | sasEgp.maxMacroExecutionDepth |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | 1 |
| Example values | - 10<br>- 20 |

## Maximum macro loop counter

Value to define maximum number of times every macro loop can be executed (to prevent hanging SAS scanner on infinite loops).

| Parameter name | Description |
|---|---|
| Prompt | Maximum macro loop counter |
| Symbol | sasEgp.maxMacroLoopCounter |
| Type | STRING |
| Required | No |
| Default value | 1 |
| Example values | - 10<br>- 20 |

## PatternReplacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | sasEgp.patternReplacerFile |
| Type | FILE |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

## Programs excluded from skipping macro calls

List of programs which should omit skip macro enforcement with sasSkipMacroCalls

| Parameter name | Description |
|---|---|
| Prompt | Programs excluded from skipping macro calls |
| Symbol | sasEgp.programExcludedFromMacroSkipping |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - SasProgram.sas |

## SAS-Script directory paths to local filesystem paths

List of mappings from directory location paths in SAS scripts to directory paths in local filesystem.

| Parameter name | Description |
|---|---|
| Prompt | SAS-Script directory paths to local filesystem paths |
| Symbol | sasEgp.dirToDirMappings |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - /sas/on/prod/directory/scripts/<br>- /feed-sas/input/scripts/ |

### Scripts Root Directory Path

Root directory of input files. Relative paths of input files resolve in the following location: [Input root directory]. Absolute paths from input files must start from the [Input root directory] to be included into processing. [Input root directory] will not be visible on the output objects structure.

| Parameter name | Description |
|---|---|
| Prompt | Scripts Root Directory Path |
| Symbol | sasEgp.inputRoot |
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\inputs\<br>- /opt/feed-etl/ |

### Skip macro calls

Indicates if SAS macro calls must be skipped from code generation

| Parameter name | Description |
|---|---|
| Prompt | Skip macro calls |
| Symbol | sasEgp.skipMacroCalls |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

# SAS Evaluator for JCL

A module for evaluating SAS statements embedded in JCL jobs.

The SAS Evaluator for JCL includes the following parameters:

## Auto Default Values

For MACRO variables with unknown value use its name by default value and log INFO message instead of WARN.

| Parameter name | Description |
|---|---|
| Prompt | Auto Default Values |
| Symbol | sas.autoDefaultValues |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Context

Context for sas objects (ie. datasets, views). It is used for searching as well as for creating database objects. Database objects are stored and searched by suffix. Use of a context with dot-separated segments allows the specification of default values for host, database and schema. These are only default values and are overridden by explicit qualification of object references in a query. Context must consist of three segments - (e.g., HOST.DBNAME.LIBNAME) - these segments are treated as default host and default database. Specifying other number of segments causes processing error.

| Parameter name | Description |
|---|---|
| Prompt | Context |
| Symbol | sas.sqlContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - Server01.internal.WORK<br>- Server01.external.USERLIB |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1

- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | sas.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - Ora01=ORACLE:prod-host.oraprod.core - The name of the connection is "Ora01" and "prod", "oraprod", and "core" are the respective host, database, and schema used by the connection<br>- Fund_TRGT=MSSQL:kevin@prod-host - "kevin" is the user and "prod-host" is the database of type MSSQL |

## Default variables values file

Path to the file with Macro variables values. File may start with [GLOBAL] section. Then, file must be split into FILE-sections. Each file section can contain ANY number of lines with default values like in example below. [FILE: 'filename'] VAR &A='MMDDYY' VAR &B='MMYY'

| Parameter name | Description |
|---|---|
| Prompt | Default variables values file |
| Symbol | sas.defaultValuesFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\defaultVariableValues.txt<br>- /opt/etc/defaultVariableValues.txt |

## Exclude input files

The paths of files to be excluded from processing. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Exclude input files |
| Symbol | sas.fileExcludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\sas\removeMe.sas<br>- /opt/input/sas/remove_Me_Too.sas |

## Input files

The paths of files to be processed. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | sas.fileIncludeFilter |
| Type | FILELIST |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\sql\224_magic.sas<br>- /opt/input/sql/225_more_magic.sas |

## List of temporary library patterns

The patterns of libnames where datasets are stored temporarily. In this case dataset WORK.D will have a qualification: "WORK-filename".#TMP*[UNIQUE*INTEGER]_DEach entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | List of temporary library patterns |
| Symbol | sas.temporaryLibrariesPatterns |
| Type | STRINGLIST |
| Required | No |
| Default value | WORK |
| Example values | - WORK-*<br>- TEMP-LIBRARY |

## Log each step in debug log file

Indicates if Sas Steps may be added to debug log

| Parameter name | Description |
|---|---|
| Prompt | Log each step in debug log file |
| Symbol | sas.fullDebug |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Macro definitions files

The paths to files with macro definitions. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Macro definitions files |
| Symbol | sas.mautosFiles |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\mautos\MyMacros.sas |

## Maximum loop counter

Value to define maximum number of times every loop can be executed (to prevent hanging Scanner on infinite loops).

| Parameter name | Description |
|---|---|
| Prompt | Maximum loop counter |
| Symbol | sas.maxLoopCounter |
| Type | STRING |
| Required | No |
| Default value | 100 |
| Example values | - 10<br>- 20 |

## Maximum macro execution depth

Value to define maximum size of macro call stack trace.

| Parameter name | Description |
|---|---|
| Prompt | Maximum macro execution depth |
| Symbol | sas.maxMacroExecutionDepth |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | 1 |
| Example values | - 10<br>- 20 |

## Maximum macro loop counter

Value to define maximum number of times every macro loop can be executed (to prevent hanging SAS scanner on infinite loops).

| Parameter name | Description |
|---|---|
| Prompt | Maximum macro loop counter |
| Symbol | sas.maxMacroLoopCounter |
| Type | STRING |
| Required | No |
| Default value | 1 |
| Example values | - 10<br>- 20 |

## PatternReplacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | sas.patternReplacerFile |
| Type | FILE |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

### Sas evaluator

Enable the use of this module by selecting "Process". Selecting "Skip" disables the module.

| Parameter name | Description |
|---|---|
| Prompt | Sas evaluator |
| Symbol | sas.shouldProcessSas |
| Type | MODULE_MODE |
| Required | No |
| Default value | Skip |
| Example values | - Skip<br>- Process |

### Scripts Root Directory Path

Root directory of input files. Relative paths of input files resolve in the following location: [Input root directory]. Absolute paths from input files must start from the [Input root directory] to be included into processing. [Input root directory] will not be visible on the output objects structure.

| Parameter name | Description |
|---|---|
| Prompt | Scripts Root Directory Path |
| Symbol | sas.inputRoot |
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\inputs\<br>- /opt/feed-etl/ |

# SAS Reader

Reads SAS files.

The SAS Reader includes the following parameters:

## Allow Sas Annotations

Allow Sas comment /@ @/ to be interpreted as annotation.

| Parameter name | Description |
|---|---|
| Prompt | Allow Sas Annotations |
| Symbol | sas.allowAnnotations |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Auto Default Values

For MACRO variables with unknown value use its name by default value and log INFO message instead of WARN.

| Parameter name | Description |
|---|---|
| Prompt | Auto Default Values |
| Symbol | sas.autoDefaultValues |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Context

Context for sas objects (ie. datasets, views). It is used for searching as well as for creating database objects. Database objects are stored and searched by suffix. Use of a context with dot-separated segments allows the specification of default values for host, database and schema. These are only default values and are overridden by explicit qualification of object references in a query. Context must consist of three segments - (e.g., HOST.DBNAME.LIBNAME) - these segments are treated as default host and default database. Specifying other number of segments causes processing error.

| Parameter name | Description |
|---|---|
| Prompt | Context |
| Symbol | sas.sqlContext |
| Type | STRING |
| Required | Yes |
| Default value | N/A |
| Example values | - Server01.internal.WORK<br>- Server01.external.USERLIB |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | sas.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Default remote library name

Remote library is used as a default library name for assets created within RSUBMIT block.

| Parameter name | Description |
|---|---|
| Prompt | Default remote library name |
| Symbol | sas.defaultRemoteLibrary |
| Type | STRING |
| Required | No |
| Default value | RWORK |
| Example values | - RWORK<br>- REMOTEWORK |

## Default variables values file

Path to the file with Macro variables values. File may start with [GLOBAL] section. Then, file must be split into FILE-sections. Each file section can contain ANY number of lines with default values like in example below. [FILE: 'filename'] VAR &A='MMDDYY' VAR &B='MMYY'

| Parameter name | Description |
|---|---|
| Prompt | Default variables values file |
| Symbol | sas.defaultValuesFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\defaultVariableValues.txt<br>- /opt/etc/defaultVariableValues.txt |

## Exclude input files

The paths of files to be excluded from processing. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Exclude input files |
| Symbol | sas.fileExcludeFilter |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\sas\removeMe.sas<br>- /opt/input/sas/remove_Me_Too.sas |

## Input files

The paths of files to be processed. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | sas.fileIncludeFilter |
| Type | FILELIST |

| Parameter name | Description |
|---|---|
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input\sql\224_magic.sas<br>- /opt/input/sql/225_more_magic.sas |

## Libraries to exclude

Four- to two-segment library patterns to exclude from the current resource. If the host or database is not present in the filter, their names are obtained from the SAS context. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Libraries to exclude |
| Symbol | sas.libraryExcludeFilter |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - Lib1.Exclude<br>- Exclude*.T?<br>- Db.Lib1.Exclude<br>- H.Db.Lib1.Exclude |

## Libraries to include

Four- to two-segment library patterns to include in the current resource. If the host or database is not present in the filter, their names are obtained from the SAS context. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Libraries to include |
| Symbol | sas.libraryIncludeFilter |
| Type | STRINGLIST |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Lib1.S<br>- Lib*.T?<br>- Db.Lib1.S<br>- H.Db.Lib1.S |

## List of temporary library patterns

The patterns of libnames where datasets are stored temporarily. In this case dataset WORK.D will have a qualification: "WORK-filename".#TMP*[UNIQUE*INTEGER]_DEach entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | List of temporary library patterns |
| Symbol | sas.temporaryLibrariesPatterns |
| Type | STRINGLIST |
| Required | No |
| Default value | WORK |
| Example values | - WORK-*<br>- TEMP-LIBRARY |

## Log each step in debug log file

Indicates if Sas Steps may be added to debug log

| Parameter name | Description |
|---|---|
| Prompt | Log each step in debug log file |
| Symbol | sas.fullDebug |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Macro definitions files

The paths to files with macro definitions. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Macro definitions files |
| Symbol | sas.mautosFiles |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\mautos\MyMacros.sas |

## Maximum loop counter

Value to define maximum number of times every loop can be executed (to prevent hanging Scanner on infinite loops).

| Parameter name | Description |
|---|---|
| Prompt | Maximum loop counter |
| Symbol | sas.maxLoopCounter |
| Type | STRING |
| Required | No |
| Default value | 100 |
| Example values | - 10<br>- 20 |

## Maximum macro execution depth

Value to define maximum size of macro call stack trace.

| Parameter name | Description |
|---|---|
| Prompt | Maximum macro execution depth |
| Symbol | sas.maxMacroExecutionDepth |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | 1 |
| Example values | - 10<br>- 20 |

## Maximum macro loop counter

Value to define maximum number of times every macro loop can be executed (to prevent hanging SAS scanner on infinite loops).

| Parameter name | Description |
|---|---|
| Prompt | Maximum macro loop counter |
| Symbol | sas.maxMacroLoopCounter |
| Type | STRING |
| Required | No |
| Default value | 1 |
| Example values | - 10<br>- 20 |

## PatternReplacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | sas.patternReplacerFile |
| Type | FILE |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

## Programs excluded from skipping macro calls

List of programs which should omit skip macro enforcement with sasSkipMacroCalls

| Parameter name | Description |
|---|---|
| Prompt | Programs excluded from skipping macro calls |
| Symbol | sas.programExcludedFromMacroSkipping |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - SasProgram.sas |

## SAS-Script directory paths to local filesystem paths

List of mappings from directory location paths in SAS scripts to directory paths in local filesystem.

| Parameter name | Description |
|---|---|
| Prompt | SAS-Script directory paths to local filesystem paths |
| Symbol | sas.dirToDirMappings |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - /sas/on/prod/directory/scripts/<br>- /feed-sas/input/scripts/ |

## Scripts Root Directory Path

Root directory of input files. Relative paths of input files resolve in the following location: [Input root directory]. Absolute paths from input files must start from the [Input root directory] to be included into processing. [Input root directory] will not be visible on the output objects structure.

| Parameter name | Description |
|---|---|
| Prompt | Scripts Root Directory Path |
| Symbol | sas.inputRoot |
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\inputs\<br>- /opt/feed-etl/ |

## Skip macro calls

Indicates if SAS macro calls must be skipped from code generation

| Parameter name | Description |
|---|---|
| Prompt | Skip macro calls |
| Symbol | sas.skipMacroCalls |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Supported SAS Procedures

- Append
- Arima
- Compare
- Contents
- Copy
- Corr
- Datasets
- Download
- Ds2
- Expand

- Export
- Fcmp
- Format
- Freq
- Genmod
- Hp4Score
- Hpforest
- Http
- Import
- Logistic
- Means
- Model
- Nlin
- Nlp
- Plm
- Print
- Rank
- Reg
- Report
- Score
- Sql
- Sort
- Stdize
- SurveySelect
- Tabulate
- Template
- Transpose
- Univariate
- UploadDataSet
- UploadLibrary
- X11
- Json
- Gchart
- Gslide

# SQL Evaluator

A module for evaluating SQL statements embedded in ETL processes. This prevents the breaking of data lineage at table components in a mapping.

The SQL Evaluator includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema

- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | sql.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Default variables values file

File format consist of [GLOBAL] or/and custom function section.

---

You can define variables definitions: [GLOBAL] VariableName = VariableExpression

VariableName that contains only standard alphanumeric characters does not require any quotes. If it contains any special characters, it requires double-quotes (").

Example of variables definition: [GLOBAL] a = 42 b = 7 c = a < b ? a + 7 : b - 7 // It's 49 d = e(1,2) + 1 // Expression use call to function. "User::table" = 'table' || "User::tableSuffix" // It's string table concatenated with value of User::tableSuffix variable

---

You can define functions definitions: [GLOBAL] FunctionName = FunctionExpression

FunctionName can be unquoted if contains only standard alphanumeric characters but needs to be quoted with double quoted (") if contains any special character. FunctionName must contain list of arguments. List of argument is defined inside "(" ")" characters and consists of "?" sings separated by "," (one "?" sign for each function argument).

Example of function definition: [GLOBAL] a(?) = 1 a(?,?) = 2 b(?) = a(1) + 1 // Expression use call to another function. c(?,?) = d + 2 // Expression use reference to variable.

---

Additional sections can be provided to match only specific functions. To match an overloaded function please provide placeholders for its arguments. Inside the matched section you can also reference matched function arguments. Example of custom function sections:

- [host.db.schema.func(x)]

- z=x
- [host.db.schema.func(x,y)]
- z=x+y

---

To escape special characters such as \n \t use the \ character. For example, to define E:\file \tgtParameterized.csv, enter E:\\file\\tgtParameterized.csv

---

To provide multiple default values please use the following syntax: [GLOBAL] (x)=(first-value),(second-value),... You can also correlate variables to ensure that only specific pairs of values will be processed: [GLOBAL] (x,y)=(first-x-value,first-y-value),(second-x-value,second-y-value),... Please keep in mind that providing multiple default values or correlations can significantly slow down your job.

---

On the right side of assignments you can use simple expression language that can use values of already defined variables and functions or variables from current job context.

Supported expressions (case insensitive) are:

- literals
  - integer: 42
  - string: 'str'
  - string with quoted single quote: 'str\''
  - boolean: true, false
- equals: 42 == 7 -> FALSE
- not equals: 42 != 7 -> TRUE
- concatenations: 'str1' + 'str2' -> 'str1str2'
- function calls: Replace('abc', 'b', 'd') -> 'adc'
- tertiary operator:
  - 42 == 7 ? 'a' : 'b' -> 'b'
  - 42 != 7 ? 'a' : 'b' -> 'a'

Supported functions:

- Hash(str, maxOutputLen) - MD5, maxOutputLen between 1 and 32 inclusive
  - Hash('abcdefgh', 4) -> 'E8DC'
- Replace(str, from, to)
  - Replace('abc', 'b', 'D') -> 'aDc'
- ReplaceRegexp(str, regex, replacement) - replaces with Java regular expressions
  - ReplaceRegexp('abcde', 'b.*', 'f') -> 'af'
  - ReplaceRegexp('graph_id20', 'id(\d+)', '$1') -> 'graph_20'
- StringLengthLimit(str, limit, hashSize) - returns str[0..limit-hashSize] + Hash( str, hashSize )
  - StringLengthLimit('abcc', 3, 2) -> 'a26'
  - StringLengthLimit('abcc', 4, 2) -> 'abcc'
- StringLengthLimit(str, limit) - hashSize has default size 8
  - StringLengthLimit('abcdabcdabcd', 10) -> 'abE340600C'

- RegexpMatch(pattern, str) - tests if str matches pattern
  - pattern uses standard Java 8 regular expressions: https://docs.oracle.com/javase/8/docs/api/java/util/regex/Pattern.html
  - RegexpMatch('[A-Za-z]+','Abcd') -> TRUE
- Upper(str) - returns uppercased str
  - Upper('Abc') -> 'ABC'
- Lower(str) - return lowercased str
  - Lower('aBC') -> 'abc'
- Date(text,format) - returns object representing date with provided format.
  - Formats uses standard Java 8 SimpleDateFormat: https://docs.oracle.com/javase/8/docs/api/java/text/SimpleDateFormat.html
  - It is designed for default values usage
  - ex. Date('2017-10-31', 'yyyy-MM-dd')
- Contains/ContainsIgnoreCase(stack, needle) - tests if stack string contains needle string
  - Contains('abc', 'ab') -> TRUE
  - Contains('abc', 'AB') -> FALSE
  - ContainsIgnoreCase('abc', 'AB') -> TRUE
  - Contains('stack', 'needle') -> FALSE
- RtfToText(str) - return clean text from rtf string
  - RtfToText('{\rtf1\ansi{\fonttbl\f0\fswiss Helvetica;}\f0\pard text\par }') -> text

Supported comments:

- Single line: \ comment
- Multi line: /* comment */

| Parameter name | Description |
|---|---|
| Prompt | Default variables values file |
| Symbol | sql.defaultValuesFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\defaultVariableValues.txt<br>- /opt/etc/defaultVariableValues.txt |

## Directory paths to local filesystem paths

List of mappings from directory location paths in sql scripts to directory paths in local filesystem.

| Parameter name | Description |
|---|---|
| Prompt | Directory paths to local filesystem paths |
| Symbol | sql.dirToDirMappings |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - /on/prod/directory/scripts/<br>- /opt/input/scripts/ |

## Java Script default variables values file

File format consist of [GLOBAL] or/and custom function section.

---

You can define variables definitions: [GLOBAL] VariableName = VariableExpression

VariableName that contains only standard alphanumeric characters does not require any quotes. If it contains any special characters, it requires double-quotes (").

Example of variables definition: [GLOBAL] a = 42 b = 7 c = a < b ? a + 7 : b - 7 // It's 49 d = e(1,2) + 1 // Expression use call to function. "User::table" = 'table' || "User::tableSuffix" // It's string table concatenated with value of User::tableSuffix variable

---

You can define functions definitions: [GLOBAL] FunctionName = FunctionExpression

FunctionName can be unquoted if contains only standard alphanumeric characters but needs to be quoted with double quoted (") if contains any special character. FunctionName must contain list of arguments. List of argument is defined inside "(" ")" characters and consists of "?" sings separated by "," (one "?" sign for each function argument).

Example of function definition: [GLOBAL] a(?) = 1 a(?,?) = 2 b(?) = a(1) + 1 // Expression use call to another function. c(?,?) = d + 2 // Expression use reference to variable.

---

Additional sections can be provided to match only specific functions. To match an overloaded function please provide placeholders for its arguments. Inside the matched section you can also reference matched function arguments. Example of custom function sections:

- [host.db.schema.func(x)]
- `z=x`
- [host.db.schema.func(x,y)]
- `z=x+y`

---

To escape special characters such as \n \t use the \ character. For example, to define E:\file \tgtParameterized.csv, enter E:\\file\\tgtParameterized.csv

---

On the right side of assignments you can use simple expression language that can use values of already defined variables and functions or variables from current job context.

Supported expressions (case insensitive) are:

- literals
  - integer: 42
  - string: 'str'
  - string with quoted single quote: 'str\''
  - boolean: true, false
- equals: 42 == 7 -> FALSE
- not equals: 42 != 7 -> TRUE
- concatenations: 'str1' + 'str2' -> 'str1str2'
- function calls: Replace('abc', 'b', 'd') -> 'adc'
- tertiary operator:
  - 42 == 7 ? 'a' : 'b' -> 'b'
  - 42 != 7 ? 'a' : 'b' -> 'a'

Supported functions:

- Hash(str, maxOutputLen) - MD5, maxOutputLen between 1 and 32 inclusive
  - Hash('abcdefgh', 4) -> 'E8DC'
- Replace(str, from, to)
  - Replace('abc', 'b', 'D') -> 'aDc'
- ReplaceRegexp(str, regex, replacement) - replaces with Java regular expressions
  - ReplaceRegexp('abcde', 'b.*', 'f') -> 'af'
  - ReplaceRegexp('graph_id20', 'id(\d+)', '$1') -> 'graph_20'
- StringLengthLimit(str, limit, hashSize) - returns str[0..limit-hashSize] + Hash( str, hashSize )
  - StringLengthLimit('abcc', 3, 2) -> 'a26'
  - StringLengthLimit('abcc', 4, 2) -> 'abcc'
- StringLengthLimit(str, limit) - hashSize has default size 8
  - StringLengthLimit('abcdabcdabcd', 10) -> 'abE340600C'
- RegexpMatch(pattern, str) - tests if str matches pattern
  - pattern uses standard Java 8 regular expressions: https://docs.oracle.com/javase/8/docs/api/java/util/regex/Pattern.html
  - RegexpMatch('[A-Za-z]+','Abcd') -> TRUE
- Upper(str) - returns uppercased str
  - Upper('Abc') -> 'ABC'
- Lower(str) - return lowercased str
  - Lower('aBC') -> 'abc'
- Date(text,format) - returns object representing date with provided format.
  - Formats uses standard Java 8 SimpleDateFormat: https://docs.oracle.com/javase/8/docs/api/java/text/SimpleDateFormat.html

- It is designed for default values usage
- ex. Date('2017-10-31', 'yyyy-MM-dd')

- Contains/ContainsIgnoreCase(stack, needle) - tests if stack string contains needle string
  - Contains('abc', 'ab') -> TRUE
  - Contains('abc', 'AB') -> FALSE
  - ContainsIgnoreCase('abc', 'AB') -> TRUE
  - Contains('stack', 'needle') -> FALSE
- RtfToText(str) - return clean text from rtf string
  - RtfToText('{\rtf1\ansi{\fonttbl\f0\fswiss Helvetica;}\f0\pard text\par }') -> text

Supported comments:

- Single line: \ comment
- Multi line: /* comment */

| Parameter name | Description |
|---|---|
| Prompt | Java Script default variables values file |
| Symbol | sql.javaScriptDefaultValuesFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\etc\javaScriptDefaultVariableValues.txt<br>- /opt/etc/javaScriptDefaultVariableValues.txt |

## Log warn on missing variable

Decide about logging warn message when there is usage of variable with unknown value

| Parameter name | Description |
|---|---|
| Prompt | Log warn on missing variable |
| Symbol | sql.warnOnMissingDefaultVariableDefinition |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Maximum function call depth

Defines maximum depth of function calls. When this value is exceeded further recursive calls will be excluded. It prevents from falling into infinite recursion.

| Parameter name | Description |
|---|---|
| Prompt | Maximum function call depth |
| Symbol | sql.maxFunctionCallDepth |
| Type | STRING |
| Required | No |
| Default value | 10 |
| Example values | - 10<br>- 20 |

## Maximum number of loop iterations

Defines maximum number of loop iterations. It prevents from processing statements inside loop more times than provided value.

| Parameter name | Description |
|---|---|
| Prompt | Maximum number of loop iterations |
| Symbol | sql.maxNumberOfLoopIterations |
| Type | STRING |
| Required | No |
| Default value | 100 |
| Example values | - 100<br>- 200 |

## MetaTables Include Filter

When you process PL/SQL statements, MetaDex does not analyze table content by default. If you need to use the content, for example, to process dynamic SQL statements, use the MetaTables Include Filter parameter. The parameter allows MetaDex to prompt the database for required metadata.

| Parameter name | Description |
|---|---|
| Prompt | MetaTables Include Filter |
| Symbol | sql.metaTableIncludeFilter |
| Type | STRINGLIST |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | N/A |
| Example values | - H.DB.USER.META_* - include all tables with prefix META_ from USER schema in database DB and host H<br>- *.*.USER.META_* - include all tables with prefix META_ from USER schema in any database and host |

## Order of connectionless vendors

Comma separated list of vendors. The order of vendors will be taken into account when multiple vendors are capable of query analysis

| Parameter name | Description |
|---|---|
| Prompt | Order of connectionless vendors |
| Symbol | sql.connectionlessParsersOrder |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - MSSQL,DB2,ORACLE<br>- DB2,SNOWFLAKE |

## PatternReplacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | sql.patternReplacerFile |
| Type | FILE |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

## Procedures not to be inlined

Procedure calls not to be inlined. You can expect the full qualification, which is 4 or 5 segments. When a procedure is not inlined, you can see it in the job result. Use inlined procedures for lineage-carrying procedures. Only a procedure or a function that fulfills the following conditions can be not inlined:

1. It accepts scalar parameters and returns scalar value. A scalar value is a simple (non-complex) value, for example, a text or a number. A scalar value is not a cursor or a record.
2. It is not a Table Valued Function.

All other procedures or functions are inlined, regardless of the configuration. When a procedure matches the 'Inline' and 'Skip inline' filters, it does not get inlined. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Procedures not to be inlined |
| Symbol | sql.skipInlineProcedures |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - PROC-H.*.*.*.* - matches calls of non-package procedures from host PROC-H<br>- *.*.*.*.P1 - matches calls of package-procedures named P1 |

## Procedures to be distinguished by their arguments

Procedure calls to be distinguished by their arguments. Full qualification is expected - 4 or 5 segments. Procedure distinguished by their arguments will be analysed for each set of arguments. It is valid only for not inlined procedures. When procedure is matched by 'Distinguish' and 'Not distinguish' filter, it will be not distinguished by their args. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Procedures to be distinguished by their arguments |
| Symbol | sql.distinguishProcedureInstances |
| Type | STRINGLIST |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | * |
| Example values | - PROC-H.*.*.*.* - matches calls of non-package procedures from host PROC-H<br>- *.*.*.*.P1 - matches calls of package-procedures named P1 |

## Procedures to be excluded

Procedure calls to be excluded. Full qualification is expected - 4 or 5 segments. When procedure is matched by 'Include' and 'Exclude' filter, it will be excluded from processing. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Procedures to be excluded |
| Symbol | sql.excludeProcedures |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - OWNER_X.*.* - excludes all calls of procedures from host OWNER_X<br>- *.*.*.*.P1 - excludes all calls of procedures named P1<br>- *.DEBUG* - excludes all calls of procedures from DB which name starts with DEBUG<br>- *.???_DEBUG - excludes all calls of procedures from DB with name starting with 3 any characters followed by _DEBUG |

## Procedures to be included

Procedure calls to be included. Full qualification is expected - 4 or 5 segments. When procedure is matched by 'Include' and 'Exclude' filter, it will be excluded from processing. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Procedures to be included |
| Symbol | sql.includeProcedures |
| Type | STRINGLIST |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | * |
| Example values | - OWNER_X.*.* - includes all calls of procedures from host OWNER_X<br>- *.*.*.*.P1 - includes all calls of procedures named P1<br>- *.DEBUG* - includes all calls of procedures from DB which name starts with DEBUG<br>- *.???_DEBUG - includes all calls of procedures from DB with name starting with 3 any characters followed by _DEBUG |

## Procedures to be inlined

Procedure calls to be inlined. Full qualification is expected - 4 or 5 segments. When a procedure is inlined, you cannot see it in the job result. It is designed to be used for helper procedures. When a procedure matches the 'Inline' and 'Skip inline' filters, it does not get inlined. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Procedures to be inlined |
| Symbol | sql.inlineProcedures |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - PROC-H.*.*.*.* - matches calls of non-package procedures from host PROC-H<br>- *.*.*.*.P1 - matches calls of package-procedures named P1 |

## Procedures to be not distinguished by their arguments

Procedure calls to be not distinguished by their arguments. Full qualification is expected - 4 or 5 segments. Procedure not distinguished by their arguments will be analysed only once. Arguments of this procedure will have unknown values. It is valid only for not inlined procedures. When procedure is matched by 'Distinguish' and 'Not distinguish' filter, it will be not distinguished by their args. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Procedures to be not distinguished by their arguments |
| Symbol | sql.doNotDistinguishProcedureInstances |
| Type | STRINGLIST |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - PROC-H.*.*.*.* - matches calls of non-package procedures from host PROC-H<br>- *.*.*.*.P1 - matches calls of package-procedures named P1 |

## Procedures to be processed as graphs

Procedure calls to be processed as graphs. Full qualification is expected - 4 or 5 segments. Procedure generated as a graph has separate HTML visualisation and image generated. It is valid only for not inlined procedures. When a procedure matches the 'Generate graph' and 'Generate subgraph' filters, it generates as a subgraph. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Procedures to be processed as graphs |
| Symbol | sql.generateGraphsForProcedures |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - PROC-H.*.*.*.* - matches calls of non-package procedures from host PROC-H<br>- *.*.*.*.P1 - matches calls of package-procedures named P1 |

## Procedures to be processed as subgraphs

Procedure calls to be processed as subgraphs. Full qualification is expected - 4 or 5 segments. Procedure generated as a subgraph does not have separate HTML visualisation and image generated.They are attached to the caller HTML/image. It is valid only for not inlined procedures. When a procedure matches the 'Generate graph' and 'Generate subgraph' filters, it generates as a subgraph. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Procedures to be processed as subgraphs |
| Symbol | sql.generateSubgraphsForProcedures |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - PROC-H.*.*.*.* - matches calls of non-package procedures from host PROC-H<br>- *.*.*.*.P1 - matches calls of package-procedures named P1 |

### Session active user

Current user for session. Used for MSSQL processing.

| Parameter name | Description |
|---|---|
| Prompt | Session active user |
| Symbol | sql.activeUser |
| Type | STRING |
| Required | No |
| Default value | user |
| Example values | - user1<br>- user2 |

### Teradata Procedure to Default Database

Mapping Teradata's procedures to its default Database

| Parameter name | Description |
|---|---|
| Prompt | Teradata Procedure to Default Database |
| Symbol | sql.teradataProcedureToDatabaseMapping |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - H.DB.Schema.Proc1=DefaultDB<br>- ProcDb.*=TablesDb |

# SQL Loader Evaluator for JCL

Reads Oracle SQL*Loader scripts.

The SQL Loader Evaluator for JCL has no configurable parameters defined.

# SQL Loader Reader

Reads Oracle SQL*Loader files.

The SQL Loader Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Context

Context for database objects (ie. tables). It is used for creating database objects table. Database objects are stored and searched by suffix. Use of a context with dot-separated segments allows the specification of default values for host, database and schema. These are only default values and are overridden by explicit qualification of object references in a query. Context must consist of: three segments - (e.g., HOST.DBNAME.SCHEMA) - these segments are treated as default host, default database and default schema names. Specifying more than three segments causes job error.

| Parameter name | Description |
|---|---|
| Prompt | Context |
| Symbol | sqlLoader.sqlContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - db<br>- db.sciencedb<br>- db.sciencedb.astronomy |

## CONTROL files directory

Path to the directory where CONTROL files are stored.

| Parameter name | Description |
|---|---|
| Prompt | CONTROL files directory |
| Symbol | sqlLoader.ctrlFilesDirectory |

| Parameter name | Description |
|---|---|
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\input\SQLLoader\CONTROL\<br>- /opt/input/SQLLoader/CONTROL/ |

## Data files current directory

Directory where SQL*Loader script is run. It is used to create path of the input file when absolute path is not provided. If not provided - 'null' path is used for files without path.

| Parameter name | Description |
|---|---|
| Prompt | Data files current directory |
| Symbol | sqlLoader.dataCurrentDirectory |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - c:\loader\input<br>- /loader/input/ |

## Data files host name

Name of the host where File Implemented Data Resources are located. If not provided - the database host is used.

| Parameter name | Description |
|---|---|
| Prompt | Data files host name |
| Symbol | sqlLoader.dataHostName |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - dataHost-z4032<br>- sqlLoader.company.com |

## Input Mode

Defines if input files are Control and Loader files or only Control files

| Parameter name | Description |
|---|---|
| Prompt | Input Mode |
| Symbol | sqlLoader.inputMode |
| Type | MODULE_MODE |
| Required | No |
| Default value | Control and Loader file |
| Example values | - Control and Loader file<br>- Control file only |

## Input parameter files

The paths of files to be processed. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input parameter files |
| Symbol | sqlLoader.fileIncludeFilter |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input\sqlLoader\224_magic.<br>- /opt/input/fastload/225_more_magic.fl |

## PatternReplacer configuration path

The path to a JSON configuration file.

You can use the pattern replacer utility to modify input code that contains unsupported syntax or complex structures that a scanner can't parse. You can also use the pattern replacer when the source code contains highly dynamic code and code written with non-standard tools. The utility searches for patterns that you specify in the JSON file and replaces them with code that you define in the file.

**IMPORTANT:** Use caution when you modify the code with the pattern replacer. Modifications that distort the code or change code logic can lead to unexpected results.

For more information about how to create the JSON file, see Replacing Complex Patterns in Source Code for MetaDex Scanners.

| Parameter name | Description |
|---|---|
| Prompt | PatternReplacer configuration path |
| Symbol | sqlLoader.patternReplacerFile |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /home/user/replacer-config.json |

# SQL Plus Evaluator for JCL

Reads Oracle SQL*Plus scripts.

The SQL Plus Evaluator for JCL includes the following parameters:

## Spool split literal

User can specify some text literals - delimiter of column entries in output file - to change definition of output file schema. Example: in query 'Select A || '|' || B from T' the output file schema has one column, but if user define a '|' as a split literal, the output file schema has two columns - 'A' and 'B'

| Parameter name | Description |
|---|---|
| Prompt | Spool split literal |
| Symbol | sqlPlus.spoolSplitLiteral |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - ','<br>- '-'<br>- ';'<br>- '\|' |

# SSAS Reader

Reads SSAS files. Requires: - complete SSAS project files (.ds, .dsv, .cube, .dim, .dwproj); - connections to used databases to get metadata

The SSAS Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema

- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | ssas.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Input files

The paths of main files SSAS projects descriptors (*.dwproj). Directories should also contain files with extensions: *.ds, *.dsv, *.cube, *.dim . *Each entry can contain wildcard characters: "?" for single and "*"* for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | ssas.inputFiles |
| Type | FILELIST |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\RootDir\input\Analysis Services Tutorial*.*<br>- /home/dsadm/RootDir/input/Analysis Services Tutorial*.* |

## Process non-lineage objects (partitions)

Non-lineage objects like partitions are used to optimize the cube performance. Set this option to true when these objects also should be processed.

| Parameter name | Description |
|---|---|
| Prompt | Process non-lineage objects (partitions) |
| Symbol | ssas.processNonLineageObjects |
| Type | BOOLEAN |
| Required | Yes |
| Default value | false |
| Example values | - false<br>- true |

## SSAS BI Folder

SSAS BI Folder - the path of root directory in which the SSAS entities will be put in Metadata Repository. Effectively used only in IBM IGC, when EMD documents as output are in use. For other Metadata Repositories this parameter is not used.

| Parameter name | Description |
|---|---|
| Prompt | SSAS BI Folder |
| Symbol | ssas.biFolder |
| Type | STRING |
| Required | Yes |
| Default value | SSAS |
| Example values | - ssas_cubes<br>- ssas/2017/projects |

## SSAS BI Server

SSAS BI Server - the name of BI server in which the feed will be presented in MetaData Repository

| Parameter name | Description |
|---|---|
| Prompt | SSAS BI Server |
| Symbol | ssas.biServer |
| Type | STRING |
| Required | Yes |

| Parameter name | Description |
|---|---|
| Default value | SSAS_SERVER |
| Example values | - server01 |

# SSAS Tabular Reader

Reads SSAS Tabular files. Requires: - complete SSAS Tabular project files (*.bim, *.smproj, *.settings); - connections to used databases to get metadata

The SSAS Tabular Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | ssasTabular.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Input files

The paths of main files SSAS Tabular projects descriptors (*.smproj). Directories should also contain files with extensions: *.bim, *.settings*. *Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | ssasTabular.inputFiles |
| Type | FILELIST |
| Required | Yes |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Y:\RootDir\input\Analysis Services Tutorial*.smproj<br>- /home/dsadm/RootDir/input/Analysis Services Tutorial/*.smproj |

### SSAS BI Folder

SSAS BI Folder - the path of root directory in which the SSAS entities will be put in MetaData Repository

| Parameter name | Description |
|---|---|
| Prompt | SSAS BI Folder |
| Symbol | ssasTabular.biFolder |
| Type | STRING |
| Required | Yes |
| Default value | SSAS |
| Example values | - ssas_cubes<br>- ssas/2017/projects |

### SSAS BI Server

SSAS BI Server - the name of BI server in which the feed will be presented in MetaData Repository

| Parameter name | Description |
|---|---|
| Prompt | SSAS BI Server |
| Symbol | ssasTabular.biServer |
| Type | STRING |
| Required | Yes |
| Default value | SSAS_SERVER |
| Example values | - server01 |

# SSIS Processor

Reads SSIS files. Requires: - SSIS project files (.dtsx and .dtsxconfig); - connections to used databases to get metadata. Command for copying SSISPackageName SSIS Package from the SQLServer database to the specified directory on a local machine: DTUTIL /SQL SSISPackageName /COPY FILE;C:\RootDir\inputs \SSISPackageName.DTSX> /QUIET /QUIET parameter will overwrite the SSISPackageName.DTSX package if

it already exist in the specified location. Please note that dtutil tool should be installed on the same machine as MetaDex. For more options please refer to dtutil official documentation on https://docs.microsoft.com/.

The SSIS Processor includes the following parameters:

## Application host name

The name used as host name for files used by SSIS data flows.

| Parameter name | Description |
|---|---|
| Prompt | Application host name |
| Symbol | ssis.applicationHostName |
| Type | STRING |
| Required | No |
| Default value | SSIS_HOST |
| Example values | - ETL-host<br>- etl01 |

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine

- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | ssis.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Default variables values file

Path to the file with default variables values. File format consist of [GLOBAL] or/and custom function section.

---

You can define variables definitions: [GLOBAL] VariableName = VariableExpression

VariableName that contains only standard alphanumeric characters does not require any quotes. If it contains any special characters, it requires double-quotes (").

Example of variables definition: [GLOBAL] a = 42 b = 7 c = a < b ? a + 7 : b - 7 // It's 49 d = e(1,2) + 1 // Expression use call to function. "User::table" = 'table' || "User::tableSuffix" // It's string table concatenated with value of User::tableSuffix variable

---

You can define functions definitions: [GLOBAL] FunctionName = FunctionExpression

FunctionName can be unquoted if contains only standard alphanumeric characters but needs to be quoted with double quoted (") if contains any special character. FunctionName must contain list of arguments. List of argument is defined inside "(" ")" characters and consists of "?" sings separated by "," (one "?" sign for each function argument).

Example of function definition: [GLOBAL] a(?) = 1 a(?,?) = 2 b(?) = a(1) + 1 // Expression use call to another function. c(?,?) = d + 2 // Expression use reference to variable.

---

Additional sections can be provided to match only specific functions. To match an overloaded function please provide placeholders for its arguments. Inside the matched section you can also reference matched function arguments. Example of custom function sections:

- [host.db.schema.func(x)]
- z=x
- [host.db.schema.func(x,y)]
- z=x+y

---

To escape special characters such as \n \t use the \ character. For example, to define E:\file \tgtParameterized.csv, enter E:\\file\\tgtParameterized.csv

---

On the right side of assignments you can use simple expression language that can use values of already defined variables and functions or variables from current job context.

Supported expressions (case insensitive) are:

- literals
  - integer: 42
  - string: 'str'
  - string with quoted single quote: 'str\''
  - boolean: true, false
- equals: 42 == 7 -> FALSE
- not equals: 42 != 7 -> TRUE
- concatenations: 'str1' + 'str2' -> 'str1str2'
- function calls: Replace('abc', 'b', 'd') -> 'adc'
- tertiary operator:
  - 42 == 7 ? 'a' : 'b' -> 'b'
  - 42 != 7 ? 'a' : 'b' -> 'a'

Supported functions:

- Hash(str, maxOutputLen) - MD5, maxOutputLen between 1 and 32 inclusive
  - Hash('abcdefgh', 4) -> 'E8DC'
- Replace(str, from, to)
  - Replace('abc', 'b', 'D') -> 'aDc'
- ReplaceRegexp(str, regex, replacement) - replaces with Java regular expressions
  - ReplaceRegexp('abcde', 'b.*', 'f') -> 'af'

  - ReplaceRegexp('graph_id20', 'id(\d+)', '$1') -> 'graph_20'
- StringLengthLimit(str, limit, hashSize) - returns str[0..limit-hashSize] + Hash( str, hashSize )
  - StringLengthLimit('abcc', 3, 2) -> 'a26'
  - StringLengthLimit('abcc', 4, 2) -> 'abcc'
- StringLengthLimit(str, limit) - hashSize has default size 8
  - StringLengthLimit('abcdabcdabcd', 10) -> 'abE340600C'
- RegexpMatch(pattern, str) - tests if str matches pattern
  - pattern uses standard Java 8 regular expressions: https://docs.oracle.com/javase/8/docs/api/java/util/regex/Pattern.html
  - RegexpMatch('[A-Za-z]+','Abcd') -> TRUE
- Upper(str) - returns uppercased str
  - Upper('Abc') -> 'ABC'
- Lower(str) - return lowercased str
  - Lower('aBC') -> 'abc'
- Date(text,format) - returns object representing date with provided format.
  - Formats uses standard Java 8 SimpleDateFormat: https://docs.oracle.com/javase/8/docs/api/java/text/SimpleDateFormat.html
  - It is designed for default values usage
  - ex. Date('2017-10-31', 'yyyy-MM-dd')
- Contains/ContainsIgnoreCase(stack, needle) - tests if stack string contains needle string
  - Contains('abc', 'ab') -> TRUE
  - Contains('abc', 'AB') -> FALSE
  - ContainsIgnoreCase('abc', 'AB') -> TRUE
  - Contains('stack', 'needle') -> FALSE
- RtfToText(str) - return clean text from rtf string
  - RtfToText('{\rtf1\ansi{\fonttbl\f0\fswiss Helvetica;}\f0\pard text\par }') -> text

Supported comments:

- Single line: \ comment
- Multi line: /* comment */

| Parameter name | Description |
|---|---|
| Prompt | Default variables values file |
| Symbol | ssis.defaultValuesFilePath |
| Type | FILE |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Y:\RootDir\defaultVariableValues.txt<br>- /opt/RootDir/defaultVariableValues.txt |

## Disable processing of child packages

Exclude child packages processing

| Parameter name | Description |
|---|---|
| Prompt | Disable processing of child packages |
| Symbol | ssis.excludePackageTasks |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## DTSX Configuration file

DTSX package configuration file. It will be used to overwrite properties (such as InitialCatalog) of Connection Managers. This file will overwrite both MetaDex configuration and values from DTSX file. Only InitialCatalog property of Connection Manager is currently supported.

| Parameter name | Description |
|---|---|
| Prompt | DTSX Configuration file |
| Symbol | ssis.configFilePath |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\RootDir\package.dtsConfig<br>- /opt/RootDir/package.dtsConfig |

## DTSX Package to Project Param file mappings

If path of a DTSX package matches pattern given in "DTSX package" section, variables defined in the corresponding Project Param file will be included.

| Parameter name | Description |
|---|---|
| Prompt | DTSX Package to Project Param file mappings |
| Symbol | ssis.projectParamsMap |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - *.dtsx=Y:\RootDir\SSIS\Project.params<br>- *Package.dtsx=/home/RootDir/ssis/Project.params<br>- Y:\RootDir\SSIS\Package.dtsx=Y:\RootDir\SSIS\Project.params<br>- Y:\RootDir\SSIS*.dtsx=Y:\RootDir\SSIS\Project.params<br>- /home/RootDir/ssis/Package.dtsx=/home/RootDir/ssis/Project.params<br>- /home/RootDir/ssis/*.dtsx=/home/RootDir/ssis/Project.params |

## Enable processing of Event Handlers

Event Handlers defined in DTSX package will be processed.

| Parameter name | Description |
|---|---|
| Prompt | Enable processing of Event Handlers |
| Symbol | ssis.processEventHandlers |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Execute Package Task directory

SSIS using Execute Package Task can run another SSIS package. Use this parameter to show directory where packages referenced by Execute Package Task tasks are stored. Define mappings form SSIS directory with

exported packages to MetaDex directory for file connectors and mappings from connection manager name to MetaDex directory for relational connection managers

| Parameter name | Description |
|---|---|
| Prompt | Execute Package Task directory |
| Symbol | ssis.executePackageTaskMappings |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - FILE:Y:\SSIS\exportedPackages=d:\RootDir\exportedPackages<br>- DATABASE:SSIS_Flow_DB=d:\RootDir\exportedPackages |

## External Connection Manager Files

Paths of connection manager files used by the SSIS packages. (2012 version only.)

| Parameter name | Description |
|---|---|
| Prompt | External Connection Manager Files |
| Symbol | ssis.connectionManagers |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - C:\SSIS\Projects\MegaEtl\MegaDb.conmgr |

## File containing paths used by Foreach File Enumerator

Location of file containing file paths. This file will be used when processing Foreach File Enumerator to find files from defined folder. File should contain one path in one line.

Example content:

- /dir1/file1.txt
- /dir1/file2.txt
- /dir1/subdir1/file3.txt
- /dir2/file4.txt

| Parameter name | Description |
|---|---|
| Prompt | File containing paths used by Foreach File Enumerator |
| Symbol | ssis.foreachFileEnumeratorFilePath |

| Parameter name | Description |
|---|---|
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /path/to/file.txt<br>- C:\path\to\file.txt |

## Filenames mappings

These mappings will be used to resolve file path (i.e. for Excel file in Connection Manager). Part given in "From" section will be replaced (if matched) with part given in "To" column.

| Parameter name | Description |
|---|---|
| Prompt | Filenames mappings |
| Symbol | ssis.excelFilesMap |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - /export/home/RootDir/ssis/excel-files=C:\Excels<br>- Y:\RootDir\ssis\excel-files=C:\Excels |

## Instances to be excluded during processing

Expression which matches full path to instance to be excluded during processing. The path should be of the form: package followed by containers and ending with Data flow/Sql task/Script task name. Supported symbols:

- ** - match 0 or more path segments, can be used only one per path
- ? - match one character
- "" - match segment that may contains dots

Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Instances to be excluded during processing |
| Symbol | ssis.excludeInstances |
| Type | STRINGLIST |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Package.** - exclude all instances in 'Package'<br>- PackageName.**.DataFlow - exclude all instances called DataFlow in PackageName package<br>- Package.DataFlo? - last position match to any character<br>- Package."segment.with.dot".DataFlow - exclude Package.segment.dot.DataFlow<br>- *.container.DataFlow - excludes Package1.container.DataFlow, Package2.container.DataFlow |

## Log warn on missing variable

Decide to log warn message when there is usage of variable with null value

| Parameter name | Description |
|---|---|
| Prompt | Log warn on missing variable |
| Symbol | ssis.warnOnMissingVariable |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Log warn on not supported SSIS component

Decide to log warning when there is not supported SSIS component

| Parameter name | Description |
|---|---|
| Prompt | Log warn on not supported SSIS component |
| Symbol | ssis.unknownControlFlowComponentLoggingLevel |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Log warning about non-existing DTSX Configuration file

If DTSX package configuration file is missing a WARN will be logged, but this switch can be used to silence this warning .

| Parameter name | Description |
|---|---|
| Prompt | Log warning about non-existing DTSX Configuration file |
| Symbol | ssis.warnAboutNonExistingConfigFile |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Missing file paths definitions file

Path to the file with different advanced options for ssis package. Each line should contain mapping: optionName=optionValue optionName is defined as qualified component name followed by property name. Qualified component name is composed from package name, data flow name and component name. Currently two property names are supported

- FileName - defines missing file name
- DefValue - defines value for variable without value

If any part of qualified component name contains space character please scape them like this: my\ package.load*file*dataflow.read\ flat\ file Example of advanced options:

- my\ package.load_file_dataflow.read\ flat\ file.FileName=d:\ssis\inFiles\rawInput.txt
- my\ package.load_file_dataflow.read\ flat\ DefValue.User::FileNameDir=d:\ssis\inFiles\

| Parameter name | Description |
|---|---|
| Prompt | Missing file paths definitions file |
| Symbol | ssis.advancedOptionsFilePath |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\RootDir\advOpts.txt<br>- /opt/RootDir/advOpts.txt |

## Read SSIS Package configurations

Read SSIS Package configuration to override package variables expression and value. Currently we support SQL Server configuration type. Provide the configuration connection details using connections module parameter

| Parameter name | Description |
|---|---|
| Prompt | Read SSIS Package configurations |
| Symbol | ssis.readPackageConfiguration |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Recover lineage for unsupported custom component

If enabled, MetaDex will try to create lineage for unsupported custom component trying to collect missing data from adjacent components in path. Use with caution as it could create invalid lineage.

| Parameter name | Description |
|---|---|
| Prompt | Recover lineage for unsupported custom component |
| Symbol | ssis.customComponentLineageRecovery |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Scripts Root Directory Path

Root directory of input files. Relative paths of input files resolve in the following location: [Input root directory]. Absolute paths from input files must start from the [Input root directory] to be included into processing. [Input root directory] will not be visible on the output objects structure.

| Parameter name | Description |
|---|---|
| Prompt | Scripts Root Directory Path |
| Symbol | ssis.inputRoot |
| Type | DIR |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | N/A |
| Example values | - Y:\inputs\<br>- /opt/feed-etl/ |

## Sets charset to reading xml

File charset default set to UTF-8.

| Parameter name | Description |
|---|---|
| Prompt | Sets charset to reading xml |
| Symbol | ssis.defaultXmlCharset |
| Type | CHARSET |
| Required | No |
| Default value | UTF-8 |
| Example values | - UTF-8<br>- ISO-8859-1 |

## Skip disabled tasks

Do not process any packages or tasks marked as "Disabled" by SSIS.

| Parameter name | Description |
|---|---|
| Prompt | Skip disabled tasks |
| Symbol | ssis.skipDisabled |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Use database from Connection Manager

Take name of the database from Connection Manager definition instead of MetaDex configuration.

| Parameter name | Description |
|---|---|
| Prompt | Use database from Connection Manager |
| Symbol | ssis.useDbFromManagers |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Supported Components

- ADO NET Destination - Data Flow
- Aggregate - Data Flow
- Audit - Data Flow
- Cache Transform - Data Flow. Presented as pass trough transformation.
- Character Map - Data Flow
- Conditional Split - Data Flow
- Copy Column - Data Flow
- Custom Component - Data Flow. All known unimplemented custom components. Each output field has lineage relation to all input fields.
- Data Conversion - Data Flow
- Data Reader Source Adapter - Data Flow
- DataReader Source Adapter - Data Flow
- Derived Column - Data Flow
- Devart Salesforce Source - Data Flow
- Excel Destination - Data Flow
- Excel Source - Data Flow
- Export Column - Data Flow
- Flat File Destination - Data Flow
- Flat File Source - Data Flow
- Flexible File Destination - Data Flow
- Flexible File Source - Data Flow
- Fuzzy Grouping - Data Flow
- Fuzzy Lookup - Data Flow
- Import Column - Data Flow
- Lookup - Data Flow. Supported modes: CACHE, OLE_DB.

- Merge - Data Flow
- Merge Join - Data Flow
- Multicast - Data Flow
- ODBC Destination - Data Flow
- ODBC Source - Data Flow
- OLE DB Command - Data Flow
- OLE DB Destination - Data Flow
- OLE DB Source - Data Flow
- Oracle Destination - Data Flow
- Oracle Source - Data Flow
- Percentage Sampling - Data Flow
- Pivot - Data Flow
- Rational Flat File Source - Data Flow
- Recordset - Data Flow
- Row Count - Data Flow. Presented as pass trough transformation.
- Row Sampling - Data Flow. Presented as pass trough transformation.
- SAP BW Destination - Data Flow
- SAP BW Source - Data Flow
- SQL Server Destination - Data Flow
- SQL Server PDW Destination Adaptor - Data Flow
- Script Component - Data Flow. Scripts are not supported and are presented as component with fields, without transformation. WARN or ERROR log is logged, depending on parameter: "SsisReaderModule"."Log warn on not supported SSIS component".
- Slowly Changing Dimension - Data Flow
- Sort - Data Flow
- TaskFactory SalesForceDestination - Data Flow
- Term Extraction - Data Flow
- Term Lookup - Data Flow
- Union all - Data Flow
- Unpivot - Data Flow
- Data Flow Task - Control Flow
- File System Task - Control Flow
- Foreach Loop - Control Flow
- Sequence Container - Control Flow
- Execute SQL Task - Control Flow

**Note:** List of supported components are up to date to SSIS 2012+. For the older versions the list may be slightly different.

# SSIS Reader

Reads SSIS files. Supports two modes Files Mode - Provide the input files SSISDB Mode - Provide the SSISDB connection details to extract project files Command for copying SSISPackageName SSIS Package from the SQLServer database to the specified directory on a local machine: DTUTIL /SQL SSISPackageName /COPY FILE;C:\RootDir\inputs\SSISPackageName.DTSX> /QUIET /QUIET parameter will overwrite the SSISPackageName.DTSX package if it already exist in the specified location. Please note that dtutil tool should be installed on the same machine as MetaDex. For more options please refer to dtutil official documentation on https://docs.microsoft.com/.

The SSIS Reader includes the following parameters:

## Application host name

The name used as host name for files used by SSIS data flows.

| Parameter name | Description |
|---|---|
| Prompt | Application host name |
| Symbol | ssis.applicationHostName |
| Type | STRING |
| Required | Yes |
| Default value | SSIS_HOST |
| Example values | - ETL-host<br>- etl01 |

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field

- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | ssis.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Default variables values file

Path to the file with default variables values. File format consist of [GLOBAL] or/and custom function section.

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You can define variables definitions: [GLOBAL] VariableName = VariableExpression

VariableName that contains only standard alphanumeric characters does not require any quotes. If it contains any special characters, it requires double-quotes (").

Example of variables definition: [GLOBAL] a = 42 b = 7 c = a < b ? a + 7 : b - 7 // It's 49 d = e(1,2) + 1 // Expression use call to function. "User::table" = 'table' || "User::tableSuffix" // It's string table concatenated with value of User::tableSuffix variable

---

You can define functions definitions: [GLOBAL] FunctionName = FunctionExpression

FunctionName can be unquoted if contains only standard alphanumeric characters but needs to be quoted with double quoted (") if contains any special character. FunctionName must contain list of arguments. List of argument is defined inside "(" ")" characters and consists of "?" sings separated by "," (one "?" sign for each function argument).

Example of function definition: [GLOBAL] a(?) = 1 a(?,?) = 2 b(?) = a(1) + 1 // Expression use call to another function. c(?,?) = d + 2 // Expression use reference to variable.

---

Additional sections can be provided to match only specific functions. To match an overloaded function please provide placeholders for its arguments. Inside the matched section you can also reference matched function arguments. Example of custom function sections:

- [host.db.schema.func(x)]
- z=x
- [host.db.schema.func(x,y)]
- z=x+y

---

To escape special characters such as \n \t use the \ character. For example, to define E:\file \tgtParameterized.csv, enter E:\\file\\tgtParameterized.csv

---

On the right side of assignments you can use simple expression language that can use values of already defined variables and functions or variables from current job context.

Supported expressions (case insensitive) are:

- literals
  - integer: 42
  - string: 'str'
  - string with quoted single quote: 'str\''
  - boolean: true, false
- equals: 42 == 7 -> FALSE
- not equals: 42 != 7 -> TRUE
- concatenations: 'str1' + 'str2' -> 'str1str2'
- function calls: Replace('abc', 'b', 'd') -> 'adc'
- tertiary operator:
  - 42 == 7 ? 'a' : 'b' -> 'b'
  - 42 != 7 ? 'a' : 'b' -> 'a'

Supported functions:

- Hash(str, maxOutputLen) - MD5, maxOutputLen between 1 and 32 inclusive
  - Hash('abcdefgh', 4) -> 'E8DC'
- Replace(str, from, to)
  - Replace('abc', 'b', 'D') -> 'aDc'
- ReplaceRegexp(str, regex, replacement) - replaces with Java regular expressions
  - ReplaceRegexp('abcde', 'b.*', 'f') -> 'af'
  - ReplaceRegexp('graph_id20', 'id(\d+)', '$1') -> 'graph_20'
- StringLengthLimit(str, limit, hashSize) - returns str[0..limit-hashSize] + Hash( str, hashSize )
  - StringLengthLimit('abcc', 3, 2) -> 'a26'
  - StringLengthLimit('abcc', 4, 2) -> 'abcc'
- StringLengthLimit(str, limit) - hashSize has default size 8
  - StringLengthLimit('abcdabcdabcd', 10) -> 'abE340600C'
- RegexpMatch(pattern, str) - tests if str matches pattern
  - pattern uses standard Java 8 regular expressions: https://docs.oracle.com/javase/8/docs/api/java/util/regex/Pattern.html
  - RegexpMatch('[A-Za-z]+','Abcd') -> TRUE
- Upper(str) - returns uppercased str
  - Upper('Abc') -> 'ABC'
- Lower(str) - return lowercased str
  - Lower('aBC') -> 'abc'
- Date(text,format) - returns object representing date with provided format.
  - Formats uses standard Java 8 SimpleDateFormat: https://docs.oracle.com/javase/8/docs/api/java/text/SimpleDateFormat.html
  - It is designed for default values usage
  - ex. Date('2017-10-31', 'yyyy-MM-dd')
- Contains/ContainsIgnoreCase(stack, needle) - tests if stack string contains needle string
  - Contains('abc', 'ab') -> TRUE
  - Contains('abc', 'AB') -> FALSE
  - ContainsIgnoreCase('abc', 'AB') -> TRUE
  - Contains('stack', 'needle') -> FALSE
- RtfToText(str) - return clean text from rtf string
  - RtfToText('{\rtf1\ansi{\fonttbl\f0\fswiss Helvetica;}\f0\pard text\par }') -> text

Supported comments:

- Single line: \ comment

- Multi line: /* comment */

| Parameter name | Description |
|---|---|
| Prompt | Default variables values file |
| Symbol | ssis.defaultValuesFilePath |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\RootDir\defaultVariableValues.txt<br>- /opt/RootDir/defaultVariableValues.txt |

## Disable processing of child packages

Exclude child packages processing

| Parameter name | Description |
|---|---|
| Prompt | Disable processing of child packages |
| Symbol | ssis.excludePackageTasks |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## DTSX Configuration file

DTSX package configuration file. It will be used to overwrite properties (such as InitialCatalog) of Connection Managers. This file will overwrite both MetaDex configuration and values from DTSX file. Only InitialCatalog property of Connection Manager is currently supported.

| Parameter name | Description |
|---|---|
| Prompt | DTSX Configuration file |
| Symbol | ssis.configFilePath |
| Type | FILE |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - Y:\RootDir\package.dtsConfig<br>- /opt/RootDir/package.dtsConfig |

## DTSX Package to Project Param file mappings

If path of a DTSX package matches pattern given in "DTSX package" section, variables defined in the corresponding Project Param file will be included.

| Parameter name | Description |
|---|---|
| Prompt | DTSX Package to Project Param file mappings |
| Symbol | ssis.projectParamsMap |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - *.dtsx=Y:\RootDir\SSIS\Project.params<br>- *Package.dtsx=/home/RootDir/ssis/Project.params<br>- Y:\RootDir\SSIS\Package.dtsx=Y:\RootDir\SSIS\Project.params<br>- Y:\RootDir\SSIS*.dtsx=Y:\RootDir\SSIS\Project.params<br>- /home/RootDir/ssis/Package.dtsx=/home/RootDir/ssis/Project.params<br>- /home/RootDir/ssis/*.dtsx=/home/RootDir/ssis/Project.params |

## Enable processing of Event Handlers

Event Handlers defined in DTSX package will be processed.

| Parameter name | Description |
|---|---|
| Prompt | Enable processing of Event Handlers |
| Symbol | ssis.processEventHandlers |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Execute Package Task directory

SSIS using Execute Package Task can run another SSIS package. Use this parameter to show directory where packages referenced by Execute Package Task tasks are stored. Define mappings form SSIS directory with

exported packages to MetaDex directory for file connectors and mappings from connection manager name to MetaDex directory for relational connection managers

| Parameter name | Description |
|---|---|
| Prompt | Execute Package Task directory |
| Symbol | ssis.executePackageTaskMappings |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - FILE:Y:\SSIS\exportedPackages=d:\RootDir\exportedPackages<br>- DATABASE:SSIS_Flow_DB=d:\RootDir\exportedPackages |

## External Connection Manager Files

Paths of connection manager files used by the SSIS packages. (2012 version only.)

| Parameter name | Description |
|---|---|
| Prompt | External Connection Manager Files |
| Symbol | ssis.connectionManagers |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - C:\SSIS\Projects\MegaEtl\MegaDb.conmgr |

## File containing paths used by Foreach File Enumerator

Location of file containing file paths. This file will be used when processing Foreach File Enumerator to find files from defined folder. File should contain one path in one line.

Example content:

- /dir1/file1.txt
- /dir1/file2.txt
- /dir1/subdir1/file3.txt
- /dir2/file4.txt

| Parameter name | Description |
|---|---|
| Prompt | File containing paths used by Foreach File Enumerator |
| Symbol | ssis.foreachFileEnumeratorFilePath |

| Parameter name | Description |
|---|---|
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - /path/to/file.txt<br>- C:\path\to\file.txt |

## Filenames mappings

These mappings will be used to resolve file path (i.e. for Excel file in Connection Manager). Part given in "From" section will be replaced (if matched) with part given in "To" column.

| Parameter name | Description |
|---|---|
| Prompt | Filenames mappings |
| Symbol | ssis.excelFilesMap |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - /export/home/RootDir/ssis/excel-files=C:\Excels<br>- Y:\RootDir\ssis\excel-files=C:\Excels |

## Input files

List of *.dtsx files exported from Microsoft SSIS. Each file should contains at least one data flow. Only data flows are processed. Each entry can contain wildcard characters: "?" for single and ""* for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | ssis.sourceFile |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\RootDir\SSIS\ordering.dtsx<br>- /home/RootDir/ssis/ordering.dtsx |

## Instances to be excluded during processing

Expression which matches full path to instance to be excluded during processing. The path should be of the form: package followed by containers and ending with Data flow/Sql task/Script task name. Supported symbols:

- ** - match 0 or more path segments, can be used only one per path
- ? - match one character
- "" - match segment that may contains dots

Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Instances to be excluded during processing |
| Symbol | ssis.excludeInstances |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - Package.** - exclude all instances in 'Package'<br>- PackageName.**.DataFlow - exclude all instances called DataFlow in PackageName package<br>- Package.DataFlo? - last position match to any character<br>- Package."segment.with.dot".DataFlow - exclude Package.segment.dot.DataFlow<br>- *.container.DataFlow - excludes Package1.container.DataFlow, Package2.container.DataFlow |

## Log warn on missing variable

Decide to log warn message when there is usage of variable with null value

| Parameter name | Description |
|---|---|
| Prompt | Log warn on missing variable |
| Symbol | ssis.warnOnMissingVariable |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Log warn on not supported SSIS component

Decide to log warning when there is not supported SSIS component

| Parameter name | Description |
|---|---|
| Prompt | Log warn on not supported SSIS component |
| Symbol | ssis.unknownControlFlowComponentLoggingLevel |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Log warning about non-existing DTSX Configuration file

If DTSX package configuration file is missing a WARN will be logged, but this switch can be used to silence this warning .

| Parameter name | Description |
|---|---|
| Prompt | Log warning about non-existing DTSX Configuration file |
| Symbol | ssis.warnAboutNonExistingConfigFile |
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Missing file paths definitions file

Path to the file with different advanced options for ssis package. Each line should contain mapping: optionName=optionValue optionName is defined as qualified component name followed by property name. Qualified component name is composed from package name, data flow name and component name. Currently two property names are supported

- FileName - defines missing file name
- DefValue - defines value for variable without value

If any part of qualified component name contains space character please scape them like this: my\ package.load*file*dataflow.read\ flat\ file Example of advanced options:

- my\ package.load_file_dataflow.read\ flat\ file.FileName=d:\ssis\inFiles\rawInput.txt

- my\ package.load_file_dataflow.read\ flat\ DefValue.User::FileNameDir=d:\ssis\inFiles\

| Parameter name | Description |
|---|---|
| Prompt | Missing file paths definitions file |
| Symbol | ssis.advancedOptionsFilePath |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\RootDir\advOpts.txt<br>- /opt/RootDir/advOpts.txt |

## Read parameters from SSISDB

Read project and package parameters from SSISDB

| Parameter name | Description |
|---|---|
| Prompt | Read parameters from SSISDB |
| Symbol | ssis.readParametersFromSsisDb |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Read parameters from SSISDB using environment

Specify the environment to use to resolve project and package parameters from SSISDB If environment is not specified then parameters referencing environment variable will not be resolved

| Parameter name | Description |
|---|---|
| Prompt | Read parameters from SSISDB using environment |
| Symbol | ssis.readParametersFromSsisDbUsingEnvironment |
| Type | STRING |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | N/A |
| Example values | - ENV1<br>- ENV2 |

## Read SSIS Package configurations

Read SSIS Package configuration to override package variables expression and value. Currently we support SQL Server configuration type. Provide the configuration connection details using connections module parameter

| Parameter name | Description |
|---|---|
| Prompt | Read SSIS Package configurations |
| Symbol | ssis.readPackageConfiguration |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Recover lineage for unsupported custom component

If enabled, MetaDex will try to create lineage for unsupported custom component trying to collect missing data from adjacent components in path. Use with caution as it could create invalid lineage.

| Parameter name | Description |
|---|---|
| Prompt | Recover lineage for unsupported custom component |
| Symbol | ssis.customComponentLineageRecovery |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Scripts Root Directory Path

Root directory of input files. Relative paths of input files resolve in the following location: [Input root directory]. Absolute paths from input files must start from the [Input root directory] to be included into processing. [Input root directory] will not be visible on the output objects structure.

| Parameter name | Description |
|---|---|
| Prompt | Scripts Root Directory Path |
| Symbol | ssis.inputRoot |
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\inputs\<br>- /opt/feed-etl/ |

## Sets charset to reading xml

File charset default set to UTF-8.

| Parameter name | Description |
|---|---|
| Prompt | Sets charset to reading xml |
| Symbol | ssis.defaultXmlCharset |
| Type | CHARSET |
| Required | No |
| Default value | UTF-8 |
| Example values | - UTF-8<br>- ISO-8859-1 |

## Skip disabled tasks

Do not process any packages or tasks marked as "Disabled" by SSIS.

| Parameter name | Description |
|---|---|
| Prompt | Skip disabled tasks |
| Symbol | ssis.skipDisabled |
| Type | BOOLEAN |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | true |
| Example values | - true<br>- false |

## SSIS reader module mode

Specify SSIS reader module mode Use mode Files to provide input package files Requires:

- SSIS project files (.dtsx) through Input files module parameter
- [Optional] Connections manager files (.conmgr) through External Connection Manager Files module parameter
- [Optional] Project parameters file (.params) through DTSX Package to Project Param file mappings module parameter
- [Optional] Dtsx configuration file (.dtsxConfig) through DTSX Configuration file module parameter

Use mode SSISDB to extract project files from SSISDB Requires:

- SSISDB database connection name. To set the name, use SSISDB database connection name module parameter
- Data source connection for the SSISDB database. To provide the connection, use Data source connections module parameter

| Parameter name | Description |
|---|---|
| Prompt | SSIS reader module mode |
| Symbol | ssis.readerModuleMode |
| Type | MODULE_MODE |
| Required | No |
| Default value | Files |
| Example values | - Files<br>- SSISDB |

## SSISDB database connection name

Name of the SSISDB database connection. You use the connection to extract the project files from the SSISDB database You use the connection name to map to the connection registry entry with Data source connections module parameter

| Parameter name | Description |
|---|---|
| Prompt | SSISDB database connection name |
| Symbol | ssis.dbConnectionName |

| Parameter name | Description |
|---|---|
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - SSISDB<br>- SSISDB Database Connection |

## SSISDB extractor filters

Case insensitive include/exclude filters used to filter folders/projects/packages that have to be extracted. Filter should be in the format: Include|Exclude:FolderPattern:ProjectPattern:PackagePattern Use asterisk '*' to match any number of characters Use question mark '?' to match any single character

| Parameter name | Description |
|---|---|
| Prompt | SSISDB extractor filters |
| Symbol | ssis.dbExtractorFilters |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - Include:Folder1:Project1:Package1<br>- Exclude:Folder1:Project1:* |

## Use database from Connection Manager

Take name of the database from Connection Manager definition instead of MetaDex configuration.

| Parameter name | Description |
|---|---|
| Prompt | Use database from Connection Manager |
| Symbol | ssis.useDbFromManagers |
| Type | BOOLEAN |
| Required | No |
| Default value | false |
| Example values | - true<br>- false |

## Supported Components

- ADO NET Destination - Data Flow

- Aggregate - Data Flow
- Audit - Data Flow
- Cache Transform - Data Flow. Presented as pass trough transformation.
- Character Map - Data Flow
- Conditional Split - Data Flow
- Copy Column - Data Flow
- Custom Component - Data Flow. All known unimplemented custom components. Each output field has lineage relation to all input fields.
- Data Conversion - Data Flow
- Data Reader Source Adapter - Data Flow
- DataReader Source Adapter - Data Flow
- Derived Column - Data Flow
- Devart Salesforce Source - Data Flow
- Excel Destination - Data Flow
- Excel Source - Data Flow
- Export Column - Data Flow
- Flat File Destination - Data Flow
- Flat File Source - Data Flow
- Flexible File Destination - Data Flow
- Flexible File Source - Data Flow
- Fuzzy Grouping - Data Flow
- Fuzzy Lookup - Data Flow
- Import Column - Data Flow
- Lookup - Data Flow. Supported modes: CACHE, OLE_DB.
- Merge - Data Flow
- Merge Join - Data Flow
- Multicast - Data Flow
- ODBC Destination - Data Flow
- ODBC Source - Data Flow
- OLE DB Command - Data Flow
- OLE DB Destination - Data Flow
- OLE DB Source - Data Flow
- Oracle Destination - Data Flow
- Oracle Source - Data Flow
- Percentage Sampling - Data Flow
- Pivot - Data Flow
- Rational Flat File Source - Data Flow
- Recordset - Data Flow
- Row Count - Data Flow. Presented as pass trough transformation.
- Row Sampling - Data Flow. Presented as pass trough transformation.

- SAP BW Destination - Data Flow
- SAP BW Source - Data Flow
- SQL Server Destination - Data Flow
- SQL Server PDW Destination Adaptor - Data Flow
- Script Component - Data Flow. Scripts are not supported and are presented as component with fields, without transformation. WARN or ERROR log is logged, depending on parameter: "SsisReaderModule"."Log warn on not supported SSIS component".
- Slowly Changing Dimension - Data Flow
- Sort - Data Flow
- TaskFactory SalesForceDestination - Data Flow
- Term Extraction - Data Flow
- Term Lookup - Data Flow
- Union all - Data Flow
- Unpivot - Data Flow
- Data Flow Task - Control Flow
- File System Task - Control Flow
- Foreach Loop - Control Flow
- Sequence Container - Control Flow
- Execute SQL Task - Control Flow

# SSRS Reader

A module that reads Microsoft SQL Server Reporting Services (SSRS) files. To use the module, perform the following prerequisite tasks: - Get a user who can connect to the Reporting Server with the permission to view and download Reports and Shared Data Sources. - Optionally, connect to the databases that the model uses to get metadata. - To process MDX queries, get the SSAS projects used in the MDX queries.

The SSRS Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | ssrs.connections |
| Type | PROPERTIES |

| Parameter name | Description |
|---|---|
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Default values for Report Parameters

Path to the file with default variables values. File format consist of [GLOBAL] or/and custom function section.

---

You can define variables definitions: [GLOBAL] VariableName = VariableExpression

VariableName that contains only standard alphanumeric characters does not require any quotes. If it contains any special characters, it requires double-quotes (").

Example of variables definition: [GLOBAL] a = 42 b = 7 c = a < b ? a + 7 : b - 7 // It's 49 d = e(1,2) + 1 // Expression use call to function. "User::table" = 'table' || "User::tableSuffix" // It's string table concatenated with value of User::tableSuffix variable

---

You can define functions definitions: [GLOBAL] FunctionName = FunctionExpression

FunctionName can be unquoted if contains only standard alphanumeric characters but needs to be quoted with double quoted (") if contains any special character. FunctionName must contain list of arguments. List of argument is defined inside "(" ")" characters and consists of "?" sings separated by "," (one "?" sign for each function argument).

Example of function definition: [GLOBAL] a(?) = 1 a(?,?) = 2 b(?) = a(1) + 1 // Expression use call to another function. c(?,?) = d + 2 // Expression use reference to variable.

---

Additional sections can be provided to match only specific functions. To match an overloaded function please provide placeholders for its arguments. Inside the matched section you can also reference matched function arguments. Example of custom function sections:

- [host.db.schema.func(x)]
- `z=x`
- [host.db.schema.func(x,y)]
- `z=x+y`

---

To escape special characters such as \n \t use the \ character. For example, to define E:\file \tgtParameterized.csv, enter E:\\file\\tgtParameterized.csv

---

On the right side of assignments you can use simple expression language that can use values of already defined variables and functions or variables from current job context.

Supported expressions (case insensitive) are:

- literals
  - integer: 42
  - string: 'str'
  - string with quoted single quote: 'str\''
  - boolean: true, false
- equals: 42 == 7 -> FALSE
- not equals: 42 != 7 -> TRUE
- concatenations: 'str1' + 'str2' -> 'str1str2'
- function calls: Replace('abc', 'b', 'd') -> 'adc'
- tertiary operator:
  - 42 == 7 ? 'a' : 'b' -> 'b'
  - 42 != 7 ? 'a' : 'b' -> 'a'

Supported functions:

- Hash(str, maxOutputLen) - MD5, maxOutputLen between 1 and 32 inclusive
  - Hash('abcdefgh', 4) -> 'E8DC'
- Replace(str, from, to)
  - Replace('abc', 'b', 'D') -> 'aDc'
- ReplaceRegexp(str, regex, replacement) - replaces with Java regular expressions
  - ReplaceRegexp('abcde', 'b.*', 'f') -> 'af'
  - ReplaceRegexp('graph_id20', 'id(\d+)', '$1') -> 'graph_20'
- StringLengthLimit(str, limit, hashSize) - returns str[0..limit-hashSize] + Hash( str, hashSize )
  - StringLengthLimit('abcc', 3, 2) -> 'a26'
  - StringLengthLimit('abcc', 4, 2) -> 'abcc'
- StringLengthLimit(str, limit) - hashSize has default size 8
  - StringLengthLimit('abcdabcdabcd', 10) -> 'abE340600C'
- RegexpMatch(pattern, str) - tests if str matches pattern
  - pattern uses standard Java 8 regular expressions: https://docs.oracle.com/javase/8/docs/api/java/util/regex/Pattern.html
  - RegexpMatch('[A-Za-z]+','Abcd') -> TRUE
- Upper(str) - returns uppercased str
  - Upper('Abc') -> 'ABC'
- Lower(str) - return lowercased str
  - Lower('aBC') -> 'abc'
- Date(text,format) - returns object representing date with provided format.
  - Formats uses standard Java 8 SimpleDateFormat: https://docs.oracle.com/javase/8/docs/api/java/text/SimpleDateFormat.html
  - It is designed for default values usage
  - ex. Date('2017-10-31', 'yyyy-MM-dd')

- Contains/ContainsIgnoreCase(stack, needle) - tests if stack string contains needle string
  - Contains('abc', 'ab') -> TRUE
  - Contains('abc', 'AB') -> FALSE
  - ContainsIgnoreCase('abc', 'AB') -> TRUE
  - Contains('stack', 'needle') -> FALSE
- RtfToText(str) - return clean text from rtf string
  - RtfToText('{\rtf1\ansi{\fonttbl\f0\fswiss Helvetica;}\f0\pard text\par }') -> text

Supported comments:

- Single line: \ comment
- Multi line: /* comment */

| Parameter name | Description |
|---|---|
| Prompt | Default values for Report Parameters |
| Symbol | ssrs.defaultValuesFilePath |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\RootDir\defaultVariableValues.txt<br>- /opt/RootDir/defaultVariableValues.txt |

## Input files

The paths of files to be processed. Each entry can contain wildcard characters: "?" for single and "*" for multiple characters.

| Parameter name | Description |
|---|---|
| Prompt | Input files |
| Symbol | ssrs.inputFiles |
| Type | FILELIST |
| Required | No |
| Default value | N/A |
| Example values | - Y:\ssrs-xmls\monthly-sales.xml<br>- /opt/ssrs-xmls/quarterly-contracts.xml |

## Object exclude filters

List of exclude filters for SSRS reports. You can use here * and ? for any number of characters and any single character.

| Parameter name | Description |
|---|---|
| Prompt | Object exclude filters |
| Symbol | ssrs.excludeFilters |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - /SunriseXA/SCMLive/HH_S*<br>- /SunriseXA/SCMLive/HH_S_ExcludePrefix* |

## Object include filters

List of include filters for SSRS reports. You can use here * and ? for any number of characters and any single character.

| Parameter name | Description |
|---|---|
| Prompt | Object include filters |
| Symbol | ssrs.includeFilters |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - /SunriseXA/SCMLive/HH_S*<br>- /SunriseXA/SCMLive/HH_S_IncludePrefix* |

## SSRS BI Server

BI Server.

| Parameter name | Description |
|---|---|
| Prompt | SSRS BI Server |
| Symbol | ssrs.biServer |
| Type | STRING |
| Required | Yes |

| Parameter name | Description |
|---|---|
| Default value | SSRS_SERVER |
| Example values | - server01 |

## SSRS source

Choose files to read SSRS from existing xml files or databases to read SSRS from database via jdbc.

| Parameter name | Description |
|---|---|
| Prompt | SSRS source |
| Symbol | ssrs.readSource |
| Type | MODULE_MODE |
| Required | No |
| Default value | Files |
| Example values | - Files<br>- Web service |

## SSRS XML dump dir

If specified, this is the directory for dumping XML versions of SRS files.

| Parameter name | Description |
|---|---|
| Prompt | SSRS XML dump dir |
| Symbol | ssrs.dumpXmlDir |
| Type | DIR |
| Required | No |
| Default value | N/A |
| Example values | - Y:\RootDir\output\ssrs-dumps\ |

## Warn on missing report parameter value

Decide to log warn message when Report Parameter is used but not defined

| Parameter name | Description |
|---|---|
| Prompt | Warn on missing report parameter value |
| Symbol | ssrs.warnOnMissingReportParameterValue |

| Parameter name | Description |
|---|---|
| Type | BOOLEAN |
| Required | No |
| Default value | true |
| Example values | - true<br>- false |

## Web service 2010 url (ReportService2010.asmx)

The url that is used to connect to web service 2010 (ReportService2010.asmx)

| Parameter name | Description |
|---|---|
| Prompt | Web service 2010 url (ReportService2010.asmx) |
| Symbol | ssrs.wsUrl |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - https://mssql02.compactbi.local/ReportServer/ReportService2010.asmx |

## Web service domain

The domain that is used to connect to web service

| Parameter name | Description |
|---|---|
| Prompt | Web service domain |
| Symbol | ssrs.wsDomain |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - compact.org |

### Web service password

The password of user that is used to connect to web service

| Parameter name | Description |
|---|---|
| Prompt | Web service password |
| Symbol | ssrs.wsPassword |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - secret123 |

### Web service username

The name of user that is used to connect to web service

| Parameter name | Description |
|---|---|
| Prompt | Web service username |
| Symbol | ssrs.wsUsername |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - user1 |

# Talend Reader

Reads Talend files. Requires: - directory with exported Talend projects, - list of database connections used in given jobs. To export projects from Talend workspace click on: File -> Export -> General -> File System, then select [x] projects and enter a destination directory.

The Talend Reader includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Data source connections

The Data source connections parameter consists of the following parts:

- Connection key that you enter in the **Database Connection key** field
- Connection target that you enter in the **Resolved MDREPO entry** field

The database connection key is the logical identifier of a database connection. Your source technology generates it. The key can contain various special characters, as shown in the following examples:

- MssqlHost_db1
- Oracle/service
- user@machine
- machine_65535_user
- machine:65535;db=myDb
- mmsql,65535\\db

The syntax can be different for every database.If you are unsure about what values to enter, in the log file:

1. Search for "Creating missing connection for key:".
2. Copy the sequence of characters after ":", for example: MAGR-PC\SQLEXPRESS.SSIS
3. Paste this string into the **Database Connection key** field.

The MDREPO entry represents the target technology. It has the following syntax that is the same for all technologies: `<vendor name>:<username\>@<host name>.<name of the database defined in the connection registry>.<schema name>` Note that <schema name> and <username> are optional for some technologies, as shown in the following examples:

- ORACLE:user@prod-host.oraproddb.oraschema
- MSSQL:prod-host.db.schema
- ORACLE:user@host.db
- ORACLE:host.db

Other technologies might require the optional values, depending on the data from the technology that you use. For more information, see the sections appropriate for your technology in the *MetaDex Scanner Reference*.

| Parameter name | Description |
|---|---|
| Prompt | Data source connections |
| Symbol | talend.connections |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - ConnectionKey=ORACLE:user@prod-host.oraprod.oraschema - ConnectionKey - logical identifier of a database connection, "prod", "oraprod", "oraschema" and "user" are the respective host, database, default schema and user used by the connection<br>- ConnectionKey=MSSQL:user@prod-host.fund.dbo - Similar to the above example, but using MSSQL |

## Default values file path

Path to the file which contains data in following format:

- [Project.Job]
- tUniqueName.Column1=66
- tUniqueName.Column2="cards"

| Parameter name | Description |
|---|---|
| Prompt | Default values file path |
| Symbol | talend.defaultValuesFilePath |
| Type | FILE |
| Required | No |
| Default value | N/A |
| Example values | - Y:\RootDir\defaultValues.txt<br>- /opt/RootDir/defaultValues.txt |

## Job context override

List of Talend job names set in properties mapped to context (execution parameters defined for each job) which will override default context set in *.item file

| Parameter name | Description |
|---|---|
| Prompt | Job context override |
| Symbol | talend.jobContextOverride |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - tRunJob=Default<br>- tMap=Test |

## Job parameter override

These mappings can be used to override job parameter values. Specify parameter name using job qualified name and parameter name separated by colon. Wildcards '*' and '?' also can be used in job name.

| Parameter name | Description |
|---|---|
| Prompt | Job parameter override |
| Symbol | talend.parameterOverride |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - Folder1/Folder2/job:paramName=parameterValue<br>- *:parameterName=parameterValue |

## Log warn on missing variable

Decide about logging warn message when there is usage of variable with unknown value

| Parameter name | Description |
|---|---|
| Prompt | Log warn on missing variable |
| Symbol | talend.logWarnOnMissingVariable |
| Type | BOOLEAN |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | true |
| Example values | - true<br>- false |

## Talend context files map

List of Talend context files mapped to local files, allowing MetaDex to use local copies of files

| Parameter name | Description |
|---|---|
| Prompt | Talend context files map |
| Symbol | talend.contextFilesMap |
| Type | PROPERTIES |
| Required | No |
| Default value | N/A |
| Example values | - /export/home/RootDir/talend/contextfile=C:\PrmFiles\contextfile<br>- Y:\RootDir\talend\contextfile=C:\PrmFiles\file |

## Talend export directory

Path to the directory of exported Talend projects

| Parameter name | Description |
|---|---|
| Prompt | Talend export directory |
| Symbol | talend.exportDirectory |
| Type | DIR |
| Required | Yes |
| Default value | N/A |
| Example values | - Y:\input\etc\EXPORT<br>- /opt/input/etc/EXPORT |

## Talend file host name

The name of the server where the files are located

| Parameter name | Description |
|---|---|
| Prompt | Talend file host name |
| Symbol | talend.fileHost |
| Type | STRING |
| Required | Yes |
| Default value | TALEND_HOST |
| Example values | - ETL-host<br>- etl01 |

## Talend input filters

Case sensitive include/exclude filters, allowing to filter Talend jobs before processing. These filters use the following language: (Include|Exclude) : Pattern

where: Include - it's include filter Exclude - it's exclude filter Pattern - it's filter syntax supported wildcards:

- Asterisk for 0 or more characters
- Question mark for a single character

| Parameter name | Description |
|---|---|
| Prompt | Talend input filters |
| Symbol | talend.inputFilters |
| Type | STRINGLIST |
| Required | No |
| Default value | N/A |
| Example values | - Include : * # process all jobs -> default Exclude : /x/y/z # exclude all jobs from folder /x/y/z<br>Include : *prod # process all jobs ending with "prod" |

## Supported Components

- tAzureStorageConfiguration - Cloud
- tAzureStorageConnection - Cloud
- tAzureStoragePut - Cloud
- tHDFSCompare - Cloud
- tHDFSConfiguration - Cloud
- tHDFSConnection - Cloud
- tHDFSCopy - Cloud

- tHDFSDelete - Cloud
- tHDFSExist - Cloud
- tHDFSGet - Cloud
- tHDFSInput - Cloud
- tHDFSList - Cloud
- tHDFSOutput - Cloud
- tHDFSOutputRaw - Cloud
- tHDFSProperties - Cloud
- tHDFSPut - Cloud
- tHDFSRowCount - Cloud
- tS3Close - Cloud
- tS3Connection - Cloud
- tS3Get - Cloud
- tS3List - Cloud
- tS3Put - Cloud
- tSnowflakeClose - Cloud
- tSnowflakeInput - Cloud
- tSnowflakeOutput - Cloud
- tSnowflakeRow - Cloud
- tAmazonAuroraClose - Database
- tAmazonAuroraCommit - Database
- tAmazonAuroraRollback - Database
- tCreateTable - Database
- tDB2Close - Database
- tDB2Commit - Database
- tDB2Connection - Database
- tDB2Input - Database
- tDB2Output - Database
- tDB2Rollback - Database
- tDB2Row - Database
- tDB2SP - Database
- tELTMSSqlInput - Database
- tELTMSSqlMap - Database
- tELTMSSqlOutput - Database
- tELTNetezzaInput - Database
- tELTNetezzaMap - Database
- tELTNetezzaOutput - Database
- tELTOracleInput - Database
- tELTOracleMap - Database

- tELTOracleOutput - Database
- tELTTeradataInput - Database
- tELTTeradataMap - Database
- tELTTeradataOutput - Database
- tHiveClose - Database
- tHiveConnection - Database
- tHiveCreateTable - Database
- tHiveInput - Database
- tHiveLoad - Database
- tHiveOutput - Database
- tHiveRow - Database
- tJDBCClose - Database
- tJDBCCommit - Database
- tJDBCConnection - Database
- tJDBCInput - Database
- tJDBCOutput - Database
- tJDBCRollback - Database
- tMSSqlBulkExec - Database
- tMSSqlClose - Database
- tMSSqlColumnList - Database
- tMSSqlCommit - Database
- tMSSqlConnection - Database
- tMSSqlInput - Database
- tMSSqlLastInsertId - Database
- tMSSqlOutput - Database
- tMSSqlOutputBulk - Database
- tMSSqlOutputBulkExec - Database
- tMSSqlRollback - Database
- tMSSqlRow - Database
- tMSSqlSP - Database
- tMSSqlTableList - Database
- tNetezzaBulkExec - Database
- tNetezzaClose - Database
- tNetezzaCommit - Database
- tNetezzaConnection - Database
- tNetezzaInput - Database
- tNetezzaNzLoad - Database
- tNetezzaOutput - Database
- tNetezzaRollback - Database

- tNetezzaRow - Database
- tOracleBulkExec - Database
- tOracleClose - Database
- tOracleCommit - Database
- tOracleConnection - Database
- tOracleInput - Database
- tOracleOutput - Database
- tOracleOutputBulk - Database
- tOracleOutputBulkExec - Database
- tOracleRollback - Database
- tOracleRow - Database
- tOracleSP - Database
- tOracleTableList - Database
- tParseRecordSet - Database
- tRedshiftClose - Database
- tRedshiftCommit - Database
- tRedshiftConnection - Database
- tRedshiftInput - Database
- tRedshiftOutput - Database
- tRedshiftRollback - Database
- tRedshiftRow - Database
- tSnowflakeConnection - Database
- tSQLDWHBulkExec - Database
- tSQLDWHConnection - Database
- tSQLDWHInput - Database
- tSQLDWHOutput - Database
- tSQLDWHRow - Database
- tTeradataClose - Database
- tTeradataCommit - Database
- tTeradataConnection - Database
- tTeradataInput - Database
- tTeradataOutput - Database
- tTeradataRollback - Database
- tTeradataRow - Database
- tFileArchive - File
- tFileCompare - File
- tFileCopy - File
- tFileDelete - File
- tFileExist - File

- tFileInputDelimited - File
- tFileInputExcel - File
- tFileInputFullRow - File
- tFileInputPositional - File
- tFileInputRaw - File
- tFileInputXML - File
- tFileList - File
- tFileOutputDelimited - File
- tFileOutputExcel - File
- tFileOutputPositional - File
- tFileOutputRaw - File
- tFileOutputXML - File
- tFileRowCount - File
- tFileUnarchive - File
- tAddCRCRow - Processing
- tAggregateRow - Processing
- tAggregateSortedRow - Processing
- tBufferInput - Processing
- tBufferOutput - Processing
- tContextLoad - Processing
- tConvertType - Processing
- tDenormalize - Processing
- tDenormalizeSortedRow - Processing
- tDie - Processing
- tExtractDelimitedFields - Processing
- tExtractJSONFields - Processing
- tExtractPositionalFields - Processing
- tExtractRegexFields - Processing
- tFilterColumns - Processing
- tFilterRow - Processing
- tFixedFlowInput - Processing
- tFlowToIterate - Processing
- tForeach - Processing
- tHashInput - Processing
- tHashOutput - Processing
- tHttpRequest - Processing
- tInfiniteLoop - Processing
- tIterateToFlow - Processing
- tJava - Processing

- tJoin - Processing
- tLogCatcher - Processing
- tLogRow - Processing
- tLoop - Processing
- tMap - Processing
- tNormalize - Processing
- tPostjob - Processing
- tPrejob - Processing
- tReplace - Processing
- tReplaceList - Processing
- tReplicate - Processing
- tRESTClient - Processing
- tRowGenerator - Processing
- tRunJob - Processing
- tSampleRow - Processing
- tSchemaComplianceCheck - Processing
- tSendMail - Processing
- tSetGlobalVar - Processing
- tSleep - Processing
- tSortRow - Processing
- tSplitRow - Processing
- tSystem - Processing
- tUniqRow - Processing
- tUnite - Processing
- tWarn - Processing
- tWriteJSONField - Processing

# Temporary Table Names Post Processor

GraphPostProcessorModule is used for refining previously created graph, in order to provide proper data lineage. Doesn't need to be configured.

The Temporary Table Names Post Processor has no configurable parameters defined.

# Unity Lineage

A module for retrieving lineage from Unity Catalog.

The Unity Lineage includes the following parameters:

## Connection registry path

Path to the connection registry

| Parameter name | Description |
|---|---|
| Prompt | Connection registry path |
| Symbol | global.repositoryPath |
| Type | DIR_PATH |
| Required | No |
| Default value | ${SCANNERS_WORKSPACE}/MDREPO |
| Example values | - Y:\workspace\etc\MDREPO<br>- /opt/workspace/etc/MDREPO |

## Delta Lake SQL Context

Delta Lake SQL context

| Parameter name | Description |
|---|---|
| Prompt | Delta Lake SQL Context |
| Symbol | databricks.sqlContext |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - DELTALAKE-HOST.DELTALAKE-DATABASE.DEFAULT |

## Lineage from Unity

Lineage from Unity Catalog: <br>Disabled. Default option. Doesn't extract lineage from Unity Catalog. <br>Enabled for filtered assets. Extracts lineage from assets selected in filter conditions. <br>Enabled for all assets. Extracts complete lineage from Unity Catalog, including referenced resources.

| Parameter name | Description |
|---|---|
| Prompt | Lineage from Unity |
| Symbol | databricks.lineageFromUnity |
| Type | MODULE_MODE |
| Required | No |

| Parameter name | Description |
|---|---|
| Default value | Disabled |
| Example values | - Disabled<br>- Enabled for filtered assets<br>- Enabled for all assets |

## Workspace authentication token

Authentication token for the Databricks workspace access. A random string of 38 symbols generated by Databricks.

| Parameter name | Description |
|---|---|
| Prompt | Workspace authentication token |
| Symbol | databricks.authenticationToken |
| Type | SENSITIVE |
| Required | No |
| Default value | N/A |
| Example values | - dapi8be46d02f290489a1defbad48bbbbf77-3 For information about generating the authentication token, see: https://docs.databricks.com/dev-tools/api/latest/authentication.html#generate-a-personal-access-token |

## Workspace host

Host name of Databricks workspace.

| Parameter name | Description |
|---|---|
| Prompt | Workspace host |
| Symbol | databricks.workspaceHost |
| Type | STRING |
| Required | No |
| Default value | N/A |
| Example values | - databricksHost.com<br>- databricksHost.com:1234 |

APPENDIX A

# Supported Database Types

## Supported Database Types

The table below lists the database types supported by MetaDex. To use a database named in the left-hand column in a configuration file, the corresponding value from the right-hand column should be used.

The following table lists the database types supported by MetaDex.

| Database type | Identification string |
|---|---|
| Db2 | DB2 |
| Db2 ISeries | DB2ISERIES |
| Db2 zOS | DB2ZOS |
| Fabric Data Warehouse | FABRIC_WAREHOUSE |
| Fabric Data Lakehouse | FABRIC_LAKEHOUSE |
| MSSQL | MSSQL |
| NETEZZA | NETEZZA |
| Oracle | ORACLE |
| Sybase | SYBASE |
| Synapse | SYNAPSE |
| Teradata | TERADATA |
| AWS ATHENA | ATHENA |
| AWS GLUE | AWS_GLUE |
| Big Query | BIG_QUERY |
| Databricks | DATABRICKS |

| Database type | Identification string |
|---|---|
| Greenplum | GREENPLUM |
| H2 | H2 |
| SAP HANA | HANA |
| Hive | HIVE |
| Hive Cloudera | HIVE_CDP |
| IBM IMS | IBM_IMS |
| IntersystemsCache | INTERSYSTEMS_CACHE |
| MariaDB | MARIADB |
| Ms Access | MSACCESS |
| Excel | MSSQL_EXCEL |
| MySQL | MYSQL |
| POSTGRESQL | POSTGRESQL |
| Redshift | REDSHIFT |
| Sas | SAS |
| Snowflake | SNOWFLAKE |
| Azure Storage | AZURE_STORAGE |
| FTP | FTP |
| HDFS | HDFS |
| MongoDB | MONGODB |
| Amazon S3 | S3 |
| VSAM | VSAM |
| SAP | SAP |
| Generic | GENERIC |

APPENDIX B

# Supported Extra Properties

## Scanners.properties file

The scanners.properties file contains properties you can configure for the scanners. You can configure these properties in the SCANNERS_WORKSPACE/etc/scanners.properties file.

You can configure the following scanner properties in the properties file:

**scanners.cobol.copyDirectiveMaxNumberOfLines**

Determines the maximum number of copy directive lines that the MetaDex COBOL scanner can process. If exceeded, the scanner processes the copy directive that fits in the limit. The job succeeds with a warning.

Default is "150"

**scanners.connectionRegistry.testingTableAccessQueryTimeoutInSeconds**

The number of seconds before a single table access query times out in the Test Connection repository tool. Enter a timeout value in seconds.

Default is "60"

**scanners.dataStage.connectAllToAllFieldsForUnsupportedStage**

Forces the previous logics when processing unsupported stages. That logics caused that fields of unsupported components were linked with all fields of preceding components. Currently only fields with same names are connected otherwise they are not connected and have UNKNOWN expression. That logics caused graphs were very complex, logs were very large and processing time was very long. It also eventually led to java.lang.OutOfMemoryError: Java heap space issues.

Default is "false"

**scanners.debug.dumpData**

Creates extra debug data in the following location: SCANNERS_WORKSPACE/debug

Default is "false"

**scanners.debug.logTables**

Saves log information about each table to read from the Connection Registry. It contains table and column names.

Default is "false"

**scanners.debug.threadDump.frequency**

Determines the frequency of debug thread dumps in the ISO 8601 format. You can use this property when the application becomes unresponsive. For example, to set the frequency to 5 minutes, enter the following value: PT5M.

Default is "PT10M"

**scanners.debug.threadDump.maxFrames**

Defines the maximum number of frames captured in thread dumps other than the main thread. The main thread captures all frames in the thread dump. Enter 0 to capture all frames for other threads.

Default is "30"

**scanners.edc.idGenerationStrategy**

Defines the ID generation strategy. To define the strategy, you can use the following values:

- useObjectNames - -generates IDs based on object names.
- useObjectTypesAndNames - generates IDs based on object names and includes the object type name in each segment of the ID.
- useObjectTypesAndNamesWithMd5 - uses the Md5 hash algorithm to generate IDs based object names and includes the object type name in each segment of the ID.

Default is "useObjectTypesAndNamesWithMd5"

**scanners.io.encoding**

Defines the character encoding used to read and write files. If this property is not available in your job configuration in the MetaDex tool, configure it in the scanners.properties file.

Default is "UTF-8"

**scanners.io.maxCsvRecordSize**

The maximum number of characters in a record of a CSV file.

Default is "3000000"

**scanners.jcl.advancedColumnsComputing**

Enables advanced byte-level computing of data set columns.

Default is "false"

**scanners.lineage.maxFieldHierarchyExpansionDepth**

The maximum field hierarchy depth to expand an expression and resolve calculation complexity.

Default is "400"

**scanners.lineage.maxFieldHierarchySize**

The maximum field hierarchy size to expand an expression, to resolve calculation complexity, and to limit the number of relations in models such as Advanced SQL and Mainframe.

Default is "1000"

**scanners.platform.auth.domain**

The domain that you use to authenticate to the Informatica domain user management. Required if the domains.infa file specifies more than one domain.

There is no default value

**scanners.platform.auth.migration.domain-order-search**

Security domain order used to search for users/groups during the LDAP -> EDC authentication migration separated by semicolon. Values are case-insensitive.

Default is ""

**scanners.platform.auth.migration.drop-ldap-entries**

Drop all LDAP-related user/group entries if LDAP usage is no longer desired. Caution: This option removes all previously configured LDAP permissions from the database.

Default is "false"

**scanners.platform.auth.migration.fail-on-not-found**

Interrupt the application if a user/group cannot be found in a specific security domain.

Default is "true"

**scanners.platform.auth.password**

The password that you use to authenticate to the Informatica domain user management.

Default is ""

**scanners.platform.auth.security-domain**

The security domain of the user that you use to authenticate to the Informatica domain user management.

Default is ""

**scanners.platform.auth.sync-interval**

The caches synchronization interval in hours or in ISO8601 format, for example: PT30M = 30min

Default is "PT2H"

**scanners.platform.auth.username**

The username that you use to authenticate to the Informatica domain user management.

Default is ""

**scanners.platform.auth.web.admin-groups**

Allows to specify groups within EDC Security Domains that you map to MetaDex admin groups. Use the following syntax to specify a single group: 'domainName/groupName'

Default is ""

**scanners.platform.auth.web.admin-groups-separator**

Allows to specify the separator for admin groups

Default is ";"

**scanners.platform.auth.web.admin-users**

Allows to specify users within EDC Security Domains that you map to MetaDex admin users. Use the following syntax to specify a single user: 'domainName/userName'

Default is ""

**scanners.platform.auth.web.admin-users-separator**

Allows to specify the separator for admin users

Default is ";"

**scanners.platform.auth.web.repository-viewer-groups**

Allows to specify groups within EDC Security Domains that you map to MetaDex repository viewer groups. Use the following syntax to specify a single group: 'domainName/groupName'

Default is ""

**scanners.platform.auth.web.repository-viewer-groups-separator**

Allows to specify the separator for repository viewer groups

Default is ";"

**scanners.reports.html-output**

The log level of HTML reports. You can choose to see all logs or warnings and errors only. You can use the following values:

- FULL - A log-level filter for the HTML report. Generates an HTML report with all logs.
- ISSUES_ONLY - Generates an HTML report with only warnings and errors.
- NONE - Disables the generation of HTML reports.

Default is "NONE"

**scanners.reports.report-filename-pattern**

Defines the pattern of report file names. You can use the following parameters:

- %d - Finish date.
- %u - Job UUID (Universal Unique Identifier).
- %p - Job name.
- %n - Blueprint name or configuration file name.

When the operating system identifies an illegal character in the file naming rules, it replaces the character with a dash.

Default is "report-%d-%u"

**scanners.reports.xlsx-output**

Configures XLSX reports. You can use the following values:

- NONE - Disables XLSX reports
- FULL - Generates an XLSX report with all logs.

Default is "NONE"

**scanners.sas.useColumnsFromNonHeaderPutStatement**

Determines whether you can use columns from the PUT statement without a header.

Default is "false"

**scanners.security.additional-keys-to-sanitize**

Hides property values based on configured key words. Properties are hidden from job logs, server logs, and the Configuration tab of the Administration page. You can hide properties that end with a specific word, such as "bearer," or properties that contain a substring, such as "authdetails." To configure a substring, use regex. To add multiple security words, put them in a comma-separated list. For example, scanners.security.additional-keys-to-sanitize=bearer, *authdetails* The following properties that end with the configured key words are always hidden:

- password

- pass
- secret
- key
- token
- vcap services

In addition, all properties that contain the word "credentials" are hidden.

Default is ""

**scanners.server.data-locations.exclude**

Not allowed paths where only admin can upload data with a browser

Default is "${SCANNERS_WORKSPACE}/etc,${SCANNERS_WORKSPACE}/MDREPO"

**scanners.server.data-locations.include**

Allowed paths where users can upload data with a browser

Default is "${SCANNERS_WORKSPACE}"

**scanners.sql.enableSlowFetchViewDataTypesInTeradata**

Enable slow fetch view data types in Teradata

Default is "false"

**scanners.sql.procedureWithUnknownArgumentsExpandLimit**

Defines the number of arguments expanded in procedure call representations. When exceeded, only the number of arguments is shown.

Default is "20"

**scanners.sql.rowsPaginationInBigQuery**

Allow to define custom limit of rows in query pagination in Google Big Query (default: 1000)

Default is "1000"

**scanners.sql.unknownValueRepresentation**

Defines the representation of the unknown values in the Advanced PL/SQL engine.

Default is "UNKNOWN"

**scanners.sql.useExprVisitorInPLSQL**

Enable visitor based expression resolving in PL/SQL

Default is "false"

**scanners.sql.useNewNetezza**

Enable new netezza processing

Default is "true"

**scanners.sso.auth.admin-groups**

Allows to specify groups within SSO Identity Provider that you map to MetaDex admin groups. Use the following syntax to specify a single group: 'sso_id/groupName'

Default is ""

**scanners.sso.auth.admin-users**

Allows to specify SSO users that are mapped to MetaDex admin users. When specifying users case sensitivity has to be preserved. Admin user entries have to be separated with semicolon. Use the following syntax to specify users: 'sso*id/userName1;sso*id/username2'

Default is ""

**scanners.sso.auth.enabled**

Allows to enable SSO integration

Default is "false"

**scanners.sso.auth.group-attribute**

Allows to specify SSO group attribute name

There is no default value

**scanners.sso.auth.repository-viewer-groups**

Allows to specify groups within SSO Identity Provider that you map to MetaDex repository viewer groups. Use the following syntax to specify a single group: 'sso_id/groupName'

Default is ""

**scanners.util.correctTempTableNameInAssetsCorrector**

Recovers original temporary table names while correcting assets.

Default is "false"

**scanners.util.filePathLimit**

The maximum length allowed for a file path. If the path exceeds the limit, the job truncates the path and appends a unique hash to it within the character limit.

Default is "255"

**scanners.visualization.dotExecutable**

Name of the Graphviz DOT program that is present on the system's PATH.

Default is "dot"

**scanners.visualization.htmlFieldTypeLengthLimit**

The maximum field type length for an HTML visualization.

Default is "128"

**scanners.visualization.layoutMethod**

Defines the tool used to create and lay out a graph. To configure the parameter, use the following values:

- INTERNAL - a built-in default solution.
- GRAPHVIZ_NATIVE - native Graphviz software. It requires Graphviz to be installed. It also requires the DOT program to be available on the operating system's PATH.

Default is "INTERNAL"

**scanners.visualization.maxGraphNodes**

Defines the maximum number of components and fields in one graph. When exceeded, the layout computing is skipped.

Default is "50000"

**scanners.visualization.maxTimeInSeconds**

Defines the maximum number of seconds to generate a visualization. When exceeded, the process is terminated. You can decrease or increase this time as needed.

Default is "120"

**scanners.visualization.timeToDumpInSeconds**

Defines the maximum number of seconds to generate a visualization. If the visualization generation exceeds this time, and you set the scanners.debug.dumpData data dumping property, the graph is stored as a JSON file in the debug directory.

Default is "600"

**scanners.web.security.restricted-file-extensions**

A comma-separated list of file types that you must not upload. Example value: '.exe,.jsp,.js'. To allow all possible file types, set it as empty.

Default is ".exe,.jsp,.js"